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*A study on calculus of several variables,  
surfaces and calculus of variations.*

by

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## ABSTRACT

We wish to study how the usual notions of continuity and differentiability in single variable translates to the multivariable realm and how it necessarily leads to newer ideas of thinking about derivatives, tangents and normals. We explore a few important theorems pertaining to what we study.

We also wish to study the notion of surfaces in higher dimensions which will enable us to explore calculus of variations from a purely mathematical standpoint.

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## 2 Limits and continuity

We start by introducing the concept of limits and continuity in *higher dimensions*<sup>1</sup> of  $\mathbb{R}$  such as  $\mathbb{R}^2, \mathbb{R}^3$  or  $\mathbb{R}^n$ . It is intuitively clear that to define *approachability* for a functional value we must be needing a notion of *distance* in higher dimensions. Hence, we proceed with our definitions and notations for these quantities.

**Definition 2.1** (Norm in Euclidean space/Modulus): Let  $x \equiv (x^1, x^2, \dots, x^n) \in \mathbb{R}^n$  be an element. We say the *modulus* of  $x$  in  $\mathbb{R}^n$  denoted by  $|x|$  is

$$|x| = \sqrt{(x^1)^2 + (x^2)^2 + \dots + (x^n)^2} \quad (1)$$

where  $(x^1, x^2, \dots, x^n)$  is the coordinate of  $x$  w.r.t the standard basis of  $\mathbb{R}^n$ .

The norm in a space is a more general concept that will be introduced later. It is sufficient to note that the standard norm in an euclidean space is the modulus.

**Definition 2.2** (Open Set): We say that  $\Omega \subseteq \mathbb{R}^n$  is open iff  $\forall x_0 \in \Omega, \exists \delta > 0 :$

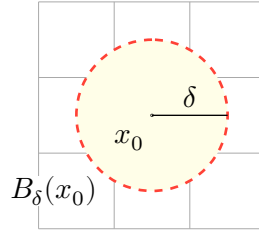
$$B_\delta(x_0) := \{x \in \mathbb{R}^n \mid |x_0 - x| < \delta\} \subset \Omega \quad (2)$$

---

<sup>1</sup>We will provide the definition for more abstract spaces later to introduce the notion of differentiability.

This is analogous to the concept of open intervals in  $\mathbb{R}$ .

The set  $B_\delta(x_0)$  is also termed as an “open ball” centered at  $x_0$  having radius  $\delta$ .



An open set (or an “open ball”) in  $\mathbb{R}^2$ .

**Definition 2.3** (Limit Point): Let  $\Omega$  be an open set in  $\mathbb{R}^n$ . We say that  $x_0 \in \mathbb{R}^n$  is a limit point or an *accumulation point* of  $\Omega$  iff

$$\forall \epsilon > 0, B_\epsilon(x_0) \cap (\Omega \setminus \{x_0\}) \neq \emptyset$$

**Definition 2.4** (Limit): Let  $f : \Omega \rightarrow \mathbb{R}$  be a function such that  $\Omega$  is open in  $\mathbb{R}$ . Let  $x_0$  be a limit point of  $\Omega$ . We say that the limit of  $f$  at  $x_0 \in \mathbb{R}^n$  exists iff  $\exists l \in \mathbb{R}$  :

$$\begin{aligned} &\forall \epsilon > 0 \exists \delta > 0 : \\ &x \in B_\delta(x_0) \Rightarrow |f(x) - f(x_0)| < \epsilon. \end{aligned}$$

We denote it by the usual notation:

$$\lim_{x \rightarrow x_0} f(x) = l.$$

**Definition 2.5** (Continuity): A function  $f : \Omega \rightarrow \mathbb{R}$  ( $\Omega \subseteq \mathbb{R}$ ) is said to be continuous at  $x_0 \in \Omega$  iff

$$\begin{aligned} &\forall \epsilon > 0 \exists \delta > 0 : \\ &x \in B_\delta(x_0) \Rightarrow |f(x) - f(x_0)| < \epsilon \end{aligned}$$

It can be clearly seen that these definitions are analogous to the ones in single variable calculus. For example, the concept of open intervals in single variable calculus has been “upgraded” to the notion of open balls in higher dimension. Rudin’s book<sup>1</sup> on Mathematical analysis was used in preparing this part of the note. Escher-Amann’s book<sup>2</sup> on Analysis is also a good read for someone learning these topics as a first-timer.

We now proceed to establish the notion of differentiability in higher dimensions. But before that, we must extend our working area to a more abstract space - called “Banach Space”.

**Example 2.1** (A continuous function): Let

$$f(x) := \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Now, to check for continuity note that when  $(x, y) \neq (0, 0)$   $f$  is a polynomial in  $x$  and  $y$  and therefore,  $f$  is trivially continuous. However, when  $(x, y) = (0, 0)$ ,

$$\begin{aligned} |f(x, y) - f(0, 0)| &= \left| \frac{xy}{\sqrt{x^2 + y^2}} - 0 \right| \\ &\leq \frac{\sqrt{x^2 + y^2} \sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}} \\ &= \sqrt{x^2 + y^2} \end{aligned}$$

We choose  $\delta = \epsilon$  and

$$\begin{aligned} |(x, y) - (0, 0)| < \delta &\Rightarrow \sqrt{x^2 + y^2} < \delta \\ &\Rightarrow |f(x, y) - f(0, 0)| < \epsilon \end{aligned}$$

Hence,  $f$  is continuous at  $(0, 0)$ .

□

**Example 2.2** (A discontinuous function): Let

$$f(x, y) := \begin{cases} \frac{xy}{(x^2+y^2)^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Now, it can be easily seen that

$$\begin{aligned} \lim_{\frac{1}{n} \rightarrow 0} f\left(\frac{1}{n}, \frac{1}{n}\right) &= \lim_{\frac{1}{n} \rightarrow 0} \frac{\frac{1}{n^2}}{\left(\frac{1}{n^2} + \frac{1}{n^2}\right)^2} \\ &= \lim_{\frac{1}{n} \rightarrow 0} \frac{n^2}{4} \\ &\neq 0 = f(0, 0) \end{aligned}$$

Therefore,  $f$  is not continuous at  $(0, 0)$ .

□

## 3 Differentiability

We first start by putting forward the definition of a metric space and norms.

### 3.1 Spaces and norms

**Definition 3.1** (Metric Space): A metric space is a pair  $(X, d)$  where  $X$  is a set and  $d : X \times X \rightarrow \mathbb{R}$  is some ‘metric’ on  $X$  (can be thought of as a distance function) such that  $\forall x, y, z \in X$ ,

- i)  $d(x, y) \geq 0$
- ii)  $d(x, y) = 0 \Leftrightarrow x = y$
- iii)  $d(x, y) = d(y, x)$
- iv)  $d(x, y) \leq d(x, z) + d(z, y)$ .

**Definition 3.2** (Norm): Let  $\mathbb{V}$  be a vector space on  $\mathbb{R}$ . A norm on  $\mathbb{V}$  is a function  $\|\cdot\| : \mathbb{V} \rightarrow \mathbb{R}$  such that  $\forall u, v \in \mathbb{V}, a \in \mathbb{R}$ ,

- i)  $\|v\| \geq 0, \quad \|v\| = 0 \Leftrightarrow v = 0$
- ii)  $\|av\| = |a| \cdot \|v\|$
- iii)  $\|u + v\| \leq \|u\| + \|v\|$ .

*Note.* A *normed linear space* is a pair  $(\mathbb{V}, \|\cdot\|)$  such that  $\mathbb{V}$  is a vector space and  $\|\cdot\|$  is a valid norm on  $\mathbb{V}$ .

A norm can induce a metric on  $\mathbb{V}$  by

$$d_{\|\cdot\|}(u, v) = \|u - v\|.$$

Now we are ready to give the definition of a Banach space.

**Definition 3.3** (Banach Space): A Banach space  $(\mathbb{V}, \|\cdot\|)$  is a complete normed space. In other words, every Cauchy sequence in  $\mathbb{V}$  converges in  $\mathbb{V}$  with respect to its norm.

We are now fully equipped to talk about differentiability in several variables (most generally, in Banach spaces). A book<sup>3</sup> on functional analysis by Erwin Kreyszig was helpful in preparing this part of the note.

## 3.2 The Fréchet derivative

**Definition 3.4** (Fréchet Differentiability): Let  $(X, \|\cdot\|_X)$  and  $(Y, \|\cdot\|_Y)$  be two normed linear spaces and let  $U \subseteq X$  and  $f : U \rightarrow Y$  be a mapping. We say that  $f$  is *fréchet differentiable* or simply differentiable at  $x_0 \in U$  iff  $\exists$  a continuous linear map

$$L_{x_0} : X \rightarrow Y \text{ such that}$$

$$\lim_{h \rightarrow 0} \frac{\|f(x_0 + h) - f(x_0) - L_{x_0}(h)\|_Y}{\|h\|_X} = 0$$

We can think of fréchet derivatives also in terms of a *continuous error function* that goes to 0 as  $x$  tends to  $x_0$ .

**Proposition 3.1** (Error Function):  $f : \Omega \subseteq X \rightarrow Y$  is differentiable at  $x_0 \in \Omega \Leftrightarrow \exists L_{x_0} \in \mathcal{L}(X, Y)$  and  $\exists E_{x_0} : \Omega \rightarrow Y$  such that  $E_{x_0}$  is continuous at  $x_0$  with  $E_{x_0}(x_0) = 0$  and  $f(x) = f(x_0) + L_{x_0}(x - x_0) + E_{x_0}(x - x_0) \forall x \in \Omega$ .

*Notation.* Suppose  $f : \Omega \rightarrow Y$  is differentiable at  $x_0 \in \Omega$ , then the linear operator  $L_{x_0} \in \mathcal{L}(X, Y)$  is called the ‘derivative’ or ‘fréchet derivative’ of  $f$  at  $x_0$  and can also be written as  $Df(x_0)$  or  $\partial f(x_0)$  or  $f'(x_0)$ .

If  $f : \Omega \rightarrow Y$  is differentiable at every point  $x \in \Omega$ , we say  $f$  is differentiable and call the map

$$Df : \Omega \rightarrow \mathcal{L}(X, Y)$$

the derivative function of  $f$ .

Now, again if  $\mathcal{L}(X, Y)$  is seen as a banach space then the second fréchet derivative can be written as the mapping

$$D^2f : \Omega \rightarrow \mathcal{L}(X, \mathcal{L}(X, Y))$$

*Note.* We say  $f$  is continuously differentiable at  $x_0$  if the linear map  $Df(x_0)$  is continuous.

### 3.3 Digression to boundedness and continuity of linear operators

**Definition 3.5** (Bounded Linear Operator): Let  $(X, \|\cdot\|_X)$  and  $(Y, \|\cdot\|_Y)$  be two normed linear spaces. We say  $T : X \rightarrow Y$  is bounded iff  $\exists c \in \mathbb{R}$  such that  $\forall x \in X$ ,

$$\|T(x)\|_Y \leq c \|x\|_X$$

Note that this is unlike how we define boundedness of a function in calculus. Boundedness of an operator does not imply boundedness of its range set. In this context we define the norm of an operator  $T$ .

$$\|T\| := \sup_{\substack{x \in X \\ \|x\|_X = 1}} \|T(x)\|_Y$$

Thus,

$$\|T(x)\|_Y \leq \|T\| \|x\|_X$$

We can also define the norm as

$$\|T\| = \sup_{\substack{x \in X \\ \|x\|_X = 1}} \|T(x)\|_Y$$

**Example 3.1:** Let  $A \equiv (a_{ij})_{r \times n}$  define an operator  $T : \mathbb{R}^n \rightarrow \mathbb{R}^r$ .

We show the boundedness of  $T$  with

$$\|x\| = \left( \sum_{i=1}^n x_i^2 \right)^{\frac{1}{2}}$$

As shown previously,

$$\|T(x)\|^2 \leq \|x\|^2 \sum_{i=1}^r \sum_{j=1}^n a_{ij}^2$$

Hence,  $\|T\| = |c|$ , where

$$c^2 = \sum_{i=1}^r \sum_{j=1}^n a_{ij}^2$$

**Proposition 3.2:** All operators on a finite dimensional normed linear space  $X$  is bounded.

**Theorem 3.1** (Boundedness implies Continuity): Let  $T : X \rightarrow Y$  be a linear operators on two normed linear spaces  $X$  and  $Y$ .  $T$  is continuous iff  $T$  is bounded.

*Proof.* The claim is trivial for the case of  $T = 0$ . Hence we assume  $T \neq 0$ .

Let  $T \neq 0$  be bounded. We choose  $\epsilon > 0$  and we take  $\delta = \frac{\epsilon}{\|T\|}$ .

Now,

$$\begin{aligned} \|x - x_0\| < \delta &\Rightarrow \|T(x) - T(x_0)\| = \|T(x - x_0)\| \\ &\leq \|T\| \|x - x_0\| \\ &< \|T\| \delta \\ &= \epsilon \end{aligned}$$

Therefore,  $T$  is continuous.

Next, let  $T$  be continuous at  $x_0$ . So,  $\forall \epsilon > 0, \exists \delta > 0$  such that

$$\|x - x_0\| < \delta \Rightarrow \|T(x) - T(x_0)\| < \epsilon$$

Also,

$$\|T(x) - T(x_0)\| = \|T(x - x_0)\|$$

We take  $x = x_0 + \frac{\delta}{\|y\|}y$  and see

$$\|x - x_0\| = \left\| \frac{\delta}{\|y\|}y \right\| = \delta$$

Now,



$$\begin{aligned} \|T(x) - T(x_0)\| &= \left\| T \frac{\delta}{\|y\|} y \right\| = \frac{\delta}{\|y\|} \|T(y)\| < \epsilon \\ \Rightarrow \frac{\|T(y)\|}{\|y\|} &< \frac{\epsilon}{\delta} \end{aligned}$$

Thus, we take  $c = \frac{\epsilon}{\delta}$  and we are done.

Now, in the definition of a fréchet derivative  $L_{x_0}$  must be a continuous linear operator.

### 3.4 The Directional Derivative

Let  $f : \Omega \rightarrow Y$ ,  $x_0 \in \Omega$  and  $v \in X \setminus \{0\}$ . Now,  $X$  is open  $\Rightarrow \exists \epsilon > 0$  such that  $x_0 + tv \in X \ \forall |t| < \epsilon$ . Therefore,

$$t \mapsto f(x_0 + tv)$$

is well defined. Now, when this is differentiable at 0 we say its derivative is the directional derivative of  $f$  at  $x_0$  in the direction of  $v$ . We denote it by

$$D_v f(x_0) := \lim_{t \rightarrow 0} \frac{f(x_0 + tv) - f(x_0)}{t}$$

**Proposition 3.3:** Let  $f : \Omega \rightarrow Y$  is differentiable at  $x_0 \in \Omega$ . Then,  $D_v f(x_0)$  exists for every  $v \in X \setminus \{0\}$  and  $D_v f(x_0) = Df(x_0) \cdot v$ .

#### 3.4.1 Partial Derivatives

We want to distinguish between the directional derivatives along the coordinate axes from that along any general direction. Especially when  $X = \mathbb{R}^n$ . These derivatives are particularly important for practical and historical reasons. We can use the notation  $\frac{\partial}{\partial x^k}$  or just simply  $\partial_k$  for the derivative along the standard basis  $e_k$ .

$$\begin{aligned} \therefore \partial_k f(x_0) &:= D_{e_k} f(x_0) \\ &= \lim_{t \rightarrow 0} \frac{f(x_0 + te_k) - f(x_0)}{t}, 1 \leq k \leq n \end{aligned}$$

We say that  $\partial_k f(x_0)$  is the partial derivative of  $f$  at  $x_0$  with respect to  $x_k$ .

The function  $f$  is said to be partially differentiable at  $x_0$  if all  $\partial_1 f(x_0), \dots, \partial_n f(x_0)$  exists and it is called continuously partially differentiable if it is partially differentiable at every point of  $\Omega$  and if  $\partial_k f : \Omega \rightarrow Y$  is continuous for  $1 \leq k \leq n$ .

*Remark:*  $f$  is partially differentiable at  $x_0$  does not necessarily imply it is continuous at  $x_0$ .

**Example 3.2** (A partially differentiable discontinuous function):

Consider the example from Example 2.2. We have

$$f(x, y) := \begin{cases} \frac{xy}{(x^2+y^2)^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Now clearly,  $\partial_1 f(0,0) = \partial_2 f(0,0) = 0$ . Therefore,  $f$  is partially differentiable at  $(0,0)$  but as we have seen previously,  $f$  is not continuous as  $(0,0)$ .

### 3.5 The jacob matrix

Let  $f \equiv (f^1, f^2, \dots, f^m) : \Omega \rightarrow \mathbb{R}^m$  with  $\Omega \subseteq \mathbb{R}^n$  be partially differentiable at  $x_0$ . We call,

$$[\partial_i f^j(x_0)] := \begin{pmatrix} \partial_1 f^1(x_0) & \partial_2 f^1(x_0) & \dots & \partial_n f^1(x_0) \\ \partial_1 f^2(x_0) & \partial_2 f^2(x_0) & \dots & \partial_n f^2(x_0) \\ \vdots & \vdots & \ddots & \vdots \\ \partial_1 f^m(x_0) & \partial_2 f^m(x_0) & \dots & \partial_n f^m(x_0) \end{pmatrix}$$

the *Jacobi Matrix* of  $f$  at  $x_0$ .

Now, if  $f$  is differentiable at  $x_0$ , then it is partially differentiable at  $x_0 \forall j \in \{1, 2, \dots, m\}$  and

$$Df(x_0) \equiv [\partial_i f^j(x_0)]$$

So, in other words the representation matrix of the derivative of  $f$  in the standard bases is the Jacobi Matrix of  $f$ .

### 3.6 The differentiability criterion

**Theorem 3.2** (Differentiability criterion): Let  $f : \Omega \rightarrow \mathbb{R}^m$  be a mapping such that  $\Omega \subseteq \mathbb{R}^n$ . Then,  $f$  has continuous partial derivatives at  $x_0$  iff  $f$  is continuously differentiable at  $x_0$ .

#### 3.6.1 Linearity of differential operators

Let  $X$  and  $Y$  be two banach spaces over  $\mathbb{K}$  and let  $f, g : \Omega \rightarrow Y$  be differentiable at  $x_0$  then for all  $\alpha \in \mathbb{K}$   $f + \alpha g$  is also differentiable at  $x_0$  and

$$\partial(f + \alpha g)(x_0) = \partial f(x_0) + \alpha \partial g(x_0)$$

*Remark.* We can state the linearity of differential operators as:

$$C^1(\Omega, Y) \subseteq C(\Omega, Y) \text{ and,} \\ \partial : C^1(\Omega, Y) \rightarrow C(\Omega, \mathcal{L}(X, Y)) \text{ is linear.}$$

We explore some of the generalizations of results from single variable calculus in several variables.

### 3.7 The chain rule

Just like in multivariable calculus we can generalize the chain rule to the several variable case. We only show the case of  $X = \mathbb{R}^n$  and  $Y = \mathbb{R}^m$  here, however it is also true for any general banach spaces.

Let  $\Omega \subseteq \mathbb{R}^n$  and  $f : \Omega \rightarrow \mathbb{R}^m$  be differentiable at  $x_0$  and  $g : \mathbb{R}^M \rightarrow \mathbb{R}^l$  be differentiable at  $y_0 = f(x_0)$ . Then,

$$h \equiv g \circ f : \Omega \rightarrow \mathbb{R}^l$$

is differentiable at  $x_0$  and

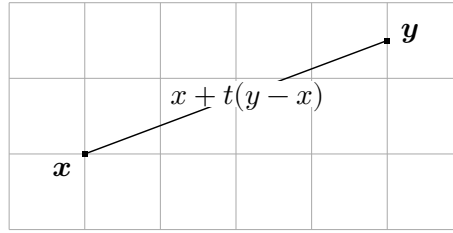
$$[\partial h(x_0)] = [\partial g(f(x_0))][\partial f(x_0)]$$

### 3.8 The mean value theorem

Similar to single variable, we can show the mean value theorem for the several variable case as well.

Let  $f : \Omega \rightarrow Y$  be differentiable. Then for some  $x, y \in \Omega$ ,

$$\begin{aligned} & \|f(x) - f(y)\| \\ & \leq \sup_{0 \leq t \leq 1} \|Df(x + t(y - x))\| \|y - x\| \end{aligned}$$



Also,  $\exists \theta \in [0, 1]$  such that,

$$\|f(x) - f(y)\| = \|Df(x + \theta(y - x))\| \|y - x\|$$

To prove this, we set  $g(t) := f(x + t(y - x))$  for  $t \in [0, 1]$ . Now, we are in the single variable world where  $f$  is differentiable and  $g$  is well defined. Now,

$$g'(t) = Df(x + t(y - x))(y - x)$$

Now, applying the mean value theorem for single variable we get,

$$\begin{aligned} \|f(y) - f(x)\| &= \|g(1) - g(0)\| \\ &\leq \sup_{0 \leq t \leq 1} \|g'(t)\| \\ &\leq \sup_{0 \leq t \leq 1} \|Df(x + t(y - x))\| \|y - x\| \end{aligned}$$

And, we get our result.

### 3.9 Differentiability of limits of sequence of functions

While discussing sequence of functions in several variable the assumption of domains and ranges being banach spaces comes in handy.

Let  $\{f_k\} \subseteq C^1(\Omega, Y)$  for  $k \in \mathbb{N}$ . Let there exist  $f : \Omega \rightarrow Y$  and  $g : \Omega \rightarrow \mathcal{L}(X, Y)$  such that

$$\begin{aligned} i) & f_k \longrightarrow f \text{ pointwise} \\ ii) & Df_k \longrightarrow g \text{ locally and uniformly} \end{aligned}$$

Then,  $f \in C^1(X, Y)$  and  $Df = g$ .

## 4 The inverse function theorem

Again, for single variable calculus we had the inverse function theorem. We state and prove the same here again and will later see that it can be easily translated to the several variable case using linearity, chain rule and mean value theorem we have earlier stated.

Let  $f : (a, b) \rightarrow \mathbb{R}$  be a differentiable function such that

$$f \in C^1(a, b) \text{ and } f'(x) \neq 0 \ \forall x \in (a, b).$$

Then,  $f$  is strictly monotone,  $J = f((a, b))$  is an interval and  $f^{-1} : J \rightarrow (a, b)$ ,  $f(x_0) \mapsto x_0$  exists and is differentiable.

*Proof.* By mean value theorem it is clear that  $f$  is strictly monotone and therefore  $J = f((a, b))$  is an interval.

Now, as  $f$  is injective (monotone) and continuous and  $J$  is an interval,  $f|_J$  is bijective and  $f^{-1} : J \rightarrow (a, b)$  exists and is continuous.

Let  $d \in J$ . Now,  $\exists c \in (a, b)$  such that  $f(c) = d$ . We have,

$$\begin{aligned} \frac{f^{-1}(y) - f^{-1}(d)}{y - d} &= \frac{x - c}{f(x) - f(c)}, y = f(x) \\ \lim_{y \rightarrow d} \frac{f^{-1}(y) - f^{-1}(d)}{y - d} &= \lim_{x \rightarrow c} \frac{x - c}{f(x) - f(c)} \\ &= \frac{1}{f'(c)} \\ &= \frac{1}{f'(f^{-1}(d))} \end{aligned}$$

$\therefore f^{-1} : J \rightarrow (a, b)$  is differentiable and,

$$(f^{-1})'(d) = \frac{1}{f'(f^{-1}(d))}$$

Now, this theorem is also true when  $f'(x_0) \neq 0$  for some  $x_0 \in (a, b)$ . In that case, we would have  $f$  to be strictly monotone for some  $\delta$ -neighborhood of  $x_0$  following the continuity of  $f'$ . Thus, restriction of  $f$  to  $N_\delta(x_0)$  would give us a bijective function:  $f|_{N_\delta(x_0)} : N_\delta(x_0) \rightarrow f(N_\delta(x_0))$ . We can now generalize this to the several variable case to get:

**Theorem 4.1** (Inverse function): Let  $\Omega \rightarrow \mathbb{R}^m$  where  $\Omega \subseteq \mathbb{R}^n$  and  $f \in C^1(\Omega, \mathbb{R}^m)$ . Also, assume that  $\det(Df(x_0)) \neq 0$  for some  $x_0 \in \Omega$ . Then,  $\exists$  open sets  $U$  and  $V$  such that  $x_0 \in U$ ,  $f(x_0) \in V$  and,

- i)  $f : U \rightarrow V$  is bijective
- ii)  $f^{-1} : V \rightarrow U \in C^1(V, U)$
- iii)  $Df^{-1}(f(x)) = (Df(x))^{-1}, \forall x \in U$

## 5 The implicit function theorem

As a motivation, let us consider the example of  $f : \mathbb{R}^2 \rightarrow \mathbb{R}, (x, y) \mapsto x^2 + y^2$ . Consider the level set  $f(x, y) = 1$ . Now assume  $a \neq \pm 1$  and  $b > 0$  and  $f(a, b) = 1$ . Then,  $\exists$  open sets  $A, B$  such that  $a \in A$  and  $b \in B$  such that for every  $x \in A$  exists  $y \in B$  (exactly one) such that  $f(x, y) =$

0. In other words,  $y$  can be implicitly defined on  $x$  by  $f$  over the region  $B$ . Now, clearly  $y = \sqrt{1 - x^2}$  here and we get *exactly one solution* for  $y$ .

However, we cannot get solutions for  $y$  in the neighborhoods of  $(1, 0)$  or  $(-1, 0)$ . These two points have in common that at  $a = \pm 1$ ,  $\partial_2 f(a, b) = 0$ . Whereas for  $a \neq \pm 1$ ,  $\partial_2 f(a, b) = 2b$ .

Now, we state the implicit function theorem as:

**Theorem 5.1** (Implicit function): Let  $f : \Omega \rightarrow \mathbb{R}$  be differentiable with  $\Omega \subseteq \mathbb{R}^2$ . Now let  $(x_0, y_0) \in \Omega$  and  $\frac{\partial f}{\partial y}(x_0, y_0) \neq 0$ . Also let  $f(x_0, y_0) = c$ . Then,  $\exists$  open set  $U \ni (x_0, y_0)$  such that

$$f^{-1}(c) \cap U = \{(x, g(x)) : f(x, g(x)) = c\}$$

where  $g$  is differentiable.

*Remark:* Let  $\Omega \subseteq \mathbb{R}^{m+n}$  be an open set and  $f \in C^1(\Omega, \mathbb{R}^n)$ . Let  $(x_0, y_0) \in \Omega$ , and  $f(x_0, y_0) = 0$ , where  $x_0 \in \mathbb{R}^m$ ,  $y_0 \in \mathbb{R}^n$ . Then, if  $\det[D_{m+k} f^j(a, b)]_{\substack{1 \leq k \leq n \\ 1 \leq j \leq n}} \neq 0$  then  $\exists$  open set  $U \subseteq \Omega$  such that  $(x_0, y_0) \in U$  and  $V \subseteq \mathbb{R}^m$  and  $g \in C^1(V, \mathbb{R}^n)$  such that

$$f^{-1}(0) \cap U = \{(x, g(x)) : x \in V\}$$

and  $g$  is differentiable.

## 6 Surfaces

We want to define the notion of a surface from a geometric standpoint without delving into rigorous topological considerations. We would then also study the notions of ‘patches’ and ‘atlases’ and introduce the concept of homeomorphisms.

We will then go on to study smoothness of surfaces by differential calculus - predominantly using the concepts and ideas we have learned so far. A lot of help was obtained from Andrew Pressley’s book<sup>4</sup> on Differential Geometry concerning this topic.

To give a simpler idea of what a surface is - It is a set  $S \subset \mathbb{R}^3$  for which the ‘vicinity’ of any point  $p \in S$  ‘looks like’ a piece in  $\mathbb{R}^2$ . We wish to define these concepts more rigorously. We shall be defining some preliminary concepts first.

**Definition 6.1** (Continuous maps): We define continuity again in a similar fashion.<sup>2</sup> Let  $X \subseteq \mathbb{R}^m$ , a map  $f : X \rightarrow \mathbb{R}^n$  is continuous at  $a \in X$  if  $\forall \epsilon > 0 \exists \delta > 0$  such that

$$u \in B_\delta(a) \cap X \Rightarrow f(u) \in B_\epsilon(f(a))$$

**Definition 6.2** (Homeomorphism): Let  $X \subseteq \mathbb{R}^m$  and  $Y \subseteq \mathbb{R}^n$ . A map  $f : X \rightarrow Y$  is called a homeomorphism if  $f$  is continuous and bijective and also its inverse map  $f^{-1} : Y \rightarrow X$  is continuous.

<sup>2</sup>However, a more useful definition of continuity is that  $\forall$  open set  $V \subseteq \mathbb{R}^n$ ,  $\exists$  an open set  $U \subseteq \mathbb{R}^m$  such that  $U \cap X = \{x \in X : f(x) \in V\}$ . In other words,  $V \subseteq \mathbb{R}^n$  is open  $\Rightarrow f^{-1}(V)$  is open.

Now, we are ready to give the definition of a surface in  $\mathbb{R}^3$ .

**Definition 6.3** (Surface): A subset  $S \subseteq \mathbb{R}^3$  is called a surface if  $\forall p \in S, \exists$  open set  $U \subseteq \mathbb{R}^2$  and open set  $W \subseteq \mathbb{R}^3$  such that  $p \in W$  and  $S \cap W$  is homeomorphic to  $U$ .

Every open subset of  $S$  is represented as  $S \cap W$  where  $W$  is open in  $\mathbb{R}^3$ .

**Definition 6.4** (Surface Patch): Let  $S$  be a surface. A homeomorphism  $\sigma : U \rightarrow S \cap W$  where  $U \subseteq \mathbb{R}^2$  and  $W \subseteq \mathbb{R}^3$  are open is called a surface patch of  $S$ . It is also called the *parameterization* of the open subset  $S \cap W$  of  $S$ .

**Definition 6.5** (Atlas): Let  $S$  be a surface and  $\{\sigma_i\}_{i \in I}$  be a set of surface patches of  $S$ . Then,  $\{\sigma_i\}_{i \in I}$  is called an atlas of  $S$  if

$$S \subseteq \bigcup_{i \in I} \sigma_i(U_i)$$

## 6.1 Examples of surfaces

We list a few generic examples of surfaces like as follows.

### 6.1.1 Plane in $\mathbb{R}^3$

Note that any plane in  $\mathbb{R}^3$  has the graph

$$\begin{aligned} S &= \{(x, y, z) \in \mathbb{R}^3 : ax + by + cz = d \text{ and } a, b, c, d \in \mathbb{R}\} \\ &= \left\{ \left( x, y, \frac{d - ax - by}{c} \right) \in \mathbb{R}^3 : (x, y) \in \mathbb{R}^2 \right\} \end{aligned}$$

Now, we define the parameterization  $\sigma : \mathbb{R}^2 \rightarrow \mathbb{R}^3$  as

$$\sigma(x, y) = \left( x, y, \frac{d - ax - by}{c} \right), (x, y) \in \mathbb{R}^2.$$

It is easy to verify that  $\sigma$  is a homeomorphism. Therefore, a plane in  $\mathbb{R}^3$  can be covered by an atlas consisting of a single surface patch.

### 6.1.2 Circular cylinder in $\mathbb{R}^3$

For simplicity let us consider the unit cylinder in  $\mathbb{R}^3$  defined as

$$S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$$

We want to define bijective maps on an open set in  $\mathbb{R}^2$ , such that it covers the entire  $S$ . One way to do it is to consider

$$\begin{aligned} U &= \{(u, v) \in \mathbb{R}^2 : 0 < u < 2\pi\} \text{ and} \\ \tilde{U} &= \{(u, v) \in \mathbb{R}^2 : -\pi < u < \pi\} \end{aligned}$$

We have the map

$$\begin{aligned}\sigma : \mathbb{R}^2 \cap (U \cup \tilde{U}) &\rightarrow \mathbb{R}^3 \\ (x, y) &\mapsto (\cos x, \sin x, y)\end{aligned}$$

Now, the surface patches of  $\sigma|_U : U \rightarrow S$  and  $\sigma|_{\tilde{U}} : \tilde{U} \rightarrow S$  are homeomorphisms and cover the entire cylinder and hence, forms an atlas.

It can again be checked that  $\sigma|_U$  and  $\sigma|_{\tilde{U}}$  are homeomorphisms.

### 6.1.3 Sphere in $\mathbb{R}^3$

An unit sphere in  $\mathbb{R}^3$  can be represented as

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$$

Any point  $p \in S^2$  can be parameterized (just as in spherical coordinates) as

$$p \equiv (\cos \theta \cos \varphi, \cos \theta \sin \varphi, \sin \theta)$$

where  $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$  and  $0 \leq \varphi \leq 2\pi$ .

But we have to decompose  $S^2$  into two surface patches  $\sigma|_U$  and  $\sigma|_{\tilde{U}}$  to ensure they map open spaces to open spaces. Here,  $U = \{(\theta, \varphi) \in \mathbb{R}^2 : -\frac{\pi}{2} < \theta < \frac{\pi}{2}, 0 < \varphi < 2\pi\}$  and

$$\begin{aligned}\sigma|_U &= (\cos \theta \cos \varphi, \cos \theta \sin \varphi, \sin \theta) \\ \sigma|_{\tilde{U}} &= (-\cos \theta \cos \varphi, -\sin \theta, -\cos \theta \sin \varphi)\end{aligned}$$

It can be verified that  $\sigma|_U$  and  $\sigma|_{\tilde{U}}$  are homeomorphisms and cover the entire sphere.

**Example 6.1** (Open subsets of a surface): Show that every open subset of a surface is a surface.

*Proof.* Let  $S \subseteq \mathbb{R}^3$  be a surface and let  $W \subseteq S$  be an open subset of  $S$ . We want to cover every point  $p \in W$  by patches in  $\mathbb{R}^2$ . Now, as  $p \in W \subseteq S$ ,  $\exists \sigma_p : U_p \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$  such that  $\sigma_p(U_p) = V_p \cap S$  where  $V_p$  is an open set in  $\mathbb{R}^3$  and  $p \in V_p \cap S$ . Now the intersection  $W \cap V_p \cap S$  is also open in  $\mathbb{R}^3$ . By definition,  $\sigma_p$  is a homeomorphism.

Hence, we have shown that  $W$  is a surface. □

## 6.2 A few properties of surfaces

We start by looking at smoothness of surfaces.

**Definition 6.6** (Smooth functions): Let  $U \subseteq \mathbb{R}^2$ , we say  $f : U \rightarrow \mathbb{R}^3$  is smooth if all the components of  $f$  have continuous partial derivatives of all order i.e.,  $f^i \in C^\infty(U, \mathbb{R}^3) \forall i \in \{1, 2, 3\}$

*Notation:* We denote

$$\frac{\partial f}{\partial x} \equiv f_x; \quad \frac{\partial f}{\partial y} \equiv f_y; \quad \frac{\partial^2 f}{\partial x^2} \equiv f_{xx}; \quad \frac{\partial^2 f}{\partial y^2} \equiv f_{yy}; \quad \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} \equiv f_{xy} = f_{yx}$$

We impose another condition of linear independence on partial derivatives of  $\sigma$  to define a ‘regular surface patch’.

**Definition 6.7** (Regular Surface Patch): A surface patch  $\sigma : U \rightarrow \mathbb{R}^3$  is called regular if it is smooth and  $\sigma_x$  and  $\sigma_y$  are linearly independent  $\forall (x, y) \in U$ . This implies that

$$\sigma_x \times \sigma_y \neq 0 \quad \forall (x, y) \in U$$

*Note.* A regular surface patch is also sometimes called an ‘allowable surface patch’.

**Example 6.2** (Plane): Another way to parameterize a plane  $S$  is by  $\sigma : \mathbb{R}^2 \rightarrow \mathbb{R}^3$  by  $\sigma(x, y) = a + xp + yq$  where  $a \in S$  and  $p$  and  $q$  are perpendicular unit vectors parallel to the plane.

Now, obviously  $\sigma_x = p$  and  $\sigma_y = q$  are linearly independent and  $\sigma$  is smooth. Hence,  $\sigma$  is regular.

**Example 6.3** (Cylinder): We parameterized a surface patch of a unit cylinder by

$$\sigma(x, y) = (\cos x, \sin x, y)$$

Now, each component of  $\sigma$  is trivially smooth and

$$\sigma_x = (-\sin x, \cos x, 0) \text{ and } \sigma_y = (0, 0, 1)$$

are linearly independent. Hence,  $\sigma$  is regular.

*Note.* We will make the assumption from now on that the surfaces we consider are *connected*, implying that any two points in  $S$  can be joined by a curve lying entirely in  $S$ .

**Proposition 6.1:** If  $f : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$  is a smooth function then its graph  $\{(x, y, f(x, y)) \in \mathbb{R}^3 : (x, y) \in \mathbb{R}^2\}$  is a smooth surface with an atlas consisting of a single regular surface patch.

*Proof.* Let  $S = \{(x, y, f(x, y)) : (x, y) \in U\}$  be the graph of a smooth function  $f$ . The map  $\sigma : U \rightarrow \mathbb{R}^3$ ,  $(x, y) \mapsto (x, y, f(x, y))$  is smooth as  $\sigma_x$ ,  $\sigma_y$  and  $\sigma_z$  are smooth as  $f$  is smooth.

Now,  $\sigma_x$ ,  $\sigma_y$  and  $\sigma_z$  are also linearly independent. Hence,  $\sigma$  forms a regular surface patch.



**Proposition 6.2:** Let  $U$  and  $\tilde{U}$  be open subsets of  $\mathbb{R}^2$  and let  $\sigma : U \rightarrow \mathbb{R}^3$  be a regular surface patch and  $\phi : \tilde{U} \rightarrow U$  be a bijective smooth mapping with inverse  $\phi^{-1} : U \rightarrow \tilde{U}$ . Then,

$$\tilde{\sigma} \equiv \sigma \circ \phi : \tilde{U} \rightarrow \mathbb{R}^3$$

is a regular surface patch.

Also, let  $(u, v) = \phi(\tilde{u}, \tilde{v})$ . Then,

$$\tilde{\sigma}_{\tilde{u}} \times \tilde{\sigma}_{\tilde{v}} = \det(J(\phi)) \sigma_u \times \sigma_v$$

where

$$J(\phi) = \begin{pmatrix} \frac{\partial u}{\partial \tilde{u}} & \frac{\partial v}{\partial \tilde{u}} \\ \frac{\partial u}{\partial \tilde{v}} & \frac{\partial v}{\partial \tilde{v}} \end{pmatrix}$$

is called the Jacobian matrix.

$\tilde{\sigma}$  can just be thought of as a reparameterization of  $\sigma$  where  $\phi$  is defined.

**Example 6.4:**  $\sigma : \mathbb{R}^2 \rightarrow \mathbb{R}^3, (x, y) \mapsto (x, y, xy)$

This is trivially smooth and surjective. When  $(x_1, y_1, x_1 y_1) = (x_2, y_2, x_2 y_2)$  then  $x_1 = x_2$  and  $y_1 = y_2$  which makes  $\sigma$  bijective and thus, regular.

**Example 6.5:**  $\sigma : \mathbb{R}^2 \rightarrow \mathbb{R}^3, (x, y) \mapsto (x, y^2, y^3)$

This is again smooth as all components are polynomials. But note that  $\sigma_x = (1, 0, 0)$  and  $\sigma_y = (0, 2y, 3y^2)$  are not linearly independent at  $y = 0$ . Hence,  $\sigma$  is regular everywhere except at  $y = 0$ . It can be now easily shown that  $\sigma$  is injective and therefore bijective on an open subset of  $\mathbb{R}^3$ .

**Example 6.6:**  $\sigma : \mathbb{R}^2 \rightarrow \mathbb{R}^3, (x, y) \mapsto (x + x^2, y, y^2)$

Note that  $\sigma(-2, y) = \sigma(1, y)$ . Hence,  $\sigma$  is not injective and therefore is not a regular surface patch.

Now that we have understood regular surface patches we wish to introduce a few more concepts of ‘smooth maps’ and ‘diffeomorphisms’.

**Definition 6.8 (Smooth Maps):** Let  $S_1$  and  $S_2$  be surfaces that are covered by single surface patches  $\sigma_1 : U_1 \rightarrow \mathbb{R}^3$  and  $\sigma_2 : U_2 \rightarrow \mathbb{R}^3$ . Now,  $\sigma_1$  and  $\sigma_2$  are bijective implies any map  $f : S_1 \rightarrow S_2$  gives rise to a map

$$\sigma_2^{-1} \circ f \circ \sigma_1 : U_1 \rightarrow U_2$$

We call  $f$  to be smooth if  $\sigma_2^{-1} \circ f \circ \sigma_1$  is smooth.

**Definition 6.9** (Diffeomorphism): A smooth map  $f : S_1 \rightarrow S_2$  is called a diffeomorphism if  $f$  is bijective and its inverse  $f^{-1} : S_2 \rightarrow S_1$  is smooth.

Two surfaces  $S_1$  and  $S_2$  are diffeomorphic if there exists a diffeomorphism between them.

**Example 6.7:** Let  $S_1$  be a half cone  $x^2 + y^2 = z^2, z > 0$  and  $S_2$  be the half plane  $\{(0, y, z) \in \mathbb{R}^3 : y > 0\}$ . We define a map from  $S_2$  to  $S_1$  by  $f(0, y, z) = (y \cos z, y \sin z, y)$ . Show that  $f$  is a local diffeomorphism but not a diffeomorphism.

To show that  $f$  is not globally diffeomorphic we note that  $f$  is periodic on  $z$ , i.e.,  $f(0, y, 0) = f(0, y, 2\pi) = f(0, y, 2n\pi)$ . Hence,  $f$  is not injective.

Now, we need to show that  $f$  is locally smooth and bijective (injective). We take an  $\epsilon$ -ball,  $\epsilon < 2\pi \in \mathbb{R}^3$  around  $(0, y_0, z_0)$ . Then,  $f|_{B_\epsilon(0, y_0, z_0)}$  is injective and hence bijective on its image.

Now, to show that  $f$  is locally smooth we compute the vectors  $f_x, f_y$  and  $f_z$ .

$$f_x = (0, 0, 0); f_y = (\cos z, \sin z, 1); f_z = (-y \sin z, y \cos z, 0)$$

Now,  $f_y$  and  $f_z$  are linearly independent, so  $f$  is a diffeomorphism. Also, the Jacobian matrix,

$$J(f) = \begin{pmatrix} 0 & \cos z & -y \sin z \\ 0 & \sin z & y \cos z \\ 0 & 1 & 1 \end{pmatrix}$$

has rank = 2 when  $y > 0$ .

Now, we go on to study tangents and normals to a surface. These are useful tools in characterizing a surface alternative to what we have been studying till now.

### 6.3 Tangents and normals

**Definition 6.10** (Tangent Vector): Let  $S \subseteq \mathbb{R}^3$  be a surface, and  $p \in S$  be a point in the surface. We say  $v$  is tangent to  $S$  if for some curve  $\gamma(t) \subset S : \gamma(t_0) = p$ , for some  $t_0 \in \mathbb{R}$ ,  $v$  is tangent to  $\gamma$  at  $t_0$ .

**Definition 6.11** (Tangent Space): The tangent space, denoted by  $T_p S$  of  $S$  at  $p$  is defined as

$$T_p S := \{v \in \mathbb{R}^3 : v \text{ is tangent to } S \text{ at } p\}$$

Let  $\sigma : U \rightarrow \mathbb{R}^3 \cap S$  be a surface patch of  $S$  such that the coordinates of  $p$  in  $U$  is  $(u_0, v_0)$ , i.e.,  $\sigma(u_0, v_0) = p$ . Now, let  $\gamma(t) \subset S$  be a curve passing through  $p$  such that  $\gamma(t_0) = p$ . Now, there will be necessarily smooth functions<sup>3</sup>  $u(t)$  and  $v(t)$  such that

$$\gamma(t) = \sigma(u(t), v(t)) \quad (3)$$

Now, the tangent space  $T_p S$  is the vector space spanned by  $\sigma_u$  and  $\sigma_v$  or more specifically  $\sigma_u(u_0, v_0)$  and  $\sigma_v(u_0, v_0)$ . This is because differentiating Eq. 3 by  $t$  we get,

$$\gamma'(t) = \sigma_u u'(t) + \sigma_v v'(t)$$

*Note.*  $\gamma'(t)$  is a tangent vector to  $p$ . Any arbitrary vector in the span of  $\sigma_u$  and  $\sigma_v$  (let's say  $\lambda\sigma_u + \mu\sigma_v$ ) can be thought of as a tangent to the curve  $\gamma(t) = \sigma(u_0 + \lambda t, v_0 + \mu t)$ .

Hence,  $\sigma_u$  and  $\sigma_v$  spans the tangent space.

We can also conclude from this that the tangent space is *two-dimensional*.

*Remark:*  $0 \in T_p S$ . This happens when the curve  $y(t) = \sigma(u_0, v_0)$ , i.e., the coordinates in  $U$  do not change with  $t$ . Also,  $\sigma_u, \sigma_v \in T_p S$ . This happens when  $\gamma(t) = \sigma(u_0 + t, v_0)$  and  $\gamma(t) = \sigma(u_0, v_0 + t)$  respectively.

Now, the question arises that where could these concepts be useful.

Note that we earlier assumed  $\sigma_u$  and  $\sigma_v$  to be linearly independent. This ensures that the surface does not collapse into 1-D and the tangent space is 2-D.

Now, the tangent places help us define the notion of derivatives of a smooth map between surfaces. Let  $f : S \rightarrow \tilde{S}$  be a smooth map. The derivative of  $f$  at  $p$  is a mapping

$$Df_p : T_p S \rightarrow T_{f(p)} \tilde{S}$$

**Definition 6.12** (Derivative of a smooth map): Let  $f : S \rightarrow \tilde{S}$  be a smooth map. Let  $w \in T_p S$  be a tangent vector to  $S$  at  $p = \gamma(t_0)$  where  $\gamma(t)$  is any curve in  $S$  passing through  $p$ . Hence,  $w = \dot{\gamma}(t_0)$ . Now,  $\tilde{\gamma} \equiv f \circ \gamma$  is the corresponding curve in  $\tilde{S}$  passing through  $f(p)$  at  $t = t_0$ . Therefore,  $\tilde{w} = \dot{\tilde{\gamma}}(t_0) \in T_{f(p)}(\tilde{S})$ .

Now, the derivative  $D_p f$  is the map

$$D_p f : T_p S \rightarrow T_{f(p)} \tilde{S}$$

such that  $D_p f(w) = \tilde{w} \forall w \in T_p S$ .

*Note.*  $D_p f(w)$  depends only on  $f, p$  and  $w$  and not on the curve  $\gamma(t)$ .

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<sup>3</sup>We will be proving this claim later.

More specifically, the derivative  $D_p f : T_p S \rightarrow T_{f(p)} \tilde{S}$  is a linear map. Let the smooth map  $f : S \rightarrow \tilde{S}$  be  $(u, v) \mapsto (\alpha(u, v), \beta(u, v))$  then the matrix of linear transformation of  $D_p f$  w.r.t the basis  $\{\sigma_u, \sigma_v\}$  of  $T_p S$  and  $\{\tilde{\sigma}_{\tilde{u}}, \tilde{\sigma}_{\tilde{v}}\}$  of  $T_{f(p)}(\tilde{S})$  is

$$\begin{pmatrix} \alpha_u & \alpha_v \\ \beta_u & \beta_v \end{pmatrix} T_p S = T_{f(p)}(\tilde{S})$$

Here,  $\tilde{u}(t) = \alpha(u(t), v(t))$  and  $\tilde{v}(t) = \beta(u(t), v(t))$ .

**Proposition 6.3:** Let  $S_1, S_2, S_3$  be surfaces and  $f_1 : S_1 \rightarrow S_2$  and  $f_2 : S_2 \rightarrow S_3$  be smooth maps then  $\forall p \in S_1$ ,

$$D_p(f_2 \circ f_1) = D_{f_1(p)} f_2 \circ D_p f_1$$

**Proposition 6.4:** If  $f : S \rightarrow \tilde{S}$  is a diffeomorphism then  $\forall p \in S$  the map  $D_p f : T_p S \rightarrow T_{f(p)} \tilde{S}$  is invertible.

However, the converse of Proposition 6.4 is not quite true. Instead, we can say if  $D_p f : T_p S \rightarrow T_{f(p)} \tilde{S}$  is invertible and if  $f$  is smooth, then  $f$  is a local diffeomorphism.

Next, we give the definition of normals to a surface.

Note that the tangent plane  $T_p S$  passes through the origin and thus is completely determined by a unit normal perpendicular to it.

Also, we note that the tangent plane is spanned by  $\sigma_u$  and  $\sigma_v$  of the patch containing  $p$ . Hence, from this we can easily define the concept of normal as -

$$N_\sigma := \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|}$$

Now, this definition is intuitive as this ensures that  $N_\sigma$  is perpendicular to the span of  $\sigma_u$  and  $\sigma_v$ . This is the standard unit normal of  $\sigma$  at  $p$ .

However, it is to be noted that  $N_\sigma$  is not quite independent of  $\sigma$ . This is also obvious as the normal vector can both point in the ‘outward’ direction as well as the ‘inward’ direction. More specifically,

$$\tilde{\sigma}_{\tilde{u}} \times \tilde{\sigma}_{\tilde{v}} = \det(J(\phi)) \sigma_u \times \sigma_v$$

where  $\phi$  is the transition map from  $\sigma$  to  $\tilde{\sigma}$ .

Hence,

$$N_{\tilde{\sigma}} = \frac{\tilde{\sigma}_{\tilde{u}} \times \tilde{\sigma}_{\tilde{v}}}{\|\tilde{\sigma}_{\tilde{u}} \times \tilde{\sigma}_{\tilde{v}}\|} = \pm \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|} = \pm N_\sigma$$

To fix this issue, we wish to give the definition of ‘orientability’. This will solidify our understanding of ‘outwardness’ and ‘inwardness’ when it comes to surfaces.

**Definition 6.13** (Orientable Surface): We say a surface  $S$  is orientable if  $\exists$  an atlas  $A$  of  $S$  with the property that for any  $\sigma_1, \sigma_2 \in A$  with transition map  $\phi$  we have  $\det(J(\phi)) > 0$ .

Now, it can be said that for an orientable surface there is a smooth choice of standard unit normal at every point  $p \in S$ . This choice of normal does not depend upon the choice of patch of  $S$  of  $p$ .

Next, we move on to study boundary of a surface.

**Example 6.8:** Find the equation of tangent plane of the following surface patches:

$$i) \sigma(u, v) = (u, v, u^2 - v^2) \text{ at } (1, 1, 0)$$

$$ii) \sigma(u, v) = (u \cosh v, u \sinh v, u^2)$$

i)

Note that,  $\sigma(1, 1) = (1, 1, 0) = p$  (say)

Now,

$$\sigma_u = \frac{\partial}{\partial u}(u, v, u^2 - v^2) = (1, 0, 2u)$$

$$\sigma_v = \frac{\partial}{\partial v}(u, v, u^2 - v^2) = (0, 1, -2v)$$

And,

$$\sigma_u(1, 1) = (1, 0, 2)$$

$$\sigma_v(1, 1) = (0, 1, -2)$$

Now, the normal vector is given by  $\frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|}$  at  $(1, 1)$ .

$$\frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|} = \frac{-2, 0, 1}{\|(-2, 0, 1)\|} = \frac{1}{\sqrt{5}}(-2, 0, 1)$$

Therefore, we the equation of plane tangent at  $(1, 1, 0)$  as:

$$\begin{aligned} \frac{1}{\sqrt{5}}[-2(x-1) + 0(y-1) + 1(z-0)] &= 0 \\ \Rightarrow z &= 2x - 2 \end{aligned}$$

ii)

We have  $\sigma(u, v) = (u \cosh v, u \sinh v, u^2)$ .

Again, note that  $\sigma(1, 0) = (1, 0, 1)$ .

Now,

$$\sigma_u = (\cosh v, \sinh v, 2u)$$

$$\sigma_v = (u \sinh v, u \cosh v, 0)$$

And,  $\sigma_u(1, 0) = (1, 0, 2)$ ;  $\sigma_v(1, 0) = (0, 1, 0)$

The normal vectors are therefore,

$$\frac{\sigma_u(1,0) \times \sigma_v(1,0)}{\|\sigma_u(1,0) \times \sigma_v(1,0)\|} = \frac{-2, 0, 1}{\|(-2, 0, 1)\|} = \frac{1}{\sqrt{5}}(-2, 0, 1)$$

Hence, the equation of tangent plane at  $(-1, 0, 1)$  is:

$$\begin{aligned} -2(x-1) + 0(y-0) + 1(z-1) &= 0 \\ \Rightarrow z &= 2x - 1 \end{aligned}$$

**Example 6.9:** Suppose two smooth surfaces  $S_1$  and  $S_2$  are diffeomorphic. Show that if  $S_1$  is orientable then  $S_2$  is also orientable.

As  $S_1$  and  $S_2$  are smooth  $\exists$  a diffeomorphism  $f : S_1 \rightarrow S_2$ .

Let us assume we have an atlas  $(U_\alpha, \varphi_\alpha)$  of  $S_1$  which is oriented. Now, if any two charts (the inverse of patches)  $\varphi_\alpha : U_\alpha \subseteq S_1 \rightarrow \mathbb{R}^2$  and  $\varphi_\beta : U_\beta \subseteq S_1 \rightarrow \mathbb{R}^2$  are overlapping then they will have a transition map:

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta) \subseteq \mathbb{R}^2$$

Now, as  $S_1$  is orientable,

$$\det(J(\varphi_\beta \circ \varphi_\alpha^{-1})) > 0$$

Now, as we have a diffeomorphism  $f : S_1 \rightarrow S_2$  we can form an atlas of  $S_2$  as  $V_\alpha := f(U_\alpha)$  and  $\psi_\alpha := \varphi_\alpha \circ f$ .

Now,  $(V_\alpha, \psi_\alpha)$  covers  $S_2$  as  $f$  is a diffeomorphism.

Now,

$$\begin{aligned} \det(J(\psi_\beta \circ \psi_\alpha^{-1})) &= \det(J((\varphi_\beta \circ f) \circ (\varphi_\alpha \circ f))) \\ &= \det(J(\varphi_\beta \circ f \circ f^{-1} \circ \varphi_\alpha^{-1})) \\ &= \det(J(\varphi_\beta \circ \varphi_\alpha^{-1})) \\ &> 0 \end{aligned}$$

Thus, we can say  $S_2$  is orientable.

## 6.4 Boundary of a surface

Let  $\Omega \subseteq \mathbb{R}^n$  be open. We have the 'closure' of  $\Omega$  as:

$$\bar{\Omega} := \{p \in \mathbb{R}^n : p \text{ is a limit point of } \Omega\}$$

Now, the boundary of  $\Omega$  is given by

$$\partial\Omega = \bar{\Omega} \setminus \Omega$$

**Definition 6.14** (Domain with  $C^k$ -boundary): Let  $\Omega \subseteq \mathbb{R}^n$  be an open set. We say that  $\Omega$  has a domain with  $C^k$ -boundary if  $\forall p \in \partial\Omega \exists R > 0$  and a  $C^k$  map  $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$  such that upon relabelling and reorienting the coordinate axes,

$$\Omega \cap B_R(p) = \{(x, y, z) \in B_R(p) : z > \phi(x, y)\}$$

In simple terms, we want to ensure that the parameterization of the boundary is  $C^k$  and it is an oriented boundary, i.e., the domain lies only on ‘one-side’ of the boundary.

**Example 6.10:** Let  $\Omega = \mathbb{R}^3 \setminus \{(x, 0, 0) : x \in \mathbb{R}\}$  then,

$$\partial\Omega = \{(x, 0, 0) : x \in \mathbb{R}\}$$

and therefore  $\Omega$  would not be a domain with a  $C^k$  boundary for any  $k$ .

Also, the same goes for

$$\Omega = \mathbb{R}^3 \setminus \{(0, 0, 0)\}$$

$$\Omega = \mathbb{R}^3 \setminus \{(x, y, 0) : x, y \in \mathbb{R}\}$$

Now, we wish to define level surfaces. These are the surfaces which are given by,

$$\begin{aligned} S &= \{(x, y, z) \in \mathbb{R}^3 : f(x, y, z) = c\} \\ &= f^{-1}(c) \subseteq \mathbb{R}^3 \end{aligned}$$

Another way to define a surface is:

$$S = \{(x, y, z) : z = \varphi(x, y)\}$$

with a map  $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}$  or  $\varphi : U \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ .

This is particularly useful for us to solve surface integrals in 3-d.

If we are to evaluate the integral  $\int_S f(x, y, z) dS$  over some surface  $S$  we can use the result

$$\int_S f(x, y, \varphi(x, y)) dS = \int_{(x, y) \in U} f(x, y, \varphi(x, y)) \sqrt{1 + (\nabla \varphi)^2} dx dy$$

## 7 Calculus of variations

Before delving into the world of calculus of variations, we wish to discuss one of its implications in physics - namely lagrangian mechanics. It is a more convenient way to study systems with a large degree of freedom. Newton’s laws does help us in formulating equations of motion for systems of all degrees of freedom. However, lagrangian mechanics provides an alternative and intuitive approach to do the same in a more ‘generalized way’.

To begin with, let a particle of general coordinates  $(q(t), \dot{q}(t), t)$  of one degree of freedom be studied.

We define the **Lagrangian**  $L(q(t), \dot{q}(t), t)$  as a smooth function of its variables.

From this, we define the **Action** as a *functional* of the path  $q(t)$  as-

$$S = \int_{t_I}^{t_F} L(q(t), \dot{q}(t), t) dt = S[q(t); t_I; t_F]$$

Where  $t_I$  and  $t_F$  are the initial and final time coordinates.

Now, the Hamilton's principle of least action states that out of all paths  $q(t)$ , that start at  $q_I$  at  $t_I$  and end at  $q_F$  at  $t_F$ , the actual path taken by the system is the one that *minimizes* (*extremizes*) the action. Therefore, the actual path taken by the system follows:

$$\partial S = 0 \quad (4)$$

*Note.* The value of  $q$  and  $\dot{q}$  at a particular instance in time are independent. However, the functions  $q(t)$  and  $\dot{q}(t)$  are not independent. So,  $\partial \dot{q}(t) = \frac{d}{dt} \partial q(t)$ .

Now, from Eq. 4 we get,

$$\begin{aligned} \partial S &= \int_{t_I}^{t_F} L(q(t) \partial q(t), \dot{q}(t) + \partial \dot{q}(t), t) dt - \int_{t_I}^{t_F} L(q(t), \dot{q}(t), t) dt \\ &= \int_{t_I}^{t_F} \left[ \frac{\partial L}{\partial q} \partial q + \frac{\partial L}{\partial \dot{q}} \partial \dot{q} + \dots \right] dt \\ &= \int_{t_I}^{t_F} \left[ \frac{\partial L}{\partial q} \partial q + \frac{\partial L}{\partial \dot{q}} \frac{d}{dt} \partial q + \dots \right] dt \\ &= \int_{t_I}^{t_F} \left[ \frac{\partial L}{\partial q} \partial q(t) + \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \partial q(t) - \partial q(t) \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} + \dots \right] dt \\ &= \int_{t_I}^{t_F} \left[ \frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right] \partial q(t) dt + \left[ \frac{\partial L}{\partial \dot{q}} \partial q(t) \right]_{t_I}^{t_F} + \dots \end{aligned}$$

Now, at the temporal boundary, the variations in  $\partial q(t)$  must vanish as  $q(t_I) = q_I$  and  $q(t_F) = q_F$  are fixed. Hence,  $\partial q(t_I) = \partial q(t_F) = 0$ .

$$\therefore \partial S = \int_{t_I}^{t_F} \left[ \frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right] \partial q(t) dt = 0$$

Now, for the above equation to hold, we must have for all variations  $\partial q(t)$ ,

$$\begin{aligned} \int_{t_I}^{t_F} \left[ \frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right] \partial q(t) dt &= 0 \\ \Rightarrow \frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} &= 0 \end{aligned}$$

This is famously called the Euler-Lagrange equation of motion. It can be expanded just a little further:



$$\begin{aligned}
\frac{\partial L}{\partial q} &= \frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}} \right) \\
&= \frac{\partial}{\partial \dot{q}} \left( \frac{dL}{dt} \right) \\
&= \frac{\partial}{\partial \dot{q}} \left( \frac{\partial L}{\partial q} \dot{q} + \frac{\partial L}{\partial \dot{q}} \dot{\dot{q}} \right) \\
\Rightarrow \frac{\partial L}{\partial q} &= \frac{\partial^2 L}{\partial q \partial \dot{q}} \dot{q} + \frac{\partial^2 L}{\partial \dot{q}^2} \dot{\dot{q}}
\end{aligned}$$

Now, proceeding to the multivariable case where we have  $n$  degrees of freedom, we see the lagrangian now becomes:

$$L \equiv L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t).$$

And we see,

$$\partial S = \sum_{i=1}^n \int_{t_I}^{t_F} \left[ \frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right] \partial \dot{q}_i(t) dt = 0$$

Now, when  $q_i$ 's can be varied independently our case of analysis becomes much simpler. We can directly conclude,

$$\begin{aligned}
&\int_{t_I}^{t_F} \left[ \frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right] \partial \dot{q}_i(t) dt = 0, \forall i \in \{1, 2, \dots, n\} \\
\Rightarrow \frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} &= 0, \forall i \in \{1, 2, \dots, n\} \\
\Rightarrow \frac{\partial L}{\partial q_i} - \sum_{j=1}^n \left( \frac{\partial^2 L}{\partial \dot{q}_i \partial q_j} \dot{q}_j + \frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} \dot{\dot{q}}_j \right) &= 0, \forall i \in \{1, 2, \dots, n\}
\end{aligned}$$

This is how calculus of variations is preliminarily taught in a basic mechanics course. However, there are more rigorous considerations that need too be taken care of while developing this theory as we shall see. A majority of the content of the following note was inspired and motivated by the work of Bruce van Brunt on Calculus of variations<sup>5</sup>. The exrercises that are solved in this note are all from the book itself.

We start with a necessary theorem:

**Theorem 7.1** (Local Extremum): Let  $f : \Omega \rightarrow \mathbb{R}, \Omega \subseteq \mathbb{R}^n$  be a mapping. We say that if  $f$  has a local extremum at  $\hat{x} \in \Omega$  then,

$$\nabla f(x) = 0$$

*Proof.* Let  $\hat{x}$  be a point of local extremum. Then,  $\exists \epsilon > 0$  such that  $f(\hat{x}) - f(x)$  does not change sign  $\forall x \in B_\epsilon(\hat{x})$ . Now, let  $\hat{x} + \epsilon \eta =: x, \eta \in \mathbb{R}^n$ . By Taylor's theorem we have,

$$\begin{aligned}
f(\hat{x} + \epsilon \eta) - f(\hat{x}) &= \epsilon(\eta \cdot \nabla f(\hat{x})) + \mathcal{O}(\epsilon^2) + \dots, \text{ and} \\
f(\hat{x} - \epsilon \eta) - f(\hat{x}) &= -\epsilon(\eta \cdot \nabla f(x)) - \mathcal{O}(\epsilon^2) - \dots
\end{aligned}$$

Now,  $\epsilon$  can be made sufficiently small such that the  $\text{sgn}(f(x) - f(\hat{x}))$  depends only on  $\text{sgn}(\eta \cdot \nabla f(x))$ . But at the local extremum,  $\text{sgn}(f(\hat{x} + \epsilon\eta)) = \text{sgn}(f(\hat{x} - \epsilon\eta))$ . Hence,  $\eta \cdot \nabla f(x) = 0, \forall \eta \in \mathbb{R}^n$ . Therefore,  $\nabla f(x) = 0$ .  $\square$

*Note.* The points at which  $\nabla f(x) = 0$  are called stationary points. However, we must note that even if  $x$  is a stationary point it is not necessarily a local extremum.

**Example 7.1** (Monkey Saddle): Consider the function

$$f(x, y) = y^3 - 3x^2y$$

Note that  $(0, 0)$  is a stationary point of  $f$ . However,  $(0, 0)$  is not a point of local extrema. To show this we can just look at the functional values along  $y$ -axis.  $f(0, y) = y^3$ . Thus,  $y > 0 \Rightarrow f > 0$  and  $y < 0 \Rightarrow f < 0$ .

Also, note that the second derivative of  $f$  at  $(0, 0)$  exists and is equal to 0.

To analyse the case with functionals is a bit trickier. Functionals belong to infinite dimensional spaces. Also, a single functional may belong to several vector spaces and once such a space has been chosen the outcome of extremization depends on which norm is being chosen for that space.

We wish to specialize to a more particular class of problem called the *fixed endpoint variational problem* where we are concerned with the functions from  $C^2[x_0, x_1]$ . We define

$$J : C^2[x_0, x_1] \longrightarrow \mathbb{R} \text{ by} \\ y \longmapsto \int_{x_0}^{x_1} f(x, y, y') dx$$

The fixed endpoint variational problem concerns itself with determining

$$y \in \mathcal{A} := \{y \in C^2[x_0, x_1] : y(x_0) = y_0, y(x_1) = y_1\}$$

.

Now note that  $\mathcal{A}$  is not closed under addition (owing to the boundary terms). Hence, we must have for each small variation of  $\hat{y} \in \mathcal{A}$  defined by  $\hat{y} = y + \epsilon\eta$ ,

$$\eta \in \mathcal{H} := \{\eta \in C^2[x_0, x_1] : \eta(x_0) = \eta(x_1) = 0\}$$

## 7.1 The first variation

Without loss of generality, by a similar argument we can say that  $\exists \epsilon > 0$  such that  $J(\hat{y})$  is a maximum and  $\|y - \hat{y}\| < \epsilon \Rightarrow J(\hat{y}) - J(y) \geq 0$ . Now, let  $y = \hat{y} + \epsilon\eta$ .

$$\begin{aligned} \therefore f(x, y, y') &= f(\hat{y} + \epsilon\eta, \hat{y}' + \epsilon\eta', x) \\ &= f(x, y, y') + \epsilon \left\{ \eta \frac{\partial f}{\partial y} + \eta' \frac{\partial f}{\partial y'} \right\} + \mathcal{O}(\epsilon) + \dots \end{aligned}$$

$$\begin{aligned}
\therefore J(\hat{y}) - J(y) &= \int_{x_0}^{x_1} f(x, \hat{y}, \hat{y}') dx - \int_{x_0}^{x_1} f(x, \hat{y} + \epsilon\eta, \hat{y}' + \epsilon\eta') dx \\
&= \epsilon \int_{x_0}^{x_1} \left( \eta \frac{\partial f}{\partial \hat{y}} + \eta' \frac{\partial f}{\partial \hat{y}'} \right) dx + \mathcal{O}(\epsilon^2) + \dots \\
&= \epsilon \delta J(\eta, \hat{y}) + \mathcal{O}(\epsilon^2) + \dots
\end{aligned}$$

The  $\delta J(\eta, \hat{y})$  is called *the first variation* in  $J$  and it determines the sign of  $J(\hat{y}) - J(y)$  for a sufficiently small  $\epsilon$ . But,  $J(\hat{y}) - J(y)$  is independent of  $\eta$ .

$$\begin{aligned}
&\therefore \delta J(\eta, \hat{y}) = 0, \forall \eta \in \mathcal{H} \\
&\Rightarrow \int_{x_0}^{x_1} \left( \eta \frac{\partial f}{\partial \hat{y}} + \eta' \frac{\partial f}{\partial \hat{y}'} \right) dx = 0 \\
&\Rightarrow \int_{x_0}^{x_1} \eta \frac{\partial f}{\partial \hat{y}} dx - \int_{x_0}^{x_1} \eta \frac{d}{dx} \left( \frac{\partial f}{\partial \hat{y}'} \right) dx + \left[ \eta \frac{\partial f}{\partial \hat{y}'} \right]_{x_0}^{x_1} \overset{0}{=} 0 \\
&\Rightarrow \int_{x_0}^{x_1} \eta \left\{ \frac{\partial f}{\partial \hat{y}} - \frac{d}{dx} \frac{\partial f}{\partial \hat{y}'} \right\} dx = 0
\end{aligned}$$

Now, as  $\hat{y} \in C^2[x_0, x_1]$  and  $f$  is smooth,

$$\begin{aligned}
E(x) &= \frac{\partial f}{\partial \hat{y}} - \frac{d}{dx} \left( \frac{\partial f}{\partial \hat{y}'} \right) \\
&= \frac{\partial f}{\partial \hat{y}} - \frac{\partial^2 f}{\partial x \partial \hat{y}} - \frac{\partial^2 f}{\partial \hat{y}^2} \hat{y}' - \frac{\partial^2 f}{\partial \hat{y}' \partial \hat{y}} \hat{y}''
\end{aligned}$$

is continuous.

Hence,

$$\begin{aligned}
&\int_{x_0}^{x_1} \eta E(x) dx = 0, \forall \eta \in \mathcal{H} \\
&\Rightarrow E(x) = 0 \\
&\Rightarrow \frac{\partial f}{\partial \hat{y}} - \frac{d}{dx} \left( \frac{\partial f}{\partial \hat{y}'} \right) = 0
\end{aligned} \tag{5}$$

We can claim this because if  $E(c) \neq 0$  for some  $x_0 < c < x_1$  then  $\exists$  interval  $(\alpha, \beta) \ni c$  such that  $E(x)$  has the same sign in  $(\alpha, \beta)$  owing to the continuity of  $E$  and

$$\begin{aligned}
\eta(x) &= \begin{cases} (x - \alpha)^3(\beta - x)^3 & \text{if } \alpha \leq x \leq \beta \\ 0 & \text{otherwise} \end{cases} \in C^2[x_0, x_1] \\
&\Rightarrow \int_{x_0}^{x_1} \eta E(x) dx \neq 0
\end{aligned}$$

would lead to a contradiction.

However, what we have arrived at is the Euler-Lagrange equation which is a second-order non-linear differential equation

$$\frac{\partial f}{\partial \hat{y}} = \frac{\partial^2 f}{\partial x \partial \hat{y}} + \frac{\partial^2 f}{\partial \hat{y}^2} \hat{y}' + \frac{\partial^2 f}{\partial \hat{y}' \partial \hat{y}} \hat{y}''$$

which is almost impossible to solve in most cases. Hence we examine a few special cases first.

**A. No explicit  $y$ –dependence.**

Suppose we wish to extremize

$$J(y) = \int_{x_0}^{x_1} f(x, y') dx$$

The Euler-Lagrange equation would give us:

$$\begin{aligned} \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right) &= 0 \\ \Rightarrow \frac{\partial f}{\partial y'} &= c_1 \end{aligned}$$

Now,  $\frac{\partial f}{\partial y'} \equiv h(x, y') = c_1 \Rightarrow y' = g(x, c_1)$ . Note that this can only be said iff  $\frac{\partial}{\partial y'} h(x, y') \neq 0$  by the implicit function theorem.

Thus, we have reduced the solution from a second order non-linear ODE to an implicit equation and quadratures.

**B. No explicit  $x$ –dependence.**

Suppose we wish to maximize

$$J(y) = \int_{x_0}^{x_1} f(y, y') dx$$

We define the function

$$H(y, y') = y' \frac{\partial f}{\partial y'} - f$$

and show that it is constant along  $\hat{y}$ .

*Proof.*

$$\begin{aligned} \frac{d}{dx} H(\hat{y}, \hat{y}') &= \frac{d}{dx} \left( \hat{y}' \frac{\partial f}{\partial \hat{y}'} - f \right) \\ &= \cancel{\hat{y}''} \frac{\partial f}{\partial \hat{y}'} - \frac{\partial f}{\partial \hat{y}} \hat{y}' - \cancel{\frac{\partial f}{\partial \hat{y}'}} \hat{y}'' - \cancel{\frac{\partial f}{\partial x}} + \hat{y}' \left( \frac{d}{dx} \frac{\partial f}{\partial \hat{y}'} \right) \\ &= \hat{y}' \left( \frac{d}{dx} \frac{\partial f}{\partial \hat{y}'} - \frac{\partial f}{\partial \hat{y}} \right) \\ &= 0 \end{aligned}$$

$\therefore H$  is constant along  $\hat{y}$

Thus, we get a first order linear differential equation for  $y$ .

$$H(y, \hat{y}) = \text{const.}$$

## 7.2 Solved Problems

A really good book to recommend to the reader pertaining to known solutions of various differential equations is an amazing work<sup>6</sup> by P.L. Sachdev.

**Exercise 7.1:** Let  $y \in S$  and  $\eta \in H$  be fixed functions. Then the quantity  $J(\hat{y} + \epsilon\eta)$  can be regarded as a function of the single real variable  $\epsilon$ . Show that the equation  $\frac{dJ}{d\epsilon} = 0$  at  $\epsilon = 0$  leads to the condition of Eq. 5 under the same hypothesis for  $f$ .

We fix  $\hat{y} \in S$  and  $\eta \in H$ . Now,  $J(\hat{y} + \epsilon\eta) \equiv g(\epsilon)$ . We see what happens when  $\frac{dg}{d\epsilon} \big|_{\epsilon=0} = 0$ . We have,

$$\begin{aligned} J(\hat{y} + \epsilon\eta) &= \int_{x_0}^{x_1} f(x, \hat{y} + \epsilon\eta, \hat{y}' + \epsilon\eta') dx \\ \Rightarrow \frac{dJ}{d\epsilon} &= \frac{d}{d\epsilon} \int_{x_0}^{x_1} f(x, \hat{y} + \epsilon\eta, \hat{y}' + \epsilon\eta') dx \\ &= \int_{x_0}^{x_1} \left[ \frac{\partial f}{\partial(\hat{y} + \epsilon\eta)} \frac{d(\hat{y} + \epsilon\eta)}{d\epsilon} + \frac{\partial f}{\partial(\hat{y}' + \epsilon\eta')} \frac{d(\hat{y}' + \epsilon\eta')}{d\epsilon} \right] dx \\ &= \int_{x_0}^{x_1} \left[ \eta \frac{\partial f}{\partial(\hat{y} + \epsilon\eta)} + \eta' \frac{\partial f}{\partial(\hat{y}' + \epsilon\eta')} \right] dx \\ \therefore \frac{dJ}{d\epsilon} \bigg|_{\epsilon=0} &= \int_{x_0}^{x_1} \left( \eta \frac{\partial f}{\partial \hat{y}} + \eta' \frac{\partial f}{\partial \hat{y}'} \right) dx = 0 \end{aligned}$$

which gives exactly the same condition as Eq. 5.

**Exercise 7.2:** Let  $J : S \rightarrow \mathbb{R}$  and  $K : S \rightarrow \mathbb{R}$  be functionals defined by

$$J(y) = \int_{x_0}^{x_1} f(x, y, y') dx, K(y) = \int_{x_0}^{x_1} g(x, y, y') dx$$

where  $f$  and  $g$  are smooth functions of the indicated arguments and  $\Omega \subseteq \mathbb{R}$ .

a) Show that for any real numbers  $A$  and  $B$ ,

$$\delta(AJ + BK)(\eta, y) = A\delta J(\eta, y) + B\delta K(\eta, y)$$

(i.e.,  $\delta$  is a linear operator), and

$$\delta(JK)(\eta, y) = K(\eta, y)\delta J(\eta, y) + J(\eta, y)\delta K(\eta, y)$$

(a product rule)

b) Suppose that  $G : \Omega \times \Omega \rightarrow \mathbb{R}$  is differentiable on  $\Omega \times \Omega$ . Show that

$$\delta G(J, K)(\eta, y) = \frac{\partial G}{\partial J} \delta J(\eta, y) + \frac{\partial G}{\partial K} \delta K(\eta, y)$$

(a “chain rule” for the  $\delta$  operator).

a) We have  $J(y) = \int_{x_0}^{x_1} f(x, y, y') dx$  and  $K(y) = \int_{x_0}^{x_1} g(x, y, y') dx$ .

$$\begin{aligned}\therefore AJ(y) &= A \int_{x_0}^{x_1} f(x, y, y') dx \\ \text{and, } BK(y) &= B \int_{x_0}^{x_1} g(x, y, y') dx\end{aligned}$$

Now,

$$\begin{aligned}\delta AJ(y, \eta) &= A \int_{x_0}^{x_1} \left( \frac{\partial f}{\partial y} \eta + \frac{\partial f}{\partial y'} \eta' \right) dx = A \delta J(y, \eta) \\ \text{and, } \delta BK(y, \eta) &= B \int_{x_0}^{x_1} \left( \frac{\partial g}{\partial y} \eta + \frac{\partial g}{\partial y'} \eta' \right) dx = B \delta K(y, \eta) \\ \therefore \delta(AJ + BK)(y, \eta) &= \int_{x_0}^{x_1} \left( \frac{\partial(Af + Bg)}{\partial y} \eta' + \frac{\partial(Af + Bg)}{\partial y'} \eta' \right) dx \\ &= A \int_{x_0}^{x_1} \left( \frac{\partial f}{\partial y} \eta + \frac{\partial f}{\partial y'} \eta' \right) dx + B \int_{x_0}^{x_1} \left( \frac{\partial g}{\partial y} \eta + \frac{\partial g}{\partial y'} \eta' \right) dx \\ &= A \delta J(y, \eta) + B \delta K(y, \eta)\end{aligned}$$

Hence, the linearity of  $\delta$  follows from the linearity of partial derivatives.

We also have,

$$\begin{aligned}\delta(JK)(y, \eta) &= \int_{x_0}^{x_1} \left( \frac{\partial(fg)}{\partial y} \eta + \frac{\partial(fg)}{\partial y'} \eta' \right) dx \\ &= \int_{x_0}^{x_1} \left[ \left( g \frac{\partial f}{\partial y} + f \frac{\partial g}{\partial y} \right) \eta + \left( g \frac{\partial f}{\partial y'} + f \frac{\partial g}{\partial y'} \right) \eta' \right] dx \\ &= \int_{x_0}^{x_1} g \left( \frac{\partial f}{\partial y} \eta + \frac{\partial f}{\partial y'} \eta' \right) dx + \int_{x_0}^{x_1} f \left( \frac{\partial g}{\partial y} \eta + \frac{\partial g}{\partial y'} \eta' \right) dx \\ &= K(y) \delta J(y, \eta) + J(y) \delta K(y, \eta)\end{aligned}$$

And, we get the product rule.

b) Note that we had  $J : S \rightarrow \Omega$  and  $K : S \rightarrow \Omega$ . Now we are given  $G : \Omega \times \Omega \rightarrow \mathbb{R}$ .

We wish to find  $\delta G(J, K)(y, \eta)$ . Note that,

$$\begin{aligned}\delta G(J, K)(y, \eta) &= \left. \frac{dG}{d\epsilon} \right|_{\epsilon=0} \quad (\text{as shown in the first problem}) \\ &= \left. \frac{\partial G}{\partial J} \cdot \frac{dJ}{d\epsilon} \right|_{\epsilon=0} + \left. \frac{\partial G}{\partial K} \cdot \frac{dK}{d\epsilon} \right|_{\epsilon=0} \\ &= \frac{\partial G}{\partial J} \delta J(y, \eta) + \frac{\partial G}{\partial K} \delta K(y, \eta)\end{aligned}$$

**Exercise 7.3:** Let  $n$  be any positive integer. Show that there exists a  $\nu \in C^n(\mathbb{R})$  such that  $\nu(x) > 0 \ \forall x \in (\alpha, \beta)$  and  $\nu = 0 \ \forall x \in \mathbb{R} \setminus (\alpha, \beta)$ .

**Exercise 7.4:** Let  $J$  be the functional defined by

$$J(y) = \int_{x_0}^{x_1} (y'^2 + y^2 + 4ye^x) dx,$$

with boundary conditions  $y(0) = 1$  and  $y(1) = 1$ . Find the extremal(s) in  $C^2[0, 1]$  for  $J$ .

Let  $f(x, y, y') = y'^2 + y^2 + 4ye^x$ . The Euler-Lagrange equation is:

$$\begin{aligned} \frac{d}{dx} \frac{\partial f}{\partial y'} - \frac{\partial f}{\partial y} &= 0 \\ \Rightarrow \frac{d}{dx}(2y') - (2y + 4e^x) &= 0 \\ \Rightarrow y'' - y &= 2e^x \end{aligned}$$

To solve the homogenous ODE:

$$y'' - y = 0$$

we note the characteristic equation is,

$$\begin{aligned} t^2 - 1 &= 0 \\ \Rightarrow t &= \pm 1 \\ \therefore y_h(x) &= c_1 e^x + c_2 e^{-x} \end{aligned}$$

Next, we evaluate the particular integral for  $(D^2 - 1)y = 2e^x$  where  $D \equiv \frac{d}{dx}$ .

The P.I. is given by:

$$\begin{aligned} y_p(x) &= \frac{1}{D^2 - 1} 2e^x = \frac{1}{(D + 1)(D - 1)} 2e^x \\ &= \frac{2xe^x}{2} \\ &= xe^x \end{aligned}$$

Hence, the final solution can be given by:

$$y(x) = h(x) + y_p(x) = c_1 e^x + c_2 e^{-x} + xe^x$$

Applying boundary conditions,

$$\begin{aligned} y(0) &= c_1 + c_2 = 0 \Rightarrow c_2 = -c_1 \\ y(1) &= c_1 e + c_2 e^{-1} + e = 1 \\ \Rightarrow c_1 e - c_1 e^{-1} + e &= 1 \\ \Rightarrow c_1 (e - e^{-1}) &= 1 - e \\ \Rightarrow c_1 &= \frac{1 - e}{e - e^{-1}} = -\frac{e}{e + 1} \\ \therefore c_2 &= \frac{e}{e + 1} \end{aligned}$$

Hence,  $y(x) = -\frac{e}{e+1}e^x + \frac{e}{e+1}e^{-x} + xe^x$  is the extremal for  $J$ .

**Exercise 7.5:** Consider the functional defined by

$$J(y) = \int_{x_0}^{x_1} x^4 y'^2 dx.$$

a) Show that no extremals in  $C^2[-1, 1]$  exist which satisfy the boundary conditions  $y(-1) = -1, y(1) = 1$ .

b) Without resorting to Euler-Lagrange equation, prove that  $J$  cannot have a local minimum in the set

$$S = \{y \in C^2[-1, 1] : y(-1) = -1 \text{ and } y(1) = 1\}$$

Let  $f(x, y, y') = x^4 y'^2$ . By Euler-Lagrange equation we get,

$$\begin{aligned} \frac{d}{dx} 2x^4 y' &= 0 \\ \Rightarrow 2x^4 y' &= c \\ \Rightarrow y' &= \frac{c}{2x^4} \\ \therefore y &= -\frac{c}{6x^3} + d \end{aligned}$$

Now, applying boundary conditions,

$$\begin{aligned} y(1) &= -\frac{c}{6} + d = 1 \\ \text{and } y(-1) &= \frac{c}{6} + d = -1 \\ \therefore 2d &= 0 \Rightarrow d = 0 \\ \text{and } -\frac{c}{6} &= 1 \Rightarrow c = -6 \\ \therefore y(x) &= \frac{1}{x^3} \end{aligned}$$

Now, clearly  $y \notin C^2[-1, 1]$  as at  $x = 0$   $y$  is not continuous and undefined. Therefore,  $\nexists C^2[-1, 1]$  solution for the boundary condition.

b) We wish to prove the same without resorting to the Euler-Lagrange equation. Let,

$$S = \{y \in C^2[-1, 1] : y(-1) = -1, y(1) = 1\}$$

The first variation must vanish if  $y$  is an extremum.



$$\begin{aligned}
&\Rightarrow \delta J(\eta, y) = \int_{x_0}^{x_1} 2x^4 y' \eta' dx = 0 \\
&\Rightarrow \int_{x_0}^{x_1} \eta \frac{d}{dx} (x^4 y') dx = 0 \\
&\Rightarrow \frac{d}{dx} x^4 y' = 0 \\
&\Rightarrow x^4 y' = c \\
&\Rightarrow y' = \frac{c}{x^4}
\end{aligned}$$

Thus, we arrive at the same result as before.

### 7.3 Functionals with higher order derivatives

We shall see in due course of time that extremization problem for the functional of the form:

$$J(y) = \int_{x_0}^{x_1} f(x, y, y', y'') dx$$

must have solutions in a more restricted space in order for our tools for solving these problems to be properly applied. We will work with the boundary conditions:

$$y(x_0) = y_0, y'(x_0) = y'_0; y(x_1) = y_1, y'(x_1) = y'_1.$$

For some  $n \in \mathbb{N}$ , the value of which will be determined later, we must have the solution set:

$$S := \{y \in C^n[x_0, x_1] : y(x_0) = y_0, y'(x_0) = y'_0, y(x_1) = y_1, y'(x_1) = y'_1\}$$

and the small variation in  $y$  must be from the set:

$$\mathcal{H} := \{\eta \in C^n[x_0, x_1] : \eta(x_0) = \eta'(x_0) = \eta(x_1) = \eta'(x_1) = 0\}.$$

Now, suppose  $J$  has a local extremum in  $S$  at  $\hat{y} \in S$ . Proceeding in the same way, let  $y = \hat{y} + \epsilon \eta, \eta \in \mathcal{H}$ . We have,

$$J(y) - J(\hat{y}) = \epsilon \int_{x_0}^{x_1} \left( \eta \frac{\partial f}{\partial \hat{y}} + \eta' \frac{\partial f}{\partial \hat{y}'} + \eta'' \frac{\partial f}{\partial \hat{y}''} \right) dx + \mathcal{O}(\epsilon^2) + \dots$$

So, we get the first variation as:

$$\delta J(\hat{y}, \eta) = \int_{x_0}^{x_1} \left( \eta \frac{\partial f}{\partial \hat{y}} + \eta' \frac{\partial f}{\partial \hat{y}'} + \eta'' \frac{\partial f}{\partial \hat{y}''} \right) dx \quad (6)$$

Again, by similar logic, at the extremum,

$$\delta J(\hat{y}, \eta) = 0$$

Now, we integrate by parts to further simplify Eq. 6

$$\begin{aligned}
\int_{x_0}^{x_1} \eta'' \frac{\partial f}{\partial \hat{y}''} dx &= \eta' \frac{\partial f}{\partial \hat{y}''} \Big|_{x_0}^{x_1} - \int_{x_0}^{x_1} \left( \eta' \frac{d}{dx} \frac{\partial f}{\partial \hat{y}''} \right) dx \\
&= -\eta \frac{d}{dx} \frac{\partial f}{\partial \hat{y}''} \Big|_{x_0}^{x_1} + \int_{x_0}^{x_1} \left( \eta \frac{d^2}{dx^2} \frac{\partial f}{\partial \hat{y}''} \right) dx \\
&= \int_{x_0}^{x_1} \left( \eta \frac{d^2}{dx^2} \frac{\partial f}{\partial \hat{y}''} \right) dx
\end{aligned}$$

Also,

$$\begin{aligned}
\int_{x_0}^{x_1} \eta' \frac{\partial f}{\partial \hat{y}'} dx &= \eta \frac{\partial f}{\partial \hat{y}'} \Big|_{x_0}^{x_1} - \int_{x_0}^{x_1} \eta \frac{d}{dx} \frac{\partial f}{\partial \hat{y}'} dx \\
&= - \int_{x_0}^{x_1} \left( \eta \frac{d}{dx} \frac{\partial f}{\partial \hat{y}'} \right) dx
\end{aligned}$$

Thus,

$$\begin{aligned}
\delta J(\hat{y}, \eta) &= 0 \\
\Rightarrow \int_{x_0}^{x_1} \eta \left\{ \frac{\partial f}{\partial \hat{y}} - \frac{d}{dx} \frac{\partial f}{\partial \hat{y}'} + \frac{d^2}{dx^2} \frac{\partial f}{\partial \hat{y}''} \right\} dx &= 0
\end{aligned}$$

Now let

$$E(x) = \frac{\partial f}{\partial \hat{y}} - \frac{d}{dx} \frac{\partial f}{\partial \hat{y}'} + \frac{d^2}{dx^2} \frac{\partial f}{\partial \hat{y}''}$$

In order for  $E$  to be continuous we must have  $y \in C^4[x_0, x_1]$ . Hence, we choose  $n = 4$ . We thus get,

$$\frac{\partial f}{\partial \hat{y}} - \frac{d}{dx} \left( \frac{\partial f}{\partial \hat{y}'} \right) + \frac{d^2}{dx^2} \left( \frac{\partial f}{\partial \hat{y}''} \right) = 0$$

## 7.4 Functionals with more than one dependent variable

So far we have been studying functionals with only one dependent variable  $y$  with  $x$  as the independent variable. However, variational problems typically depend on several dependent variables. For example, in classical mechanics the motion of a particle is dependent upon three dependent variables  $(x(t), y(t), z(t))$  with ' $t$ ' as the independent parameter. We now wish to derive the Euler-Lagrange equation(s) for functionals that depend on several dependent variables.

So, to begin with we can say our functional  $J$  is a mapping

$$J : \mathbb{C}^2[t_0, t_1] \rightarrow \mathbb{R}$$

where  $\mathbb{C}^2[t_0, t_1]$  denotes the set of all functions  $\vec{q} : [t_0, t_1] \rightarrow \mathbb{R}^n$  such that

$$\begin{aligned}
\vec{q} &\equiv (q_1, q_2, \dots, q_n) \text{ and} \\
q_k &\in C^2[t_0, t_1] \quad \forall k \in \{1, 2, \dots, n\}
\end{aligned}$$

Now, to work with the functional  $J$ , we must define a norm on  $\mathbb{C}^2[t_0, t_1]$ . Again, the choice of norm is subjective and depends upon its application. We define,

$$\|\vec{q}\| := \max_{k=1,2,\dots,n} \sup_{t \in [t_0, t_1]} |q_k(t)|$$

Thus, we have set up our vector space of dependent variables. Now, our solutions set becomes (for the fixed endpoint variational problem):

$$S := \{\vec{q} \in \mathbb{C}^2[t_0, t_1] : \vec{q}(t_0) = \vec{q}_0, \vec{q}(t_1) = \vec{q}_1\}$$

and the ‘perturbation variable’ comes from the space

$$\mathcal{H} := \{\vec{\eta} \in \mathbb{C}^2[t_0, t_1] : \vec{\eta}(t_0) = \vec{\eta}(t_1) = 0\}$$

We can now compute for  $\vec{q} = \hat{\vec{q}} + \epsilon \vec{\eta}$

$$\begin{aligned} J(\vec{q}) - J(\hat{\vec{q}}) &= \int_{t_0}^{t_1} L(t, \hat{\vec{q}}, \epsilon \vec{\eta} \cdot \dot{\hat{\vec{q}}} + \epsilon \dot{\vec{q}}) dt - \int_{t_0}^{t_1} L(t, \hat{\vec{q}}, \dot{\hat{\vec{q}}}) dt \\ &= \epsilon \int_{t_0}^{t_1} \sum_{k=1}^n \left( \eta_k \frac{\partial L}{\partial q_k} + \dot{\eta}_k \frac{\partial L}{\partial \dot{q}_k} \right) dt + \mathcal{O}(\epsilon^2) + \dots \end{aligned}$$

Thus, the first variation becomes:

$$\delta J(\vec{\eta}, \hat{\vec{q}}) = \int_{t_0}^{t_1} \sum_{k=1}^n \left( \eta_k \frac{\partial L}{\partial q_k} + \dot{\eta}_k \frac{\partial L}{\partial \dot{q}_k} \right) dt$$

By a similar argument, we say that if  $J$  has an extremal at  $\hat{\vec{q}}$ ,

$$\delta J(\vec{\eta}, \hat{\vec{q}}) = 0$$

Note that,  $(\eta_1, 0, \dots, 0), (0, \eta_2, 0, \dots, 0), \dots, (0, 0, \dots, \eta_n) \in \mathcal{H}$ . So we can directly conclude:

$$\begin{aligned} \int_{t_0}^{t_1} \left( \eta_1 \frac{\partial L}{\partial q_1} + \dot{\eta}_1 \frac{\partial L}{\partial \dot{q}_1} \right) dt &= 0 \\ \int_{t_0}^{t_1} \left( \eta_2 \frac{\partial L}{\partial q_2} + \dot{\eta}_2 \frac{\partial L}{\partial \dot{q}_2} \right) dt &= 0 \\ &\vdots \\ \int_{t_0}^{t_1} \left( \eta_n \frac{\partial L}{\partial q_n} + \dot{\eta}_n \frac{\partial L}{\partial \dot{q}_n} \right) dt &= 0 \end{aligned}$$

Therefore we get,

$$\begin{aligned} \frac{\partial L}{\partial \hat{q}_1} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\hat{q}}_1} &= 0 \\ \frac{\partial L}{\partial \hat{q}_2} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\hat{q}}_2} &= 0 \\ &\vdots \\ \frac{\partial L}{\partial \hat{q}_n} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\hat{q}}_n} &= 0 \end{aligned}$$

Hence, we get a system of  $n$  second order differential equations.  $\hat{\vec{q}}$  satisfies this system iff  $\delta J(\hat{\vec{q}}, \vec{\eta}) = 0 \forall \vec{\eta} \in \mathcal{H}$ .

In other way, if  $\hat{\vec{q}}$  is an extremal of  $J$  in  $S$  then,

$$\frac{\partial L}{\partial \hat{q}_k} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\hat{q}}_k} = 0 \quad \forall k \in \{1, 2, \dots, n\}$$

## 7.5 Solved Problems

**Exercise 7.6:** Find the general solution for the extremals to the functionals defined by:

$$J(y) = \int_{x_0}^{x_1} (y''^2 - y^2 + 2yx^3) dx$$

We have the fourth order Euler-Lagrange equation:

$$\begin{aligned} \frac{d^2}{dx^2} \frac{\partial f}{\partial \hat{y}''} - \frac{d}{dx} \frac{\partial f}{\partial \hat{y}'} + \frac{\partial f}{\partial \hat{y}} &= 0 \\ \Rightarrow \frac{d^2}{dx^2} (2\hat{y}'') - \frac{d}{dx} (0) + (-2\hat{y} + 2x^3) &= 0 \\ \Rightarrow \hat{y}^{(4)} - \hat{y} + x^3 &= 0 \\ \Rightarrow \hat{y}^{(4)} - \hat{y} &= -x^3 \end{aligned}$$

We have the characteristic equation:

$$\begin{aligned} r^4 - 1 &= 0 \\ \Rightarrow r &= \pm 1, \pm i \end{aligned}$$

Therefore, solution to the homogenous equation  $\hat{y}^{(4)} - \hat{y} = 0$  is

$$\hat{y}_h(x) = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x, c_k \in \mathbb{C} \quad \forall k = 1, 2, 3, 4$$

TO evaluate the particular integral, we evaluate (taking  $\frac{d}{dx} \equiv D$ )

$$\begin{aligned} \hat{y}_p(x) &= -\frac{1}{D^4 - 1} x^3 = (1 - D^4)^{-1} x^3 \\ &= (1 + D^4 + D^8 + \dots) x^3 \\ &= x^3 \end{aligned}$$

Hence, the final solution is:

$$\hat{y}_h(x) + \hat{y}_p(x) = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x + x^3$$

**Exercise 7.7:** Show that when the integrand does not contain  $x$  explicitly,

$$H(y, y', y'') = y'' \frac{\partial f}{\partial y''} - y' \left( \frac{d}{dx} \frac{\partial f}{\partial y''} - \frac{\partial f}{\partial y'} \right) - f = \text{constant}$$

For an extremal  $y \in C^4[x_0, x_1]$  we have,

$$\begin{aligned}
\frac{d}{dx}H(y, y', y'') &= \frac{d}{dx} \left( y'' \frac{\partial f}{\partial y''} \right) - \frac{d}{dx} \left( y' \left( \frac{d}{dx} \frac{\partial f}{\partial y''} - \frac{\partial f}{\partial y'} \right) \right) - \frac{d}{dx} f \\
&= \cancel{y'''} \frac{\partial f}{\partial y''} + y'' \cancel{\frac{d}{dx}} \frac{\partial f}{\partial y''} - y'' \cancel{\frac{d}{dx}} \frac{\partial f}{\partial y''} + \cancel{y''} \frac{\partial f}{\partial y'} \\
&\quad - y' \frac{d^2}{dx^2} \frac{\partial f}{\partial y''} + y' \frac{d}{dx} \frac{\partial f}{\partial y'} - \frac{\partial f}{\partial y'} y' - \cancel{\frac{\partial f}{\partial y'} y''} - \cancel{\frac{\partial f}{\partial y''} y'''} \\
&= -y' \left[ \frac{d^2}{dx^2} \frac{\partial f}{\partial y''} + \frac{d}{dx} \frac{\partial f}{\partial y'} - \frac{\partial f}{\partial y'} y' \right] \\
&= 0
\end{aligned}$$

Hence,  $H(y, y', y'')$  is constant along an extremal.

**Exercise 7.8:** For the functional  $J$  defined by

$$J(y) = \int_{x_0}^{x_1} y' \sqrt{1 + y'^2} \, dx$$

find an extremal satisfying  $y(0) = 0, y'(0) = 0, y(1) = 1, y'(1) = 2$ .

As the integrand does not depend on  $y$  explicitly, we have for an extremal  $y$

$$\begin{aligned}
\frac{d}{dx} \frac{\partial f}{\partial y''} - \frac{\partial f}{\partial y'} &= \text{const.} \\
\Rightarrow \frac{d}{dx} \left( \frac{y' y''}{\sqrt{1 + y'^2}} \right) - \sqrt{1 + y'^2} &= c_1
\end{aligned}$$

**Exercise 7.9:**  $J$  is a functional of the form

$$J(y) = \int_{x_0}^{x_1} (A(x, y, y') y'' + B(x, y, y')) dx$$

where  $A$  and  $B$  are smooth functions. Show that the Euler-Lagrange equation for this functional is at most a second order differential equation. Any solution can satisfy at most two arbitrary conditions.

The Euler-Lagrange equation is:

$$\begin{aligned}
& \frac{d^2}{dx^2} \frac{\partial f}{\partial y''} - \frac{d}{dx} \frac{\partial f}{\partial y'} + \frac{\partial f}{\partial y} = 0 \\
& \Rightarrow \frac{d^2}{dx^2} A(x, y, y') - \frac{d}{dx} \left( y'' \frac{\partial A}{\partial y'} \right) + \frac{\partial B}{\partial y'} + y'' \frac{\partial A}{\partial y} + \frac{\partial B}{\partial y} = 0 \\
& \Rightarrow \frac{d}{dx} \left( \frac{\partial A}{\partial x} + \frac{\partial A}{\partial y} y' + \frac{\partial A}{\partial y'} y'' \right) - \frac{d}{dx} \left( y'' \frac{\partial A}{\partial y'} \right) + \frac{\partial B}{\partial y'} + y'' \frac{\partial A}{\partial y} + \frac{\partial B}{\partial y} = 0 \\
& \Rightarrow \frac{d}{dx} \left( \frac{\partial A}{\partial x} + \frac{\partial A}{\partial y} y' \right) + \frac{\partial B}{\partial y'} + \frac{\partial B}{\partial y} + y'' \frac{\partial A}{\partial y} = 0 \\
& \Rightarrow \frac{d}{dx} \frac{\partial A}{\partial x}(x, y, y') + y' \frac{d}{dx} \frac{\partial A}{\partial y}(x, y, y') + \frac{\partial A}{\partial y'}(x, y, y') y'' + \\
& y'' \frac{\partial A}{\partial y}(x, y, y') + \frac{\partial B}{\partial y'}(x, y, y') + \frac{\partial B}{\partial y}(x, y, y') = 0
\end{aligned}$$

In the above equation it is clear that we do not get any term higher than 2<sup>nd</sup> order. Hence, any solution can satisfy at most two arbitrary constants.

**Exercise 7.10:** Define the functionals  $J$  and  $K$  by:

$$\begin{aligned}
J(y) &= \int_{x_0}^{x_1} f(x, y, y', y'') dx \\
K(y) &= \int_{x_0}^{x_1} F(x, y, y', y'') dx
\end{aligned}$$

where for some smooth function  $G$ ,

$$F(x, y, y', y'') = f(x, y, y', y'') + \frac{d}{dx} G(x, y, y')$$

Show that any extremal of  $J$  is also an extremal of  $K$ .

Observe that,

$$\begin{aligned}
K(y) &= \int_{x_0}^{x_1} F(x, y, y', y'') dx = \int_{x_0}^{x_1} \left( f(x, y, y', y'') + \frac{d}{dx} G(x, y, y') \right) dx \\
&= \int_{x_0}^{x_1} f(x, y, y', y'') dx + \\
&\quad \{G(x_1, y(x_1), y'(x_1)) - G(x_0, y(x_0), y'(x_0))\} \\
&= \int_{x_0}^{x_1} f(x, y, y', y'') dx + C
\end{aligned}$$

Therefore, having fixed boundary values ensures that

$$K(y) = J(y) + C$$

In other words, any extremal  $y$  for  $J$  must therefore also be an extremal for  $K$ .

Thus, adding a total differential to the integrand does not change the extremals for the functionals.

**Exercise 7.11:** Let  $J$  be a functional of the form

$$J(y) = \int_{x_0}^{x_1} f(x, y, y'', \dots, y^{(n)}) dx$$

a) Formulate the fixed endpoint variational problem for this functional and prove any smooth extremal must satisfy

$$(-1)^n \frac{d^n}{dx^n} \frac{\partial f}{\partial y^{(n)}} + (-1)^{n-1} \frac{d^{n-1}}{dx^{n-1}} \frac{\partial f}{\partial y^{(n-1)}} + \dots + \frac{\partial f}{\partial y} = 0$$

What are the assumptions you make on the function  $f$  and the function space?

b) Let  $f$  be of the form  $A(x, y, y', \dots, y^{(n-1)})y^{(n)} + B(x, y, y', \dots, y^{(n-1)})$ .

What is the maximum order the Euler-Lagrange equation can be?

The functional we wish to extremize is

$$J(y) = \int_{x_0}^{x_1} f(x, y, y'', \dots, y^{(n)}) dx$$

Let

$$\eta \in \left\{ \eta \in C^n[x_0, x_1] : \eta(x_0) = \eta(x_1) = \eta'(x_0) = \eta'(x_1) = \dots = \eta^{(n-1)}(x_0) = \eta^{(n-1)}(x_1) = 0 \right\}$$

The value of  $k$  is to be determined later.

The solution space for this problem can be given by

$$\mathcal{S} := \left\{ y \in C^k[x_0, x_1] : y(x_0) = y_0, y(x_1) = y_1, \dots, y^{(n-1)}(x_0) = y_0^{(n-1)}, y^{(n-1)}(x_1) = y_1^{(n-1)} \right\}$$

Let  $\hat{y} \in \mathcal{S}$  be an extremal for  $J$  and let  $y = \hat{y} + \epsilon \eta \in \mathcal{S}$ . Now,

$$J(y) - J(\hat{y}) = \epsilon \int_{x_0}^{x_1} \left( \eta \frac{\partial f}{\partial \hat{y}} + \eta' \frac{\partial f}{\partial \hat{y}'} + \dots + \eta^{(n)} \frac{\partial f}{\partial \hat{y}^{(n)}} \right) dx + \mathcal{O}(\epsilon^2) + \dots$$

Hence, the first variation is given by

$$\delta J(\hat{y}) = \int_{x_0}^{x_1} \left( \eta \frac{\partial f}{\partial \hat{y}} + \eta' \frac{\partial f}{\partial \hat{y}'} + \dots + \eta^{(n)} \frac{\partial f}{\partial \hat{y}^{(n)}} \right) dx \quad (7)$$

We wish to prove the result of integration by parts for higher derivatives.

*Claim:*

$$\int_{x_0}^{x_1} \eta^{(k)} g(x, y, y', \dots, y^{(n)}) dx = (-1)^k \int_{x_0}^{x_1} \eta \frac{d^k}{dx^k} g(x, y, y', \dots, y^{(n)}) dx, \forall g \in C^k(\mathbb{R}^{n+1}), \forall k, n \in \mathbb{N}$$

We prove the same by induction.

Base case:  $k = 1$ .

$$\begin{aligned} \int_{x_0}^{x_1} \eta' g(x, y, y', \dots, y^{(n)}) dx &= \eta g(x, y, y', \dots, y^{(n)}) \Big|_{x_0}^{x_1} - \int_{x_0}^{x_1} \eta \frac{d}{dx} g(x, y, y', \dots, y^{(n)}) dx \\ &= - \int_{x_0}^{x_1} \eta \frac{d}{dx} g(x, y, y', \dots, y^{(n)}) dx \end{aligned}$$

Inductive case: Assume the statement is true for  $k = i - 1$ .

We have,

$$\int_{x_0}^{x_1} \eta^{(i-1)} g(x, y, y', \dots, y^{(n)}) dx = (-1)^{i-1} \int_{x_0}^{x_1} \eta \frac{d^{i-1}}{dx^{i-1}} g(x, y, y', \dots, y^{(n)}) dx$$

Now,

$$\begin{aligned} \int_{x_0}^{x_1} \eta^{(i)} g(x, y, y', \dots, y^{(n)}) dx &= \eta^{(i-1)} g(x, y, y', \dots, y^{(n)}) \Big|_{x_0}^{x_1} - \int_{x_0}^{x_1} \eta^{(i-1)} \frac{d}{dx} g(x, y, y', \dots, y^{(n)}) dx \\ &= (-1)^{i-1+1} \int_{x_0}^{x_1} \eta \frac{d^{i-1+1}}{dx^{i-1+1}} g(x, y, y', \dots, y^{(n)}) dx \\ &= (-1)^i \int_{x_0}^{x_1} \eta \frac{d^i}{dx^i} g(x, y, y', \dots, y^{(n)}) dx \end{aligned}$$

Thus the statement holds for  $k = i$ . Hence, our claim was true.

Now, we can conclude from Eq. 7,

$$\delta J(\hat{y}, \eta) = \int_{x_0}^{x_1} \eta \left( (-1)^n \frac{d^n}{dx^n} \frac{\partial f}{\partial y^{(n)}} + (-1)^{n-1} \frac{d^{n-1}}{dx^{n-1}} \frac{\partial f}{\partial y^{(n-1)}} + \dots + \frac{\partial f}{\partial y} \right) dx = 0$$

Now, we set  $k = 2n$ , i.e.,  $\hat{y} \in C^{2n}[x_0, x_1]$  such that

$$E(x) = (-1)^n \frac{d^n}{dx^n} \frac{\partial f}{\partial y^{(n)}} + (-1)^{n-1} \frac{d^{n-1}}{dx^{n-1}} \frac{\partial f}{\partial y^{(n-1)}} + \dots + \frac{\partial f}{\partial y}$$

is continuous and we have,

$$(-1)^n \frac{d^n}{dx^n} \frac{\partial f}{\partial y^{(n)}} + (-1)^{n-1} \frac{d^{n-1}}{dx^{n-1}} \frac{\partial f}{\partial y^{(n-1)}} + \dots + \frac{\partial f}{\partial y} = 0$$

Hence, we are done.  $\square$

b) Note that the integrand is a linear function in  $y^{(n)}$  and the Euler-Lagrange equation is:

$$\sum_{i=0}^n (-1)^i \frac{d^i}{dx^i} \frac{\partial f}{\partial y^{(i)}} = 0$$

the maximum order of the term  $(-1)^n \frac{d^n}{dx^n} \frac{\partial f}{\partial y^{(n)}}$  is  $2n + 1$  this is because the integrand being linear in  $y^{(n)}$  the maximum order of the partial derivative can be  $n - 1$ . Now, all the other terms have a maximum order of the partial derivatives as  $n$  but the order of the



total derivative  $\frac{d^i}{dx^i}$  is less than  $n$ , i.e.,  $i < n$ . Hence, the maximum order of the the other terms also do not exceed  $2n - 1$ .

Thus, the maximum order of the Euler-Lagrange equation can be  $2n - 1$  and it can satisfy at most  $2n - 1$  arbitrary constraints.

**Exercise 7.12:** Let  $q \equiv (q_1, q_2)$  and let  $L(t, q, \dot{q}) = \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2) - gq_2$ .

a) Find the extremals for the functionals  $J$  defined by

$$J(q) = \int_{t_0}^{t_1} L(t, q, \dot{q}) dt$$

b) Verify that

$$H = \sum_{k=1}^n \dot{q}_k \frac{\partial L}{\partial \dot{q}_k} - L = \text{constant}$$

is satisfied along an extremal.

We have a set of two Euler-Lagrange equations for the system:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_k} - \frac{\partial L}{\partial q_k} = 0, \quad k = 1, 2$$

Hence we get,

$$\begin{aligned} \frac{d}{dt} \dot{q}_1(t) = 0 &\Rightarrow \frac{d^2}{dt^2} q_1(t) = 0 \\ &\Rightarrow q_1(t) = a_1 t + b_1 \end{aligned}$$

and,

$$\begin{aligned} \frac{d}{dt} \dot{q}_2(t) = -g &\Rightarrow \frac{d^2}{dt^2} q_2(t) = -g \\ &\Rightarrow q_2(t) = -\frac{g}{2} t^2 + a_2 t + b_2 \end{aligned}$$

where  $a_1, a_2, b_1, b_2 \in \mathbb{R}$  are arbitrary constants.

b) We have,

$$\begin{aligned} H &= \sum_{k=1}^2 \dot{q}_k \frac{\partial L}{\partial \dot{q}_k} - L = \dot{q}_1 \cdot \dot{q}_1 + \dot{q}_2 \cdot \dot{q}_2 - \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2) \\ &= gq_2 + \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2) \end{aligned}$$

Now along an extremal,

$$\begin{aligned}
\frac{d}{dt}H &= \frac{d}{dt} \left( gq_2 + \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2) \right) \\
&= g\dot{q}_2 + \dot{q}_1 \frac{d^2}{dt^2} q_1(t) + \dot{q}_2 \frac{d^2}{dt^2} q_2(t) \\
&= g\dot{q}_2 + \dot{q}_1 \cdot 0 + \dot{q}_2(-g) \\
&= 0
\end{aligned}$$

Hence,  $H$  is constant along an extremal.  $\square$

**Exercise 7.13:** Prove that

$$H = \sum_{k=1}^n \dot{q}_k \frac{\partial L}{\partial \dot{q}_k} - L$$

is constant along an extremal when  $L$  does not depend on  $t$  explicitly. We wish to show that  $\frac{d}{dt}H = 0$  along an extremal.

Note that,

$$\begin{aligned}
\frac{d}{dt}H &= \frac{d}{dt} \left( \sum_{k=1}^n \dot{q}_k \frac{\partial L}{\partial \dot{q}_k} - L \right) \\
&= \sum_{k=1}^n \left( \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_k} \dot{q}_k + \dot{q}_k \frac{\partial L}{\partial \dot{q}_k} \right) - \sum_{k=1}^n \left( \frac{\partial L}{\partial q_k} \dot{q}_k + \frac{\partial L}{\partial \dot{q}_k} \dot{\dot{q}}_k \right) \\
&= \sum_{k=1}^n \left( \frac{\partial L}{\partial q_k} \dot{q}_k + \dot{q}_k \frac{\partial L}{\partial \dot{q}_k} \right) - \sum_{k=1}^n \left( \frac{\partial L}{\partial q_k} \dot{q}_k + \frac{\partial L}{\partial \dot{q}_k} \dot{\dot{q}}_k \right) - \frac{\partial L}{\partial t} \\
&= -\frac{\partial L}{\partial t} \\
&= 0
\end{aligned}$$

$\square$

**Exercise 7.14:** Let  $q \equiv (q_1, q_2)$  and  $L(t, q, \dot{q}) = \sqrt{\dot{q}_1^2 + q_2^2 \dot{q}_2^2} - kq_2$  where  $k \in \mathbb{R}$  is a constant. Find the extremals for the functional  $J$  defined by

$$J(q) = \int_{t_0}^{t_1} L(t, q, \dot{q}) dt$$

As we see, the integrand is not dependent on  $t$  explicitly. Hence, we can say that  $H = \sum_{k=1}^n \dot{q}_k \frac{\partial L}{\partial \dot{q}_k} - L$  is a constant along extremals.

$$\begin{aligned}
H &= \sum_{k=1}^n \dot{q}_k \frac{\partial L}{\partial \dot{q}_k} - L \\
&= \dot{q}_1 \frac{\partial L}{\partial \dot{q}_1} + \dot{q}_2 \frac{\partial L}{\partial \dot{q}_2} - L \\
&= \frac{1}{2} \frac{2\dot{q}_1 \dot{q}_1}{\sqrt{\dot{q}_1^2 + q_2^2 \dot{q}_2^2}} + \frac{1}{2} \frac{2q_2^2 \dot{q}_2 \dot{q}_2}{\sqrt{\dot{q}_1^2 + q_2^2 \dot{q}_2^2}} - \sqrt{\dot{q}_1^2 + q_2^2 \dot{q}_2^2} + kq_2 \\
&= kq_2 \\
\Rightarrow q_2 &= \frac{H}{k}
\end{aligned}$$

Therefore, the Euler-Lagrange equation now becomes:

$$\begin{aligned}
\frac{d}{dt} \frac{\dot{q}_1}{\sqrt{\dot{q}_1^2 + q_2^2 \dot{q}_2^2}} &= 0 \\
\Rightarrow \text{sgn}(\dot{q}_1) &= \text{constant}
\end{aligned}$$

Thus, the extremising function is  $q_1$  is any monotone  $C^2$  function and  $q_2 = \frac{H}{k}$  is a constant.

**Exercise 7.15:** Let  $\Xi = \Xi(t, q)$  be any smooth function and let

$$\Theta(t, q, \dot{q}) = \frac{\partial \Xi}{\partial t} \sum_{k=1}^n \frac{\partial \Xi}{\partial q_k} \dot{q}_k$$

a) [Degenerate Case]: Prove that Euler-Lagrange for any smooth function  $q$  is satisfied for the functional:

$$A(q) = \int_{t_0}^{t_1} \Theta(t, q, \dot{q}) dt$$

b) Let

$$J(q) = \int_{t_0}^{t_1} L(t, q, \dot{q}) dt$$

and

$$K(q) = \int_{t_0}^{t_1} (L + \Theta)(t, q, \dot{q}) dt$$

Show that  $q$  is an extremal for  $J$  if and only if  $q$  is an extremal for  $K$ .

a) To get the Euler-Lagrange equation for  $A$ ,

$$\frac{\partial \Theta}{\partial q_k} - \frac{\partial^2 \Xi}{\partial t \partial q_k} + \sum_{i=1}^n \frac{\partial^2 \Xi}{\partial q_i \partial q_k} \dot{q}_i$$

and

$$\frac{\partial \Theta}{\partial \dot{q}_k} = \frac{\partial^2 \Xi}{\partial t \partial \dot{q}_k} \overset{0}{\cancel{\dot{q}_k}} + \frac{\partial \Xi}{\partial q_k}$$

Now,

$$\frac{d}{dt} \frac{\partial \Theta}{\partial \dot{q}_k} = \frac{d}{dt} \frac{\partial \Xi}{\partial q_k} = \sum_{i=1}^n \frac{\partial^2 \Xi}{\partial q_i \partial \dot{q}_k} \dot{q}_i + \frac{\partial^2 \Xi}{\partial t \partial \dot{q}_k} = \frac{\partial \Theta}{\partial q_k}$$

Hence, the Euler-Lagrange equation reduces to an identity.

b) We have,

$$K(q) = \int_{t_0}^{t_1} (L + \Theta)(t, q, \dot{q}) dt$$

Now,

$$q \text{ is smooth} \Rightarrow q \text{ is an extremal of } A(q) = \int_{t_0}^{t_1} \Theta(t, q, \dot{q}) dt$$

owing to the Euler-Lagrange equation being an identity. Hence, if  $q$  is an extremal of  $J$  then it is also an extremal of  $J(q) + A(q) = K(q)$ .

This can also be argued in the way that  $\delta A = 0$  identically. Hence, if  $q$  satisfies  $\delta J = 0$  then  $q$  also satisfies

$$\delta J + \delta A = \delta K = 0$$

Therefore,  $q$  is an extremal of  $J$  if and only if it is an extremal of  $K$ .

## 7.6 The Isoperimetric problem.

*Generally, problems involving extremization of some functional does not come cheap. It is generally associated with one or more constraints on the solution. We state as an example and study the isoperimetric problem:*

Let  $J : C^2[x_0, x_1] \rightarrow \mathbb{R}$  be a functional of the form

$$J(y) = \int_{x_0}^{x_1} f(x, y, y') dx$$

where  $f$  is a smooth function of its variables. The isoperimetric problem consists of determining extremals of  $J$  satisfying boundary conditions  $y(x_0) = y_0$  and  $y(x_1) = x_1$  and a constraint<sup>4</sup> of the form:

$$I(y) = \int_{x_0}^{x_1} g(x, y, y') dx = L$$

It is clear that the presence of constraints further limits the set  $S$  in which we search for solutions.

It is in our interest to study a simpler version of an isoperimetric problem before moving to the one stated above.

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<sup>4</sup>We will discuss later what is means to have constraints of other forms.

**Recall:** *The implicit function theorem.*

Let  $g : \Omega \rightarrow \mathbb{R}$  be a function defined on the neighborhood  $\Omega \subseteq \mathbb{R}^2$  of the point  $(x_0, y_0)$  such that

$$g(x_0, y_0) = 0$$

Let  $g$  have continuous partial derivatives with respect to  $y$  on  $\Omega$ . If

$$\left. \frac{\partial g}{\partial y} \right|_{(x_0, y_0)} \neq 0$$

then we can say there exists neighborhoods  $I_{x_0} \subset \mathbb{R}$  of  $x_0$  and  $I_{y_0} \subset \mathbb{R}$  of  $y_0$  and a function  $\varphi : I_{x_0} \rightarrow \mathbb{R}$  such that

- i)  $\varphi(x_0) = y_0$
- ii)  $(x, \varphi(x)) \in \Omega$  and  $g(x, \varphi(x)) = 0 \quad \forall x \in I_{x_0}$
- iii)  $\varphi$  is unique and continuous

Moreover, if  $\frac{\partial g}{\partial x}$  exists and is continuous in  $\Omega$  then  $\varphi$  is differentiable in  $I_{x_0}$  and,

$$\varphi'(x) = -\frac{\frac{\partial g}{\partial x}}{\frac{\partial g}{\partial y}}$$

### 7.6.1 The Lagrange multiplier

Let us suppose now that  $f$  and  $g$  are smooth functions. We aim to find a necessary condition for  $f$  to have a local extremum subject to

$$g(x, y) = 0$$

We also assume that

$$\nabla g(x, y) \neq 0$$

Now by  $\nabla g(x, y) \neq 0$  we can conclude that the curve  $\gamma$  defined by  $g(x, y) = 0$  is a smooth parameterization.

$$r(t) = (x(t), y(t)), t \in I$$

and  $r'(t) \neq 0 \quad \forall t \in I$ .

Hence, a necessary condition for  $f$  to have a local extremum on  $\gamma$  at  $(x(t), y(t))$  is:

$$\frac{d}{dt} f(x(t), y(t)) = \frac{\partial f}{\partial x} x'(t) + \frac{\partial f}{\partial y} y'(t) = 0 \quad (8)$$

Now from  $g(x(t), y(t)) = 0$  we can say,

$$\frac{d}{dt} g(x(t), y(t)) = \frac{\partial g}{\partial x} x'(t) + \frac{\partial g}{\partial y} y'(t) = 0 \quad (9)$$

Now,  $\nabla g(x, y) = 0 \Rightarrow$  at least one of  $\frac{\partial g}{\partial x}$  or  $\frac{\partial g}{\partial y}$  is non-zero. Without loss of generality, assume  $\frac{\partial g}{\partial y} \neq 0$ .

From Eq. 9,

$$y'(t) = -\frac{\frac{\partial g}{\partial x}}{\frac{\partial g}{\partial y}} x'(t)$$

Putting this in Eq. 8 we thus get,

$$\frac{x'(t)}{\frac{\partial g}{\partial y}} \left\{ \frac{\partial f}{\partial x} \frac{\partial g}{\partial y} - \frac{\partial f}{\partial y} \frac{\partial g}{\partial x} \right\} = 0 \quad (10)$$

Now, as we have already assumed  $\frac{\partial g}{\partial y} \neq 0$ , we can describe the curve as  $r(t) = (x(t), \varphi(x(t)))$  where  $\varphi$  is smooth. And,  $x'(t) = 0 \Rightarrow r'(t) = (0, 0)$  would lead to a contradiction. Hence, we conclude  $x'(t) \neq 0$ .

Thus, from Eq. 10 we have,

$$\frac{\partial f}{\partial x} \frac{\partial g}{\partial y} - \frac{\partial f}{\partial y} \frac{\partial g}{\partial x} = 0$$

Therefore,

$$\nabla f \times \nabla g = 0$$

Now, note that  $|\nabla f \times \nabla g| = |\nabla f| \cdot |\nabla g| \cdot \sin \theta$ . So, either  $\theta = 0$  or  $\nabla f = 0$ . In either case we have,

$$\nabla f = \lambda \nabla g$$

i.e.,  $\nabla f$  is parallel to  $\nabla g$ . Thus, Eq. 8 reduces to

$$\nabla(f - \lambda g) = 0$$

Here, the constant  $\lambda$  is called *The Lagrange Multiplier*.

We formalize the Lagrange multiplier rule as:

**Theorem 7.2** (Lagrange Multiplier): Let  $\Omega \subseteq \mathbb{R}^2$  be a region and let  $f : \Omega \rightarrow \mathbb{R}$  and  $g : \Omega \rightarrow \mathbb{R}$  be smooth. If  $f$  has a local extremum at  $x \in \Omega$  subject to  $g(x, y) = 0$  and  $\nabla g(x, y) \neq 0$  then there is a number  $\lambda$  such that

$$\nabla(f(x) - \lambda g(x)) = 0$$

Now, we can even generalize the domain of  $f$  and the number of constraints associated to even higher dimensions. There is nothing special about 2 as a dimension.

Let  $x \equiv (x_1, x_2, \dots, x_n) \in \Omega \subseteq \mathbb{R}^n$  and  $f : \Omega \rightarrow \mathbb{R}$  be smooth. When  $n > 2$  it is possible to impose more than one constraint. Let  $m < n$ . We wish to find the local extremum of  $f$  in  $\Omega$  subject to

$$g_k(x) = 0$$

Where  $k = 1, 2, \dots, m$  and  $g_k : \Omega \rightarrow \mathbb{R}$ 's are smooth for all  $k$ .

We have seen what happens when  $n = 2$  and  $m = 1$ . The tangent lines of  $f$  and  $g$  coincide at extrema. In higher dimensions however, we have the analogue of this condition.

The tangent space (hyperplane) of  $f$  at a critical point is contained in the tangent space defined by  $g_k$  at  $x$ . Geometrically, this also means that the normal vector  $\nabla f(x)$  lies in the normal space  $N_{g(x)}$  spanned by  $\nabla g_k(x)$ .

In other words,  $\nabla f(x)$  is linearly independent on  $\{\nabla g_k(x) : k = 1, 2, \dots, m\}$

$\therefore \exists \lambda_1, \lambda_2, \dots, \lambda_m$  such that,

$$\nabla f(x) = \sum_{k=1}^m \lambda_k \nabla g_k(x)$$

i.e.,

$$\nabla \left( f(x) - \sum_{k=1}^m \lambda_k g_k(x) \right) = 0$$

Now let  $M(x)$  be the  $m \times n$  matrix such that,

$$M(x) := \begin{pmatrix} \nabla g_1(x) \\ \nabla g_2(x) \\ \vdots \\ \nabla g_m(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \cdots & \frac{\partial g_1}{\partial x_n} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \cdots & \frac{\partial g_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_m}{\partial x_1} & \frac{\partial g_m}{\partial x_2} & \cdots & \frac{\partial g_m}{\partial x_n} \end{pmatrix}$$

and  $M_f(x)$  be the  $(m+1) \times n$  augmented matrix

$$M_f(x) := \begin{pmatrix} M(x) \\ \nabla f(x) \end{pmatrix}$$

. As  $\nabla f$  is linearly dependent on  $\nabla g_k$  we must have,

$$\text{rank}(M_f(x)) \leq \text{rank}(M(x))$$

We also state that if  $(x, y)$  is a local extremum but  $\nabla g(x, y) = 0$  then the existence of the Lagrange multiplier is not guaranteed. This kind of a problem is called an *abnormal problem*. Note that  $\nabla g(x, y) = 0$  means that we cannot invoke the implicit function theorem directly to solve for  $x$  in terms of  $y$  and vice versa. Geometrically, it means that the level set  $g = 0$  may not be a smooth curve in the neighborhood of  $(x, y)$ . It does not necessarily mean that the curve has a singularity at  $(x, y)$ , [which would make the tangent undefined]. It can also mean that there might be a “corner” at the curve at  $(x, y)$  or the curve might self-intersect. It can also happen that  $(x, y)$  is an isolated point in the solution of  $g = 0$ .

We now state the theorem that inculcates the abnormal problem as well.

**Theorem 7.3:** Let  $\Omega \subseteq \mathbb{R}^2$  (or  $\mathbb{R}^n$ ) be a region and  $f : \Omega \rightarrow \mathbb{R}$  and  $g : \Omega \rightarrow \mathbb{R}$  be smooth. If  $f$  has a local extremum at  $(x, y) \in \Omega$  subject to  $g(x) = 0$  then  $\exists$  numbers  $\lambda_0$  and  $\lambda_1$  not both zero such that

$$\nabla(\lambda_0 f(x, y) - \lambda_1 g(x, y)) = 0 \tag{11}$$

If  $\nabla g(x, y) \neq 0$ , the problem is normal and we choose  $\lambda_0 = 1$ .

Now, if the problem is abnormal then we can reduce the condition to  $\lambda_0 \nabla f(x, y) = 0$  and if  $\nabla f(x, y) \neq 0$  then  $\lambda_0 = 0$  and  $\lambda_1$  remains undetermined. If  $\nabla f(x, y) = 0$  then any  $\lambda_0, \lambda_1 \in \mathbb{R}$  would satisfy Eq. 11.

We are now ready to move on to the isoperimetric problem again. Recall that  $J : C^2[x_0, x_1] \rightarrow \mathbb{R}$  is a functional such that,

$$J(y) = \int_{x_0}^{x_1} f(x, y, y') dx$$

where  $f$  is smooth and we wish to extremize  $J$  with respect to the constraint

$$I(y) = \int_{x_0}^{x_1} g(x, y, y') dx = L$$

and the boundary conditions  $y(x_0) = y_0$ ;  $y(x_1) = y_1$ .

To tackle this problem we proceed in a similar fashion as before. Suppose that  $J(\hat{y})$  is an extremal. We need to consider and analyse small variations of the functional  $J$ . Now, taking  $y = \hat{y} + \epsilon \eta$  would not always guarantee that  $y$  would satisfy the constraint. Instead, we choose  $y = \hat{y} + \epsilon_1 \eta_1 + \epsilon_2 \eta_2$  such that  $\eta_1$  is arbitrary and  $\epsilon_2 \eta_2$  is chosen in a way that  $y$  satisfies the constraint.

The term  $\epsilon_2 \eta_2$  can be seen as a “correction term”.

We are also faced with another concern in our analysis. Consider the example of

$$I(y) = \int_{x_0}^{x_1} \sqrt{1 + y'^2} dx = \sqrt{2}$$

In this case,  $\sqrt{2}$  is an extremal value and only  $\hat{y}(x) = x$  satisfies this constraint. Therefore  $\hat{y}$  cannot be varied.

We call such  $\hat{y}$  a *rigid extremal*.

We will now formally characterize the isoperimetric problem to study these pathologies clearly.

Let  $J$  have an extremal at  $\hat{y}$  subject to the isoperimetric constraint (and boundary conditions). We consider

$$I(y) = \int_{x_0}^{x_1} g(x, \hat{y} + \epsilon_1 \eta_1 + \epsilon_2 \eta_2, \hat{y}' + \epsilon_1 \eta_1' + \epsilon_2 \eta_2') dx$$

For a fixed  $\{\eta_1, \eta_2\}$ , we can express  $I(y) \equiv A(\epsilon_1, \epsilon_2)$ . Now,  $g$  being smooth makes  $A$  a smooth function of  $\epsilon_1, \epsilon_2$ .

Now, as  $\hat{y}$  satisfies the constraint, we have,

$$A(0, 0) = L$$

Now if  $\nabla A(0, 0) \neq 0$ , we can again invoke the implicit function theorem to say that in a neighborhood of  $(0, 0)$  [i.e., for  $\max\{\epsilon_1, \epsilon_2\}$  sufficiently small] we have (without loss of generality),

$$\epsilon_1 = \epsilon_1(\epsilon_2)$$

such that

$$A(\epsilon_1(\epsilon_2), \epsilon_2) = L$$



But if  $\hat{y}$  is a rigid extremal then  $\nabla A(0,0) = 0$ . For now, we only suppose  $\nabla A(0,0) \neq 0$ .

To extend our analysis we need to transform  $J(y)$  to a function on Euclidean space so that we can use our known tools. Let,

$$B(\epsilon_1, \epsilon_2) \equiv J(y) = \int_{x_0}^{x_1} f(x, \hat{y} + \epsilon_1 \eta_1 + \epsilon_2 \eta_2, \hat{y}' + \epsilon_1 \eta_1' + \epsilon_2 \eta_2') dx$$

Now, we have completely transformed the problem to our known problem of lagrangian multipliers.

Note that  $J$  has an extremum at  $\hat{y}$  subject to  $I(\hat{y}) = L$ . Hence,  $B$  must have an extremum at  $(0,0)$  subject to  $A(\epsilon_1, \epsilon_2) - L = 0$ . Therefore, for any critical point,

$$\nabla(B(\epsilon_1, \epsilon_2) - \lambda(A(\epsilon_1, \epsilon_2) - L)) = 0$$

Now,  $(0,0)$  is a critical point. Therefore,

$$\left. \frac{\partial}{\partial \epsilon_1} (B(\epsilon_1, \epsilon_2) - \lambda(A(\epsilon_1, \epsilon_2))) \right|_{(\epsilon_1, \epsilon_2)=(0,0)} \quad (12)$$

$$\left. \frac{\partial}{\partial \epsilon_2} (B(\epsilon_1, \epsilon_2) - \lambda(A(\epsilon_1, \epsilon_2))) \right|_{(\epsilon_1, \epsilon_2)=(0,0)} \quad (13)$$

Now,

$$\begin{aligned} \left. \frac{\partial}{\partial \epsilon_1} B(\epsilon_1, \epsilon_2) \right|_{(\epsilon_1, \epsilon_2)=(0,0)} &= \left. \frac{\partial}{\partial \epsilon_1} \int_{x_0}^{x_1} f(x, \hat{y} + \epsilon_1 \eta_1 + \epsilon_2 \eta_2, \hat{y}' + \epsilon_1 \eta_1' + \epsilon_2 \eta_2') dx \right|_{(\epsilon_1, \epsilon_2)=(0,0)} \\ &= \left. \int_{x_0}^{x_1} \frac{\partial}{\partial \epsilon_1} f(x, \hat{y} + \epsilon_1 \eta_1 + \epsilon_2 \eta_2, \hat{y}' + \epsilon_1 \eta_1' + \epsilon_2 \eta_2') dx \right|_{(\epsilon_1, \epsilon_2)=(0,0)} \\ &= \int_{x_0}^{x_1} \left( \eta_1 \frac{\partial f}{\partial \hat{y}} + \eta_1' \frac{\partial f}{\partial \hat{y}'} \right) dx \end{aligned}$$

Integrating by parts,

$$\left. \frac{\partial}{\partial \epsilon_1} B(\epsilon_1, \epsilon_2) \right|_{(\epsilon_1, \epsilon_2)=(0,0)} = \int_{x_0}^{x_1} \eta_1 \left( \frac{\partial f}{\partial \hat{y}} - \frac{d}{dx} \frac{\partial f}{\partial \hat{y}'} \right) dx$$

Similarly,

$$\left. \frac{\partial}{\partial \epsilon_1} A(\epsilon_1, \epsilon_2) \right|_{(\epsilon_1, \epsilon_2)=(0,0)} = \int_{x_0}^{x_1} \eta_1 \left( \frac{\partial g}{\partial \hat{y}} - \frac{d}{dx} \frac{\partial g}{\partial \hat{y}'} \right) dx$$

Hence, we write Eq. 12 as:

$$\int_{x_0}^{x_1} \eta_1 \left\{ \frac{\partial f}{\partial \hat{y}} - \frac{d}{dx} \frac{\partial f}{\partial \hat{y}'} - \lambda \left( \frac{\partial g}{\partial \hat{y}} - \frac{d}{dx} \frac{\partial g}{\partial \hat{y}'} \right) \right\} dx = 0$$

Consider,  $f - \lambda g \equiv F$ . Now,

$$\frac{d}{dx} \frac{\partial F}{\partial \hat{y}'} - \frac{\partial F}{\partial \hat{y}} = 0 \quad (14)$$

Note that Eq. 14 can only be written because  $\eta_1$  is the completely arbitrary function from  $\mathcal{H}$ . Thus, Eq. 14 is our required Euler-Lagrange equation for this problem. However, the interesting point here is that from Eq. 13 we can derive by a similar argument that

$$\int_{x_0}^{x_1} \eta_2 \left( \frac{d}{dx} \frac{\partial F}{\partial \hat{y}'} - \frac{\partial F}{\partial \hat{y}} \right) dx = 0 \quad (15)$$

But if Eq. 14 is satisfied then Eq. 15 will also be satisfied.

*What happens when  $\hat{y}$  is a rigid extremal?*

When  $\hat{y}$  is a rigid extremal, we have  $\nabla A(0, 0) = 0$ . Which means,

$$\left. \frac{\partial}{\partial \epsilon_1} A(\epsilon_1, \epsilon_2) \right|_{(\epsilon_1, \epsilon_2) = (0, 0)} = \int_{x_0}^{x_1} \eta_1 \left( \frac{\partial g}{\partial \hat{y}} - \frac{d}{dx} \frac{\partial g}{\partial \hat{y}'} \right) dx = 0 \quad (16)$$

and,

$$\left. \frac{\partial}{\partial \epsilon_2} A(\epsilon_1, \epsilon_2) \right|_{(\epsilon_1, \epsilon_2) = (0, 0)} = \int_{x_0}^{x_1} \eta_2 \left( \frac{\partial g}{\partial \hat{y}} - \frac{d}{dx} \frac{\partial g}{\partial \hat{y}'} \right) dx = 0 \quad (17)$$

Now, Eq. 16 is valid for any arbitrary  $\eta_1$ . Hence,

$$\frac{\partial g}{\partial \hat{y}} - \frac{d}{dx} \frac{\partial g}{\partial \hat{y}'} = 0$$

*Therefore, it becomes clear that rigid extremals for isoperimetric problems are characterized by functions that are extremals for the functional that defines the isoperimetric condition.*

We can now summarize our analysis by the theorem:

**Theorem 7.4** (Isoperimetric Problem): Let  $J : C^2[x_0, x_1] \rightarrow \mathbb{R}$  defined by

$$J(y) = \int_{x_0}^{x_1} f(x, y, y') dx$$

have an extremal at  $\hat{y}$  subject to  $\hat{y}(x_0) = y_0$  and  $\hat{y}(x_1) = y_1$  and

$$I(y) = \int_{x_0}^{x_1} g(x, y, y') dx = L.$$

Then,  $\exists$  two numbers  $\lambda_0, \lambda_1 \in \mathbb{R}$  not both zero such that

$$\frac{d}{dx} \frac{\partial K}{\partial \hat{y}'} - \frac{\partial K}{\partial \hat{y}} = 0$$

where  $K \equiv \lambda_0 f - \lambda_1 g$ . If  $\hat{y}$  is not a rigid extremal [not an extremal for  $I$ ], we can take  $\lambda_0 = 1$ . If  $\hat{y}$  is rigid we take  $\lambda_0 = 0$  unless  $\hat{y}$  is also an extremal for  $J$  in which case  $\lambda_0$  and  $\lambda_1$  remains undetermined.

**Exercise 7.16:** For the functional  $J$  defined by

$$J(y) = \int_0^1 y' \sqrt{1 + y''^2} dx,$$

find an extremal satisfying the conditions  $y(0) = 0, y'(0) = 0, y(1) = 1$  and  $y'(1) = 2$ .

Proceeding in the ways we did for the previous exercises will lead to complicated and absurd ODE's for us to solve. Instead a first approach that can be taken to solve this problem is substitution.

Taking  $z = y'$ , we obtain the Euler-Lagrange equation:

$$cz'^2 - z^2 + 1 = 0 \text{ [where } c \text{ is an arbitrary constant]}$$

Now, with the boundary conditions  $z(0) = 0, z(1) = 2$  we can see with some calculation that this has no solutions.

Now, we must notice that we can impose an isoperimetric condition from the boundary conditions given. i.e.,

$$\int_0^1 z dx = 1$$

Therefore, we now have to extremize the functional:

$$\hat{J}(z) = \int_0^1 (z\sqrt{1 + z'^2} - \lambda z) dx$$

There is no explicit  $x$  -dependence for the above functional. Hence, we have

$$\begin{aligned} H &= \frac{zz'^2}{\sqrt{1 + z'^2}} - z\sqrt{1 + z'^2} + \lambda z \\ &= \lambda z - \frac{z}{\sqrt{1 + z'^2}} = \text{const.} = c \text{ (say)} \end{aligned}$$

Now, for  $x = 0$ , we must have  $c = 0$ . Therefore,

$$\begin{aligned} z' &= \sqrt{\frac{1}{\lambda^2} - 1} = \text{const.} \\ \Rightarrow z(x) &= c_1 x + c_2 \end{aligned}$$

Applying boundary conditions we get,

$$c_1 = 2 \text{ and } c_2 = 0$$

Hence,

$$z(x) = 2x$$

Thus,

$$y(x) = \int_0^x z(t) dt = x^2$$

Our extremized solution is thus,  $y(x) = x^2$ .

## 7.7 Calculus of variations on higher dimensions

So far, we have been discussing calculus of variations on single variables, i.e., only one independent variable was present in the parameters of smooth function ' $f$ '. Now, we wish to study the case when more than one independent variables are present. In this study, whatever we had learnt about multivariable calculus will come in handy.

Let  $\Omega$  be a bounded open set in  $\mathbb{R}^n$ . Consider  $u : \Omega \rightarrow \mathbb{R}$ . Define:

$$J(u) = \int_{\Omega} f(x, u(x), \nabla u(x)) dx$$

where  $f : \bar{\Omega} \times \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}$  is a smooth function of its parameters. We are interesting to calculate (if exists)

$$\min_{u \in \mathcal{A}_g} J(u).$$

where

$$\mathcal{A}_g := \left\{ u : \bar{\Omega} \rightarrow \mathbb{R} : u \in C^1(\Omega) \text{ and } u(x) = g(x), \forall x \in \partial\Omega \right\}$$

Suppose,

$$\min_{u \in \mathcal{A}_g} J(u) = J(u_0), u_0 \in \mathcal{A}_g$$

**Definition 7.1** (Compact Support): Let  $f : X \rightarrow \mathbb{R}$  be defined on a Banach space (or a topological space)  $X$ . The support of  $f$  denoted by  $\mathcal{S}(f)$  is defined as

$$\mathcal{S}(f) := \overline{\{x \in X : f(x) \neq 0\}}$$

A function is said to have compact support if  $\mathcal{S}(f)$  is compact. Moreover, if  $f$  is smooth and is compactly supported we say,

$$f \in C_c^\infty(X)$$

Let  $\eta \in C_c^\infty(\Omega)$ . Then, for any  $\epsilon \in \mathbb{R}$ ,  $u_0 + \epsilon\eta \in \mathcal{A}_g$  and  $\epsilon = 0$  is a minimum of  $h(\epsilon) := J(u_0 + \epsilon\eta)$ .

$$\begin{aligned} &\Rightarrow h'(0) = 0 \\ &\Rightarrow \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} J(u_0 + \epsilon\eta) = 0 \\ &\Rightarrow \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \int_{\Omega} f(x, u_0 + \epsilon\eta, \nabla u_0 + \epsilon\nabla\eta) dx = 0 \\ &\Rightarrow \int_{\Omega} \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} f(x, u_0 + \epsilon\eta, \nabla u_0 + \epsilon\nabla\eta) dx = 0 \end{aligned}$$

Let us name the variables in the parameter of  $f$  as  $f(x, y, z)$ . We then have,

$$\begin{aligned}
& \int_{\Omega} \left( \cancel{\frac{\partial f}{\partial x} \frac{dx}{d\epsilon}} + \frac{\partial f}{\partial y} (x, u_0 + \epsilon \eta, \nabla u_0 + \epsilon \nabla \eta) \frac{dy}{d\epsilon} + \nabla_z f(x, u_0 + \epsilon \eta, \nabla u_0 + \epsilon \nabla \eta) \frac{\partial z}{\partial \epsilon} \right) \Big|_{\epsilon=0} = 0 \\
& \Rightarrow \int_{\Omega} \left[ \frac{\partial f}{\partial y} (x, u_0, \nabla u_0) \cdot \eta + \nabla_z f(x, u_0, \nabla u_0) \cdot \nabla \eta \right] dx = 0
\end{aligned}$$

Note that  $\eta$  is not supported on the boundary of  $\Omega$ . Therefore, with integration by parts of the second term in the above expression the boundary terms cancel over  $\Omega$ . We have,

$$\int_{\Omega} \left[ \frac{\partial f}{\partial y} (x, u_0, \nabla u_0) \cdot \eta - \sum_{i=1}^n \frac{\partial}{\partial x_i} \left( \frac{\partial f}{\partial z_i} (x, u_0, \nabla u_0) \right) \cdot \eta \right] = 0$$

which is true for all  $\eta \in C_c^\infty(\Omega)$  and we conclude,

$$-div(\nabla_z f(x, u_0(x), \nabla u_0(x))) + \frac{\partial f}{\partial y}(x, u_0, \nabla u_0) = 0 \quad (18)$$

This is the Euler-Lagrange equation for functionals defined on more than one independent variables.

### 7.7.1 Dirichlet principle

Consider the equation:

$$\begin{aligned}
-\nabla^2 u(x) &= f(x), x \in \Omega \\
u(x) &= g(x), x \in \partial\Omega
\end{aligned} \quad (19)$$

and the functional:

$$J(u) = \frac{1}{2} \int_{\Omega} |\nabla u(x)|^2 dx - \int_{\Omega} f(x)u(x) dx$$

and

$$\mathcal{A}_g := \left\{ u : \bar{\Omega} \rightarrow \mathbb{R} : u \in C^2(\bar{\Omega}) \text{ and } u(x) = g(x), x \in \partial\Omega \right\}$$

Then we have the following theorem:

**Theorem 7.5** (Dirichlet):  $u \in \mathcal{A}_g$  solves Eq. 19 if and only if  $u$  minimizes  $J$  in  $\mathcal{A}_g$ .

*Proof.* Let  $u \in \mathcal{A}_g$  be a minimizer. We have,

$$\begin{aligned}
& \frac{d}{d\epsilon} \Big|_{\epsilon=0} \left[ \frac{1}{2} \int_{\Omega} |\nabla u(x) + \epsilon \nabla \eta|^2(x) dx - \int_{\Omega} f \cdot (u + \epsilon \eta)(x) dx \right] = 0 \\
& \Rightarrow \int_{\Omega} \nabla u(x) \cdot \nabla \eta(x) dx - \int_{\Omega} (f\eta)(x) dx = 0 \\
& \Rightarrow - \int_{\Omega} ((\nabla^2 u + f)\eta)(x) dx = 0 \\
& \Rightarrow (\nabla^2 u + f)(x) = 0 \\
& \Rightarrow -\nabla^2 u(x) = f(x)
\end{aligned}$$

To prove the converse, we assume that  $u$  is a solution of Eq. 19 and we show that  $J(u) \leq J(v)$ ,  $\forall v \in \mathcal{A}_g$ . We have,

$$-\int_{\Omega} \nabla^2 u(x) = f(x), x \in \Omega$$

Let  $v \in \mathcal{A}_g$  be arbitrary. Then,  $v - u \in C^2(\Omega)$  and  $v - u = 0$  on  $\partial\Omega$ . Now,

$$-\int_{\Omega} \nabla^2 u(u - v)(x) dx = \int_{\Omega} f(u - v)(x) dx$$

Applying integration by parts, the boundary terms cancel for  $(u - v)$  and we get,

$$\begin{aligned} \int_{\Omega} (\nabla u(\nabla u - \nabla v))(x) dx &= \int_{\Omega} (fu)(x) dx - \int_{\Omega} (fv)(x) dx \\ \Rightarrow \int_{\Omega} |\nabla u|^2(x) dx - \int_{\Omega} (\nabla u \nabla v)(x) dx &= \int_{\Omega} (fu)(x) dx - \int_{\Omega} (fv)(x) dx \\ \Rightarrow \int_{\Omega} |\nabla u|^2(x) dx - \int_{\Omega} (fu)(x) dx &= \int_{\Omega} (\nabla u \nabla v)(x) dx - \int_{\Omega} (fv)(x) dx \\ &\leq \frac{1}{2} \int_{\Omega} (|\nabla u|^2(x) + |\nabla v|^2(x)) dx - \int_{\Omega} (fv)(x) dx \\ &\quad [\text{By Cauchy-Schwartz inequality}] \\ \Rightarrow \frac{1}{2} \int_{\Omega} |\nabla u|^2(x) dx - \int_{\Omega} (fu)(x) dx &\leq \frac{1}{2} \int_{\Omega} |\nabla v|^2(x) dx - \int_{\Omega} (fv)(x) dx \\ &\Rightarrow J(u) \leq J(v) \end{aligned}$$

And we are done. □

**Example 7.2:** Let

$$\begin{aligned} -\nabla^2 u(x) &= (f \circ u)(x), x \in \Omega \\ u(x) &= 0, x \in \partial\Omega \end{aligned} \tag{20}$$

$f : \mathbb{R} \rightarrow \mathbb{R}$  be a smooth function. Define  $\mathcal{A} := \{u \in C^2(\bar{\Omega}) : u(x) = 0 \text{ for } x \in \partial\Omega\}$ . Also define,

$$F(t) = \int_0^t f(s) dx$$

and

$$J(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2(x) dx - \int_{\Omega} (F \circ u)(x) dx$$

Show that the Euler-Lagrange equation for  $J$  is Eq. 20.

*Solution:* We wish to calculate the first variation in  $J$  and which should vanish along an extremal of  $J$ . For some  $t \in \mathbb{R}$  and  $\eta \in C_c^\infty(\bar{\Omega})$ ,

$$\therefore \left. \frac{d}{dt} \right|_{t=0} J(u + t\eta) = 0$$

Then,

$$\left. \frac{d}{dt} \right|_{t=0} \left[ \frac{1}{2} \int_{\Omega} |\nabla(u + t\eta)|^2(x) dx - \int_{\Omega} (F \circ (u + t\eta))(x) dx \right] = 0$$

Now,

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=0} \frac{1}{2} \int_{\Omega} |\nabla(u + t\eta)|^2(x) dx &= \frac{1}{2} \int_{\Omega} \left. \frac{d}{dt} \right|_{t=0} |\nabla(u + t\eta)|^2(x) dx \\ &= \int_{\Omega} (\nabla u \nabla \eta)(x) dx \end{aligned}$$

and

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=0} \int_{\Omega} (F \circ (u + t\eta))(x) dx &= \int_{\Omega} \left. \frac{d}{dt} \right|_{t=0} (F \circ (u + t\eta))(x) dx \\ &= \int_{\Omega} \frac{\partial F}{\partial (u + t\eta)}(u + t\eta) \Big|_{t=0} \cdot \frac{d(u + t\eta)}{dt} dx \\ &= \int_{\Omega} (f \circ u)(x) \eta(x) dx \\ \therefore \int_{\Omega} (\nabla u \nabla \eta)(x) dx - \int_{\Omega} (f \circ u)(x) \eta(x) dx &= 0 \end{aligned}$$

Again, integrating by parts we have,

$$\begin{aligned} & - \int_{\Omega} ((\nabla^2 u) \eta)(x) dx - \int_{\Omega} (f \circ u)(x) \eta(x) dx = 0 \\ \Rightarrow & - \int_{\Omega} (\nabla^2 u(x) + (f \circ u)(x)) \eta(x) dx = 0 \\ \Rightarrow & \nabla^2 u(x) + (f \circ u)(x) = 0 \\ \Rightarrow & -\nabla^2 u(x) = (f \circ u)(x) \end{aligned}$$

Thus we have Eq. 20 as the Euler-Lagrange equation.

**Example 7.3:** Let

$$J(u) = \frac{1}{p} \int_{\Omega} |\nabla u(x)|^p dx - \int_{\Omega} (F \circ u)(x) dx, \quad 1 < p < \infty \quad (21)$$

We wish to find the Euler-Lagrange equation of the functional in Eq. 21. Setting

$$\left. \frac{d}{dt} \right|_{t=0} J(u + t\eta) = 0$$

We note that,

$$\begin{aligned}
\left. \frac{d}{dt} \right|_{t=0} \frac{1}{p} \int_{\Omega} |\nabla u(x) + t \nabla \eta(x)|^p dx &= \frac{1}{p} \int_{\Omega} \left. \frac{d}{dt} \right|_{t=0} |\nabla u(x) + t \nabla \eta(x)|^p dx \\
&= \frac{1}{p} \int_{\Omega} \binom{p}{1} |\nabla u(x)|^{p-2} \nabla u(x) \cdot \nabla \eta(x) dx \\
&= \int_{\Omega} |\nabla u(x)|^{p-2} \nabla u(x) \cdot \nabla \eta(x) dx
\end{aligned}$$

Thus we get in a similar way as the previous example,

$$\begin{aligned}
&\int_{\Omega} |\nabla u(x)|^{p-2} \nabla u(x) \cdot \nabla \eta(x) dx - \int_{\Omega} (f \circ u)(x) \eta(x) dx = 0 \\
\Rightarrow \int_{\Omega} -\operatorname{div}(|\nabla u(x)|^{p-2} \nabla u(x)) \cdot \eta(x) dx &= \int_{\Omega} (f \circ u)(x) \eta(x) dx = 0 \\
\Rightarrow -\operatorname{div}(|\nabla u(x)|^{p-2} \nabla u(x)) &= (f \circ u)(x)
\end{aligned}$$

And, this is our required Euler-Lagrange equation.  $\square$

### 7.7.2 Proof of Dirichlet principle using Fréchet differentiation

We wish to establish the same result from earlier using the theories of Fréchet derivatives we have learnt in Section 3.2. We are working with

$$\begin{aligned}
-\nabla^2 u(x) &= (f \circ u)(x), x \in \Omega \\
u(x) &= 0, x \in \partial\Omega
\end{aligned}$$

and

$$J(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2(x) dx - \int_{\Omega} (F \circ u)(x) dx$$

where

$$F(t) = \int_0^t f(s) ds$$

Now, let

$$X = \{u \in C^2(\bar{\Omega}) : u(x) = 0 \ \forall x \in \partial\Omega\}$$

Now,  $X$  is a Banach space with the norm

$$\|u\|_X := \sup_{x \in \Omega} \{|u(x)|, |u_i(x)|, |u_{i,j}(x)|\}_{i,j=1,2,\dots,n}$$

where  $u_i(x)$  refers to the partial derivative of  $u$  in the  $i^{\text{th}}$  coordinate. Now, we wish to find the Fréchet derivative of  $J$  at  $u$ . We have to calculate  $L_u \equiv DJ(u) : X \rightarrow \mathbb{R}$  such that,

$$\frac{|J(u + \eta) - J(u) - L_u(\eta(x))|}{\|\eta\|_X} \rightarrow 0 \text{ as } \eta \rightarrow 0$$

$$J(u + \eta) = \frac{1}{2} \int_{\Omega} |\nabla u(x) + \nabla \eta(x)|^2 dx - \int_{\Omega} (F \circ (u + \eta))(x) dx$$



Now, there exists a function  $\theta : \bar{\Omega} \rightarrow (0, 1)$  such that

$$\begin{aligned} J(u + \eta) &= \frac{1}{2} \left[ \int_{\Omega} |\nabla u|^2(x) dx + \int_{\Omega} |\nabla \eta|^2(x) dx + 2 \int_{\Omega} (\nabla u \cdot \nabla \eta)(x) dx \right] \\ &\quad - \int_{\Omega} \left[ (F \circ u)(x) dx + (F' \circ u)(x) \eta(x) + \frac{1}{2} ((F'' \circ (u + \theta \eta)) \eta^2)(x) \right] dx \\ &\quad \text{[By Taylor expansion]} \end{aligned}$$

Therefore,

$$\begin{aligned} J(u + \eta) &= \frac{1}{2} \int_{\Omega} |\nabla u|^2(x) dx - \int_{\Omega} (F \circ u)(x) dx + \int_{\Omega} (\nabla u \cdot \nabla \eta)(x) dx - \int_{\Omega} (f \circ u)(x) dx \\ &\quad + \int_{\Omega} |\nabla \eta|^2(x) dx + \int_{\Omega} ((f' \circ (u + \theta \eta)) \eta^2)(x) dx \\ &= J(u) + \int_{\Omega} (\nabla u \cdot \nabla \eta)(x) dx - \int_{\Omega} (f \circ u)(x) dx \\ &\quad + \int_{\Omega} |\nabla \eta|^2(x) dx + \int_{\Omega} ((f' \circ (u + \theta \eta)) \eta^2)(x) dx \end{aligned}$$

Choose,  $L_u(\eta) = \int_{\Omega} (\nabla u \cdot \nabla \eta)(x) dx - \int_{\Omega} (f \circ u)(x) dx$ . Note that,  $L_u$  is clearly a linear operator. We get,

$$|J(u + \eta) - J(u) - L_u(\eta)| = \left| \int_{\Omega} |\nabla \eta|^2(x) dx \right| + \left| \int_{\Omega} (f' \circ (u + \theta \eta)) \eta^2(x) dx \right| \quad (22)$$

We wish to analyse the *R.H.S* of Eq. 22. Observe that,

$$\begin{aligned} |\nabla \eta|^2(x) &= \sum_{i=1}^n |\eta_i(x)|^2 \\ &\leq n \|\eta\|_X^2 \end{aligned}$$

and

$$\left| \int_{\Omega} (f' \circ (u + \theta \eta)) \eta^2(x) dx \right| \leq \int_{\Omega} |(f' \circ (u + \theta \eta)) \eta^2(x) dx|$$

Note that,  $u, \eta$  and  $\theta$  are bounded functions. This is because  $u$  is continuous and defined on a compact set while  $\eta$  is compactly supported. This makes  $f' \circ (u + \theta \eta)$  bounded as  $f$  is smooth.

$$\begin{aligned} \therefore \left| \int_{\Omega} (f' \circ (u + \theta \eta)) \eta^2(x) dx \right| &\leq \int_{\Omega} |(f' \circ (u + \theta \eta)) \eta^2(x) dx| \\ &\leq C |\eta(x)|^2 \text{ [for some } C \in \mathbb{R}] \\ &\leq C \|\eta\|_X^2 \end{aligned}$$

Therefore we can conclude

$$\begin{aligned} |J(u + \eta) - J(u) - L_u(\eta)| &= \mathcal{O}(\|\eta\|_X^2) \\ \Rightarrow \frac{|J(u + \eta) - J(u) - L_u(\eta)|}{\|\eta\|_X} &= \mathcal{O}(\|\eta\|_X) \rightarrow 0 \text{ as } \eta \rightarrow 0 \end{aligned}$$

Thus,  $L_u$  is the Fréchet derivative of  $J$  at  $u$  - i.e.,

$$DJ(u)(\eta) = \int_{\Omega} (\nabla u \cdot \nabla \eta)(x) dx - \int_{\Omega} (f \circ u)(x) dx$$

Now, we are set to prove Dirichlet principle. First let  $u$  be a solution to Eq. 20. Then,

$$\begin{aligned} -\nabla^2 u(x) &= (f \circ u)(x) \\ u(x) &= 0 \text{ on } \partial\Omega \end{aligned}$$

$$\Rightarrow - \int_{\Omega} \nabla^2 u(x) \eta(x) dx = \int_{\Omega} (f \circ u)(x) \eta(x) dx$$

Now,  $\eta \in C_c^\infty(\Omega)$ ,

$$\Rightarrow \int_{\Omega} \nabla u(x) \nabla \eta(x) dx = \int_{\Omega} (f \circ u)(x) \eta(x) dx$$

To prove the converse, suppose  $u$  is a critical point of  $J$ . Then,  $u \in \mathcal{A}$ ,  $DJ(u)(\eta) = 0 \ \forall \eta \in C_c^\infty(\Omega)$ .

$$\begin{aligned} \Rightarrow \int_{\Omega} \nabla u(x) \nabla \eta(x) dx - \int_{\Omega} (f \circ u)(x) \eta(x) dx &= 0 \\ \Rightarrow - \int_{\Omega} (\nabla^2 u(x)) \eta(x) dx - \int_{\Omega} (f \circ u)(x) \eta(x) dx &= 0 \\ \Rightarrow \int_{\Omega} (\nabla^2 u(x) + (f \circ u)(x)) \eta(x) dx &= 0 \\ \Rightarrow \nabla^2 u(x) + (f \circ u)(x) &= 0 \\ \Rightarrow -\nabla^2 u(x) &= (f \circ u)(x) \end{aligned}$$

$\therefore u$  is a solution to Eq. 20. And we are done.  $\square$

*Note:* At the step involving Taylor expansion we had to control the higher order terms of  $\eta$  to ensure they converge. It was done just by using the fact that there exists an approximation in the order of  $\eta^2$  that satisfies the equation. A single term in  $\eta^2$  can be ‘controlled’ easily since we have already made the necessary assumptions of boundedness of  $u$  and  $\eta$  and smoothness of  $f'$ . This would not have been possible if instead we had expanded  $(F \circ (u + \eta))(x)$  by an infinite sum.

We conclude our analysis by the following theorem now:

**Theorem 7.6:**  $u_0$  is a solution to

$$\begin{aligned} -\nabla^2 u(x) &= (f \circ u)(x), x \in \Omega \\ u(x) &= 0, x \in \partial\Omega \end{aligned}$$

if and only if  $u_0$  is a critical point of

$$\begin{aligned} J : X &\rightarrow \mathbb{R} \\ u &\mapsto \frac{1}{2} \int_{\Omega} |\nabla u|^2(x) dx - \int_{\Omega} (F \circ u)(x) dx \end{aligned}$$

Hence, our initial problem of finding solutions to Eq. 20 essentially boils down to finding critical points of  $J$ .

The new question is how we can check whether a critical point of this kind exists. We will only state as an example a result that talks about the existence of such a critical point for  $J$  in a specific setting.

**Example 7.4** (Palais-Smale condition): Let  $X$  be a Banach space and  $J : X \rightarrow \mathbb{R}$  be a functional.  $J \in C^1(X)$  and  $J(0) = 0, J(x_0) \leq 0$  for some  $x_0 \neq 0$ . Further assume  $\exists R > 0$ , s.t.  $0 < R < \|x_0\|_X$  and  $\alpha > 0$ , s.t.,

$$\inf_{\|x\|_X = R} J(x) \geq \alpha.$$

Let

$$\mathcal{A} = \{\gamma : [0, 1] \rightarrow X \mid \gamma(0) = 0 \text{ and } \gamma(1) = x_0\}$$

and

$$\inf_{\gamma \in \mathcal{A}} \sup_{t \in [0, 1]} J(\gamma(t)) = \beta$$

We say that  $J$  satisfies the ‘Palais-Smale’ condition at level  $\beta$  if for every sequence  $\{x_n\} \subseteq X$  satisfying  $J(x_n) \rightarrow \beta$  and  $J'(x_n) \rightarrow 0 \exists$  a convergent subsequence  $\{x_{n_k}\} \subseteq X$ .

Now, if  $J$  satisfies the Palais-Smale condition at level  $\beta$  then  $\beta$  is a critical value, i.e.,  $\exists x \in X$  such that  $J(x) = \beta$  and  $DJ(x) = 0$ .

## 7.8 The Hamiltonian

In the previous sections we have seen how Newtonian mechanics can be replaced by an alternative approach to studying nature: Lagrangian mechanics via calculus of variations. Now, we have also seen how then the Lagrangian has no explicit  $x$ —dependence we have a constant quantity:  $H$ . Now, this quantity carries some deeper significance and that is precisely what we will be studying.

### 7.8.1 Total energy of a system

We will be showing that for a classical system with the Lagrangian  $L = T - V$ , the quantity  $H$  (which we will call the Hamiltonian) is nothing but the total energy of the system:  $H = T + V$ . To establish the same we wish to talk a little bit about Euler’s formula

A multivariable function  $f$  is called a homogenous function of degree  $k$  if:

$$f(\lambda q_1, \lambda q_2, \dots, \lambda q_n) = \lambda^k f(q_1, q_2, \dots, q_n) \quad (23)$$

Take an example of a function  $g(x, y) = \frac{3x^2y + y^3}{xy}$  and note that  $g(\lambda x, \lambda y) = \lambda g(x, y)$ . Therefore,  $g$  is a homogenous function of first degree.

Now, differentiating Eq. 23 w.r.t.  $\lambda$  at  $(\lambda q_1, \lambda q_2, \dots, \lambda q_n)$  we obtain,

$$\begin{aligned} \left. \frac{d}{d\lambda} f(\lambda q_1, \lambda q_2, \dots, \lambda q_n) \right|_{(\lambda q_1, \lambda q_2, \dots, \lambda q_n)} &= \frac{d}{d\lambda} \lambda^k f(q_1, q_2, \dots, q_n) \\ \Rightarrow \frac{\partial f}{\partial(\lambda x_1)} \frac{d(\lambda x_1)}{d\lambda} + \frac{\partial f}{\partial(\lambda x_2)} \frac{d(\lambda x_2)}{d\lambda} + \dots + \frac{\partial f}{\partial(\lambda x_n)} \frac{d(\lambda x_n)}{d\lambda} \Big|_{(\lambda q_1, \lambda q_2, \dots, \lambda q_n)} &= k \lambda^{k-1} f(q_1, q_2, \dots, q_n) \end{aligned}$$

Now, putting  $\lambda = 1$  in the above equation we have,

$$x_1 \frac{\partial f}{\partial x_1} + x_2 \frac{\partial f}{\partial x_2} + \dots + x_n \frac{\partial f}{\partial x_n} = kf$$

This relation will be useful for us to prove our statement.

Note that,  $T = \frac{1}{2}m(\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_3^2)$  is a homogenous function of degree 2 in the generalised velocities. Also,  $V$  is independent of generalised velocities. Therefore, we must have,  $\frac{\partial L}{\partial \dot{q}_k} = \frac{\partial T}{\partial \dot{q}_k}$ .

Now,

$$\begin{aligned} H &= \sum_{k=1}^n p_k \dot{q}_k - L \\ &= \sum_{k=1}^n \frac{\partial L}{\partial \dot{q}_k} \dot{q}_k - L \\ &= \sum_{k=1}^n \frac{\partial T}{\partial \dot{q}_k} \dot{q}_k - T + V \\ &= 2T - T + V \\ &= T + V \end{aligned}$$

Thus, the Hamiltonian just gives us the total energy of the system when the Lagrangian is  $T - V$ .

This is significant because the fact the Lagrangian explicitly independent of  $t$  implies that  $H$  is conserved can be interpreted as “Time translation invariance leads to conservation of energy.”

We will try to generalize this statement even further later in this notes.

### 7.8.2 Legendre transformation

We have often encountered transformations from one coordinate system to another. For example, transformation from cartesian coordinate system to polar coordinate system is of the form:

$$\begin{aligned} r &= \sqrt{x^2 + y^2}, \\ \theta &= \tan^{-1}\left(\frac{y}{x}\right) \end{aligned}$$

Now, this is an example of a *point transformation*. However, we wish to look at a different kind of transformations called *contact transformations*. These type of transformations involve derivatives of the dependent variable with the independent variable. One of the most ‘famous’ contact transformation is the **Legendre transformation**. This is the transformation that essentially takes us from the Euler-Lagrange world to the Hamilton-Jacobi world.

The transformation is defined for a smooth function  $y : [x_0, x_1] \rightarrow \mathbb{R}$  by:

$$\begin{aligned} p &= y'(x), \\ H(p) &= px - y(x) \end{aligned}$$

However, to ensure that  $(x, y(x))$  can be defined implicitly by  $(p, H(p))$  we need to make one more assumption:  $y''(x) \neq 0$ .

This transformation itself has a remarkable property of involution. Note that,

$$\begin{aligned}
H'(p) &= \frac{dH}{dp} = \frac{d}{dp}(px) - \frac{d}{dp}y(x) \\
&= +p \frac{dx}{dp} + x - \frac{dy}{dx} \frac{dx}{dp} \\
&= x + (p - y'(x)) \frac{dx}{dp} \\
&= x
\end{aligned}$$

Also note that,

$$\begin{aligned}
H'(p)p - H(p) &= xp - (y(x) - px) \\
&= y(x)
\end{aligned}$$

Thus, we have shown that applying Legendre transformation to  $(p, H(p))$  just gets us back to  $(x, y(x))$ . The transformation is its own inverse and hence is an involution.

We are however more interesting in functions of the form  $f(x, y, y')$ . In this case, we regard  $x, y(x)$  as *passive variables* and  $y'$  as an *active variable*. This is because  $y'$  participates directly in the transformation. We define,

$$\begin{aligned}
p &= \frac{\partial f}{\partial y'}, \\
H(x, y, p) &= py' - f(x, y, y')
\end{aligned}$$

We also make the same assumption from before:  $\frac{\partial^2 f}{\partial y'^2} \neq 0$ . Using the same arguments as before, it can be shown that this is also an involution.

For Lagrangian of several variables  $L \equiv L(t, \mathbf{q}, \dot{\mathbf{q}})$ . We define the Legendre transformation as:

$$\begin{aligned}
p_k &= \frac{\partial L}{\partial \dot{q}_k}, \\
H(t, \mathbf{q}, \mathbf{p}) &= \sum_{k=1}^n p_k \dot{q}_k - L(t, \mathbf{q}, \dot{\mathbf{q}})
\end{aligned}$$

The function  $H$  above is called the **HAMILTONIAN** for the Lagrangian  $L$ . The set of variables  $\mathbf{p}$  is called the **generalised momenta** while the coordinate system  $(t, \mathbf{q}, \mathbf{p})$  is called the generalised coordinates. The set of all points defined by  $(\mathbf{q}, \mathbf{p})$  is called the **phase space** of the system.

We can again invoke the implicit function theorem to say that  $\dot{q}_k$  variables can be solved from the phase space provided that the Jacobian (also called the determinant of the Hessian matrix) is non-zero, i.e.,

$$\frac{\partial(p_1, p_2, \dots, p_n)}{\partial(\dot{q}_1, \dot{q}_2, \dots, \dot{q}_n)} \neq 0$$

However, once we have transformed to the Hamiltonian system we wish to work the transformed version of the Euler-Lagrange equation as well. For that, we will be focussing on developing a new set of equations called the Hamilton-Jacobi equation.

Note that, the Legendre transformation  $[(\dot{\mathbf{q}}, L) \rightarrow (\mathbf{p}, H)]$  is an involution. So,

$$p_k = \frac{\partial L}{\partial \dot{q}_k} \Rightarrow \dot{q}_k = \frac{\partial H}{\partial p_k} \tag{24}$$

Also, from the Euler-Lagrange equation we get:

$$\begin{aligned}\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_k} - \frac{\partial L}{\partial q_k} &= 0 \\ \Rightarrow \dot{p}_k &= \frac{\partial L}{\partial q_k}\end{aligned}$$

Note that  $q_k$  is a passive variable in the Legendre transformation. Hence,

$$\frac{\partial L}{\partial q_k} = -\frac{\partial H}{\partial q_k}$$

Thus,

$$\dot{p}_k = -\frac{\partial H}{\partial q_k} \quad (25)$$

Thus, Eq. 24 and Eq. 25 together make up a system of  $2n$  equations and is called the **Hamiltonian system** while the equations themselves are called **Hamilton's equations**.

### 7.8.3 Symplectic maps

We already have constructed our Hamiltonian system of the generalised coordinates  $(\mathbf{q}, \mathbf{p})$  such that,

$$\begin{aligned}\dot{q}_k &= \frac{\partial H}{\partial p_k} \\ \dot{p}_k &= -\frac{\partial H}{\partial q_k}\end{aligned}$$

Now, a **symplectic map** (also known as a **canonical transformation**) is a transformation of the form

$$\begin{aligned}Q_k &= Q_k(t, \mathbf{q}, \mathbf{p}) \\ P_k &= P_k(t, \mathbf{q}, \mathbf{p})\end{aligned}$$

such that it produces a new Hamiltonian system  $(\hat{H})$

$$\begin{aligned}\dot{Q}_k &= \frac{\partial \hat{H}}{\partial P_k} \\ \dot{P}_k &= -\frac{\partial \hat{H}}{\partial Q_k}.\end{aligned}$$

Hence, a symplectic map is a transformation of the phase space that preserves the Hamiltonian structure.

Consider the two functionals:

$$\begin{aligned}J(\mathbf{q}) &= \int_{t_0}^{t_1} L(t, \mathbf{q}, \dot{\mathbf{q}}) dt, \\ \hat{J}(\mathbf{Q}) &= \int_{t_0}^{t_1} \hat{L}(t, \mathbf{Q}, \dot{\mathbf{Q}}) dt\end{aligned}$$

where,

$$L(t, \mathbf{q}, \dot{\mathbf{q}}) = \sum_{k=1}^n p_k \dot{q}_k - H(t, \mathbf{q}, \mathbf{p}),$$

$$\hat{L}(t, \mathbf{Q}, \dot{\mathbf{Q}}) = \sum_{k=1}^n P_k \dot{Q}_k - H(t, \mathbf{Q}, \mathbf{P})$$

Now, we say that  $J$  and  $\hat{J}$  are variationally equivalent if they produce the same set of extremals. Now, a symplectic map is just a transformation from the phase space  $(\mathbf{q}, \mathbf{p})$  to  $(\mathbf{Q}, \mathbf{P})$  such that the resulting functionals  $J$  and  $\hat{J}$  are variationally equivalent.

One method of constructing a symplectic map is by adding a total differential term to the Lagrangian. We have already seen before that adding any total differential term does not change the extremals of the functional. Hence, we proceed in the same way with a smooth function  $\Phi$ :

$$\sum_{k=1}^n p_k \dot{q}_k - H(t, \mathbf{q}, \mathbf{p}) = \sum_{k=1}^n P_k \dot{Q}_k - \hat{H}(t, \mathbf{Q}, \mathbf{P}) + \frac{d}{dt} \Phi(t, \mathbf{q}, \mathbf{p}).$$

This transformation obviously produces variationally equivalent functionals. We use the transformation function to express  $\Phi \equiv \Phi(t, \mathbf{q}, \mathbf{Q})$  and get:

$$\frac{d}{dt} \Phi(t, \mathbf{q}, \mathbf{Q}) = \sum_{k=1}^n p_k \dot{q}_k - \sum_{k=1}^n P_k \dot{Q}_k + \hat{H}(t, \mathbf{Q}, \mathbf{P}) - H(t, \mathbf{q}, \mathbf{p})$$

But also,

$$\frac{d}{dt} \Phi(t, \mathbf{q}, \mathbf{Q}) = \sum_{k=1}^n \left( \frac{\partial \Phi}{\partial q_k} \dot{q}_k + \frac{\partial \Phi}{\partial Q_k} \dot{Q}_k \right) + \frac{\partial \Phi}{\partial t}$$

Comparing coefficients we get,

$$p_k = \frac{\partial \Phi}{\partial q_k},$$

$$P_k = -\frac{\partial \Phi}{\partial Q_k}$$

and

$$\hat{H}(t, \mathbf{Q}, \mathbf{P}) = H(t, \mathbf{q}, \mathbf{p}) + \frac{\partial \Phi}{\partial t}$$

Note that  $\Phi$  can be considered a function of any two variables from either phase spaces to get different relations pertaining to the transformed Hamiltonian function. An exercise regarding the same has been solved later.

Now, our problem boils down to finding the smooth function  $\Phi$  to solve the Hamilton's system. The function  $\Phi$  in the above equations is called a **generating function**.

### ***The Hamilton-Jacobi Equation:***

We wish to work with symplectic maps to produce simpler Hamiltonian systems that are easier to solve. Let's say we find a generating function that leads to a Hamiltonian such that  $\hat{H} = 0$ . Therefore,

$$\begin{aligned}\dot{Q}_k &= \frac{\partial \hat{H}}{\partial P_k} = 0 \\ \dot{P}_k &= -\frac{\partial \hat{H}}{\partial Q_k} = 0\end{aligned}$$

Thus,

$$\begin{aligned}Q_k &= \alpha_k \\ P_k &= \beta_k\end{aligned}$$

where  $\alpha_k, \beta_k$  are constants.

Now, from the transformed equation, we have,

$$H(t, \mathbf{q}, \mathbf{p}) + \frac{d}{dt}\Phi = \hat{H}(t, \mathbf{Q}, \mathbf{P}) = 0$$

We eliminate the  $p_k$  variables in the above expression to get:

$$H\left(t, q_1, q_2, \dots, q_n, \frac{\partial \Phi}{\partial q_1}, \frac{\partial \Phi}{\partial q_2}, \dots, \frac{\partial \Phi}{\partial q_n}\right) + \frac{\partial \Phi}{\partial t} = 0 \quad (26)$$

Eq. 26 is then called the Hamilton-Jacobi equation. We solve this equation to get the generating function  $\Phi$  and then solve the simpler Hamiltonian system which variationally equivalent to the old one.

#### 7.8.4 Noether's theorem

Let  $J$  be a functional of the form

$$J(y) = \int_{x_0}^{x_1} f(x, y, y', \dots, y^{(n)}) dx.$$

If there exists a function  $\varphi(x, y, y', \dots, y^{(k)})$  such that

$$\frac{d}{dx}\varphi(x, y, y', \dots, y^{(k)}) = 0$$

then the above equation is called a  $k^{\text{th}}$  order conservation law for  $J$ . For example, we have already encountered

$$H = y' \frac{\partial f}{\partial y'} - f = \text{const.}$$

when there is no explicit  $x$ —dependence on  $f$ . This is a first order conservation law for  $f$ .

If there exists such a conservation law then,

$$\varphi(x, y, y', \dots, y^{(k)}) = \text{const.}$$

is known as a **first integral** to the Euler-Lagrange equation.

If we are to find conservation laws for a system we must find the corresponding transformations under which the system is invariant. This was famously stated and proved by the German mathematician Emmy Noether (1882 - 1935). She was able to show that every symmetry (invariant transformation) comes associated with a conservation law. We have seen a similar transformation before - time translation invariance that leads to energy conservation. We will now further generalize the statement to any general invariant transformation.



Let

$$J(y) = \int_{x_0}^{x_1} f(x, y, y') dx$$

Let us consider a one-parameter family of transformation:

$$X = \theta(x, y; \epsilon), \quad Y = \psi(x, y; \epsilon) \quad (27)$$

such that  $\epsilon = 0$  corresponds to the identity transformation. Take as an example the translation transformation:

$$\begin{aligned} X &= x + \epsilon, Y = y, \text{ or} \\ X &= x, Y = y + \epsilon \end{aligned}$$

or the rotation transformation:

$$X = x \cos \epsilon + y \sin \epsilon, Y = -x \sin \epsilon + y \cos \epsilon$$

We denote by  $Y_\epsilon(X)$  the function of  $Y$  w.r.t  $X$  to differentiate it from the transformation relation.

The integrand  $f(x, y, y')$  of the functional  $J$  is said to be variationally invariant over the interval  $[x_0, x_1]$  under transformations  $\theta$  and  $\psi$  if, for all  $\epsilon$  sufficiently small, in any subinterval  $[a, b] \subseteq [x_0, x_1]$  we have

$$\int_a^b f(x, y(x), y'(x)) dx = \int_{a_\epsilon}^{b_\epsilon} f(X, Y_\epsilon(X), \dot{Y}_\epsilon(X)) dX$$

for all smooth functions  $y$  on  $[a, b]$ . Here,

$$a_\epsilon = \theta(a, y(a); \epsilon), \quad b_\epsilon = \psi(b, y(b); \epsilon)$$

In the above case, the transformation is then regarded as a variational symmetry of  $J$ .

The transformation in Eq. 27 can be expanded using Taylor's expansion as:

$$\begin{aligned} X &= \theta(x, y; 0) + \epsilon \left. \frac{\partial \theta}{\partial \epsilon} \right|_{(x, y; 0)} + \mathcal{O}(\epsilon^2), \\ Y &= \psi(x, y; 0) + \epsilon \left. \frac{\partial \psi}{\partial \epsilon} \right|_{(x, y; 0)} + \mathcal{O}(\epsilon^2) \end{aligned}$$

We define,

$$\begin{aligned} \xi(x, y) &:= \left. \frac{\partial \theta}{\partial \epsilon} \right|_{(x, y; 0)}, \\ \eta(x, y) &:= \left. \frac{\partial \psi}{\partial \epsilon} \right|_{(x, y; 0)} \end{aligned}$$

We call the functions  $\xi$  and  $\eta$  the infinitesimal generators for the transformation in Eq. 27. Note that, for  $|\epsilon|$  small we have,

$$\begin{aligned} X &\approx x + \epsilon \xi, \\ Y &\approx y + \epsilon \eta \end{aligned}$$

We are now ready to state Noether's theorem:

**Theorem 7.7** (Noether): Suppose that  $f(x, y, y')$  is variationally invariant on  $[x_0, x_1]$  under the transformation in Eq. 27 with infinitesimal generators  $\xi$  and  $\eta$ . Then,

$$\eta \frac{\partial f}{\partial y'} + \xi \left( f - \frac{\partial f}{\partial y'} y' \right) = \text{const.}$$

along an extremal of

$$J(y) = \int_{x_0}^{x_1} f(x, y, y') dx$$

The conservation law can also be written as:

$$\eta p - \xi H = \text{const.}$$

We have seen before how time translation invariance leads to conservation of energy. Now, suppose  $L$  is variationally invariant under the transformation

$$T = t, Q_1 = q_1 + \epsilon, Q_k = q_k$$

Then,  $\xi = \eta_2 = \eta_3 = 0$  but  $\eta_1 = 1$ . In this case from Noether's theorem we get:

$$p_1 = \frac{\partial L}{\partial \dot{q}_1} = m \dot{q}_1 = \text{const.}$$

This signifies that *space translation invariance leads to conservation of momentum*.

### 7.8.5 Solved Examples

**Exercise 7.17:** Find the Hamiltonian  $H$  for the functionals with integrands:

a)  $f(x, y, y') = y\sqrt{1 + y'^2}$  (the catenary)

b)  $f(x, y, y') = \sqrt{\frac{1+y'^2}{y}}$  (the brachistochrone)

a) The Hamiltonian is defined by the Legendre transformation:  $H = py' - f$ . We have,

$$p = \frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{1 + y'^2}}$$

Thus we get,

$$H = py' - f = \frac{y'}{\sqrt{1 + y'^2}} y' - y\sqrt{1 + y'^2} = -\frac{1}{\sqrt{1 + y'^2}}$$

b) Again, we calculate

$$p = \frac{\partial f}{\partial y'} = \frac{1}{\sqrt{y}} \frac{y'}{\sqrt{1 + y'^2}}$$

And we get,

$$H = py' - f = \frac{y'^2}{\sqrt{y(1+y'^2)}} - \sqrt{\frac{1+y'^2}{y}} = -\frac{1}{\sqrt{y(1+y'^2)}}$$

**Exercise 7.18:** Show that the Hamiltonian for the motion of a single particle of mass  $m$  is:

- a)  $H = \frac{1}{2m}(p_1^2 + p_2^2 + p_3^2) + V(t, \mathbf{q})$  in cartesian coordinates;
- b)  $H = \frac{1}{2m}\left(p_1^2 + \frac{p_2^2}{q_1^2} + p_3^2\right) + V(t, \mathbf{q})$  in cylindrical coordinates with parameters  $(r, \varphi, z)$ ;
- c)  $H = \frac{1}{2m}\left(p_1^2 + \frac{p_2^2}{q_1^2} + \frac{p_3^2}{q_1^2 \sin^2 q_2}\right) + V(t, \mathbf{q})$  in spherical coordinates with parameters  $(r, \theta, \varphi)$ .

a) The kinetic energy of a particle in cartesian coordinates is given by

$$T = \frac{1}{2m}|\mathbf{p}|^2 = \frac{1}{2m} \sum_{k=1}^3 p_k^2$$

Also, momentum is given by  $p_k = m\dot{q}_k$ . This can be verified from that fact that  $T = \frac{1}{2}m(\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_3^2)$  and  $p_k = \frac{\partial L}{\partial \dot{q}_k} = \frac{\partial T}{\partial \dot{q}_k} = m\dot{q}_k$ . Therefore,  $\dot{q}_k = \frac{p_k}{m}$ .

The Lagrangian of the system is therefore:

$$L = \frac{1}{2m} \sum_{k=1}^3 p_k^2 - V(t, \mathbf{q})$$

And the Hamiltonian is:

$$\begin{aligned} H &= \sum_{k=1}^3 p_k \dot{q}_k - L = \sum_{k=1}^3 p_k \frac{p_k}{m} - \left( \frac{1}{2m} \sum_{k=1}^3 p_k^2 - V(t, \mathbf{q}) \right) \\ &= \frac{1}{m} \sum_{k=1}^3 p_k^2 - \frac{1}{2m} \sum_{k=1}^3 p_k^2 + V(t, \mathbf{q}) \\ &= \frac{1}{2m} (p_1^2 + p_2^2 + p_3^2) + V(t, \mathbf{q}) \end{aligned}$$

□

b) The relation between cylindrical coordinates  $(r, \varphi, z)$  and cartesian coordinates  $(x, y, z)$  is given by:

$$x = r \cos \varphi, y = r \sin \varphi, z = z$$

The position vector in cylindrical coordinates is  $\mathbf{r} = r\hat{r} + z\hat{z}$ , while the velocity vector is given by  $\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{r}\hat{r} + r\dot{\varphi}\hat{\varphi} + \dot{z}\hat{z}$ . Hence,

$$\begin{aligned}
T &= \frac{1}{2}m|\mathbf{v}|^2 = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2 + \dot{z}^2) \\
\Rightarrow p_r &= \frac{\partial T}{\partial \dot{r}} = m\dot{r}, \quad p_\varphi = \frac{\partial T}{\partial \dot{\varphi}} = mr^2\dot{\varphi}, \quad p_z = m\dot{z} \\
\Rightarrow \dot{r} &= \frac{p_r}{m}, \quad \dot{\varphi} = \frac{p_\varphi}{mr^2}, \quad \dot{z} = \frac{p_z}{m} \\
\Rightarrow T &= \frac{1}{2m} \left( p_r^2 + \frac{p_\varphi^2}{r^2} + p_z^2 \right)
\end{aligned}$$

And we have,

$$\begin{aligned}
H &= p_r\dot{r} + p_\varphi\dot{\varphi} + p_z\dot{z} - T + V(t, \mathbf{q}) \\
&= \frac{1}{m} \left( p_r^2 + \frac{p_\varphi^2}{r^2} + p_z^2 \right) - \frac{1}{2m} \left( p_r^2 + \frac{p_\varphi^2}{r^2} + p_z^2 \right) + V(t, \mathbf{q}) \\
&= \frac{1}{2m} \left( p_r^2 + \frac{p_\varphi^2}{r^2} + p_z^2 \right) + V(t, \mathbf{q})
\end{aligned}$$

□

c) Spherical coordinates and cartesian coordinates are related as

$$x = r \sin \theta \cos \varphi, \quad y = r \sin \theta \sin \varphi, \quad z = r \cos \theta$$

The position vector is given by  $\mathbf{r} = r\hat{r}$  while the velocity vector is given by  $\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\dot{\varphi}\hat{\varphi}$ . The kinetic energy is given by:

$$\begin{aligned}
T &= \frac{1}{2}m|\mathbf{v}|^2 = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\varphi}^2) \\
\Rightarrow p_r &= m\dot{r}, \quad p_\theta = mr^2\dot{\theta}, \quad p_\varphi = mr^2\sin^2\theta\dot{\varphi} \\
\Rightarrow \dot{r} &= \frac{p_r}{m}, \quad \dot{\theta} = \frac{p_\theta}{mr^2}, \quad \dot{\varphi} = \frac{p_\varphi}{mr^2\sin^2\theta} \\
\Rightarrow T &= \frac{1}{2m} \left( p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\varphi^2}{r^2\sin^2\theta} \right)
\end{aligned}$$

The Hamiltonian thus becomes,

$$\begin{aligned}
H &= p_r\dot{r} + p_\theta\dot{\theta} + p_\varphi\dot{\varphi} - T + V(t, \mathbf{q}) \\
&= \frac{1}{m} \left( p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\varphi^2}{r^2\sin^2\theta} \right) - \frac{1}{2m} \left( p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\varphi^2}{r^2\sin^2\theta} \right) + V(t, \mathbf{q}) \\
&= \frac{1}{2m} \left( p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\varphi^2}{r^2\sin^2\theta} \right) + V(t, \mathbf{q})
\end{aligned}$$

□

**Exercise 7.19:** The Lagrangian for a linear harmonic oscillator is

$$L(t, q, \dot{q}) = \frac{1}{2m}m\dot{q}^2 - \frac{1}{2}kq,$$

where  $m$  is mass and  $k$  is a restoring force coefficient. Show that

$$H(t, q, p) = \frac{1}{2m}(p^2 + m^2\omega^2q^2)$$

where

$$\omega = \sqrt{\frac{k}{m}}.$$

Derive and solve the Euler-Lagrange equation assuming  $m$  and  $k$  are constants. Derive Hamilton's equations and verify that the solution obtained for the Euler-Lagrange equation is also a solution for Hamilton's equation.

We first compute the momentum:  $p = \frac{\partial L}{\partial \dot{q}} = m\dot{q} \Rightarrow \dot{q} = \frac{p}{m}$ . Then by Legendre transformation,

$$\begin{aligned} H = p\dot{q} - L &= p\left(\frac{p}{m}\right) - \left(\frac{1}{2}m\left(\frac{p}{m}\right)^2 - \frac{1}{2}kq^2\right) \\ &= \frac{1}{2m}(p^2 + m^2\omega^2q^2) \end{aligned}$$

The Euler-Lagrange equation to the functional is:

$$\begin{aligned} \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} &= 0 \\ \Rightarrow m\ddot{q} + kq &= 0 \\ \Rightarrow \ddot{q} + \omega^2q &= 0 \end{aligned}$$

The general solution to which is:

$$q(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t), \quad c_1, c_2 \in \mathbb{C} \text{ are arbitrary constants.}$$

From Hamilton's equation we have,

$$\begin{aligned} \dot{q} = \frac{\partial H}{\partial p} &= \frac{p}{m} \text{ and } \dot{p} = -\frac{\partial H}{\partial q} = -\frac{kq}{m} \\ \Rightarrow \ddot{q} &= \frac{\dot{p}}{m} = -\frac{k}{m^2}q \\ \Rightarrow \ddot{q} + \omega^2q &= 0 \end{aligned}$$

Thus, we get the same ODE from Hamilton's equations.

**Exercise 7.20:** For any smooth functions  $\Phi(t, \mathbf{q}, \mathbf{p})$  and  $\Theta(t, \mathbf{q}, \mathbf{p})$ , the Poisson bracket is defined by

$$[\Phi, \Theta] = \sum_{k=1}^n \left( \frac{\partial \Phi}{\partial q_k} \frac{\partial \Theta}{\partial p_k} - \frac{\partial \Phi}{\partial p_k} \frac{\partial \Theta}{\partial q_k} \right).$$

Let  $H$  be the Hamiltonian function associated with a functional  $J$ , and suppose that along the extremals for  $J$

$$\Phi(\mathbf{q}, \mathbf{p}) = \text{const.}$$

The function  $\Phi$  is then called a first integral of the system. Show that

$$[\Phi, H] = 0$$

Prove the converse: if  $[\Phi, H] = 0$ , then  $\Phi$  is a first integral of the system.

We have along an extremal of  $J$ ,

$$\begin{aligned} \Phi(\mathbf{q}, \mathbf{p}) &= \text{const.} \\ \Leftrightarrow \frac{d\Phi}{dt} &= 0 \\ \Leftrightarrow \sum_{k=1}^n \left( \frac{\partial \Phi}{\partial q_k} \dot{q}_k + \frac{\partial \Phi}{\partial p_k} \dot{p}_k \right) &= 0 \\ \Leftrightarrow \sum_{k=1}^n \left( \frac{\partial \Phi}{\partial q_k} \frac{\partial H}{\partial p_k} - \frac{\partial \Phi}{\partial p_k} \frac{\partial H}{\partial q_k} \right) &= 0 \text{ [from Hamilton's equations]} \\ \Leftrightarrow [\Phi, H] &= 0 \end{aligned}$$

□

**Exercise 7.21:** Show that  $\dot{p}_k = [p_k, H]$  and  $\dot{q}_k = [q_k, H]$ .

From the definition of Poisson brackets we have:

$$\begin{aligned} [q_k, H] &= \sum_{i=1}^n \left( \frac{\partial q_k}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial q_k}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \\ &= \frac{\partial q_k}{\partial q_k} \frac{\partial H}{\partial p_k} - 0 \\ &= \dot{q}_k \end{aligned}$$

And,

$$\begin{aligned} [p_k, H] &= \sum_{i=1}^n \left( \frac{\partial p_k}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial p_k}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \\ &= 0 - \frac{\partial p_k}{\partial p_k} \frac{\partial H}{\partial q_k} \\ &= -\dot{p}_k \end{aligned}$$

□

**Exercise 7.22:** Suppose that we regard  $\Phi$  as a function of  $\mathbf{q}$  and  $\mathbf{P}$ . Show that

$$p_k = \frac{\partial \Phi}{\partial q_k}, Q_k = \frac{\partial \Phi}{\partial P_k}, \hat{H} = H + \frac{\partial \Phi}{\partial t}$$

Derive similar equations for a symplectic map if  $\Phi$  is regarded as a function of  $\mathbf{p}$  and  $\mathbf{Q}$ .

Suppose we have a generating function  $\Phi(t, \mathbf{q}, \mathbf{P})$ . We have,

$$\begin{aligned} \sum_{k=1}^n p_k \dot{q}_k - H(t, \mathbf{q}, \mathbf{p}) &= \sum_{k=1}^n P_k \dot{Q}_k - \hat{H}(t, \mathbf{Q}, \mathbf{P}) + \frac{d}{dt}(\Phi(t, \mathbf{q}, \mathbf{P}) - \mathbf{Q} \cdot \mathbf{P}) \\ &\quad [-\mathbf{Q} \cdot \mathbf{P} \text{ term is added as a legendre transformation}] \\ \Rightarrow \frac{d}{dt} \Phi(t, \mathbf{q}, \mathbf{P}) &= \sum_{k=1}^n p_k \dot{q}_k - \sum_{k=1}^n \cancel{P_k \dot{Q}_k} + \sum_{k=1}^n \cancel{P_k \dot{Q}_k} + \sum_{k=1}^n Q_k \dot{P}_k + \hat{H}(t, \mathbf{Q}, \mathbf{P}) - H(t, \mathbf{q}, \mathbf{p}) \\ \Rightarrow \frac{d}{dt} \Phi(t, \mathbf{q}, \mathbf{P}) &= \sum_{k=1}^n p_k \dot{q}_k + \sum_{k=1}^n Q_k \dot{P}_k + \hat{H}(t, \mathbf{Q}, \mathbf{P}) - H(t, \mathbf{q}, \mathbf{p}) \end{aligned}$$

But,

$$\frac{d}{dt} \Phi(t, \mathbf{q}, \mathbf{P}) = \sum_{k=1}^n \left( \frac{\partial \Phi}{\partial q_k} \dot{q}_k + \frac{\partial \Phi}{\partial P_k} \dot{P}_k \right) + \frac{\partial \Phi}{\partial t}$$

Hence, we get by matching coefficients

$$p_k = \frac{\partial \Phi}{\partial q_k}, Q_k = \frac{\partial \Phi}{\partial P_k}, \hat{H} = H + \frac{\partial \Phi}{\partial t}$$

Now, we work with the generating function  $\Phi \equiv \Phi(t, \mathbf{p}, \mathbf{Q})$ . We now have,

$$\begin{aligned} \sum_{k=1}^n p_k \dot{q}_k - H(t, \mathbf{q}, \mathbf{p}) &= \sum_{k=1}^n P_k \dot{Q}_k - \hat{H}(t, \mathbf{Q}, \mathbf{P}) + \frac{d}{dt}(\Phi(t, \mathbf{p}, \mathbf{Q}) + \mathbf{q} \cdot \mathbf{p}) \\ &\quad [\mathbf{q} \cdot \mathbf{p} \text{ term is added as a legendre transformation}] \\ \Rightarrow \frac{d}{dt} \Phi(t, \mathbf{p}, \mathbf{Q}) &= \sum_{k=1}^n \cancel{p_k \dot{q}_k} - \sum_{k=1}^n P_k \dot{Q}_k - \sum_{k=1}^n \cancel{p_k \dot{q}_k} - \sum_{k=1}^n q_k \dot{p}_k + \hat{H}(t, \mathbf{Q}, \mathbf{P}) - H(t, \mathbf{q}, \mathbf{p}) \\ \Rightarrow \frac{d}{dt} \Phi(t, \mathbf{p}, \mathbf{Q}) &= - \sum_{k=1}^n P_k \dot{Q}_k - \sum_{k=1}^n q_k \dot{p}_k + \hat{H}(t, \mathbf{Q}, \mathbf{P}) - H(t, \mathbf{q}, \mathbf{p}) \end{aligned}$$

Also, we have

$$\frac{d}{dt} \Phi(t, \mathbf{p}, \mathbf{Q}) = \sum_{k=1}^n \left( \frac{\partial \Phi}{\partial p_k} \dot{p}_k + \frac{\partial \Phi}{\partial Q_k} \dot{Q}_k \right) + \frac{\partial \Phi}{\partial t}$$

Hence, we get

$$q_k = -\frac{\partial \Phi}{\partial p_k}, P_k = -\frac{\partial \Phi}{\partial Q_k}, \hat{H} = H + \frac{\partial \Phi}{\partial t}$$

□

**Exercise 7.23:** Let  $\Phi = \sum_{k=1}^n q_k Q_k$ . Show that this generating function leads to a symplectic map that essentially interchanges the spatial variables with the momenta variables. This further shows that these variables are on the same footing in the Hamiltonian framework.

Here  $\Phi \equiv \Phi(t, \mathbf{q}, \mathbf{Q})$ . Thus, we have

$$\begin{aligned} p_k &= \frac{\partial \Phi}{\partial q_k} \\ &= Q_k \end{aligned}$$

and,

$$\begin{aligned} P_k &= -\frac{\partial \Phi}{\partial Q_k} \\ &= -q_k \end{aligned}$$

Therefore, this transformation just swaps the position and momenta variables and show these variables are on the same footing in Hamiltonian framework.

**Exercise 7.24:** a) Let  $\Phi = \sum_{k=1}^n q_k P_k$ . Show that this function merely generates the identity transformation.

b) Let  $\Phi = \sum_{k=1}^n g_k(t, \mathbf{q}) P_k$ . Show that this generating function leads to point transformations; i.e., the  $Q_k$  depend only on  $\mathbf{q}$  and  $t$ .

a) Here,  $\Phi \equiv \Phi(t, \mathbf{q}, \mathbf{P})$ . Hence,

$$\begin{aligned} p_k &= \frac{\partial \Phi}{\partial q_k} \\ &= P_k \end{aligned}$$

and,

$$\begin{aligned} Q_k &= \frac{\partial \Phi}{\partial P_k} \\ &= q_k \end{aligned}$$

So it just generates the identity transformation.

b) Here,  $\Phi \equiv \Phi(t, \mathbf{q}, \mathbf{P})$ . We again have,

$$\begin{aligned} p_k &= \frac{\partial \Phi}{\partial q_k} \\ &= P_k \frac{\partial g_k}{\partial q_k}(t, \mathbf{q}) \end{aligned}$$

and,



$$\begin{aligned} Q_k &= \frac{\partial \Phi}{\partial P_k} \\ &= g_k(t, \mathbf{q}). \end{aligned}$$

Hence,  $Q \equiv g(t, \mathbf{q})$ . Therefore this leads to point transformations in  $\mathbf{q}$ .

**Exercise 7.25:** Let

$$J(y) = \int_{x_0}^{x_1} y'^2 dx$$

Derive the Hamilton-Jacobi equation corresponding to this functional. Solve the Euler-Lagrange equation for this functional and construct a solution to the Hamilton-Jacobi equation.

*Solving the Euler-Lagrange for the functional:*

$$\begin{aligned} \frac{d}{dt} \frac{\partial L}{\partial y'} &= \frac{\partial L}{\partial y} \\ \Rightarrow 2y'' &= 0 \\ \Rightarrow y(x) &= c_1 x + c_2, \quad c_1, c_2 \in \mathbb{R} \end{aligned}$$

The generalised momentum is given by:  $p = \frac{\partial L}{\partial y'} = 2y' \Rightarrow y' = \frac{p}{2}$ .

Now, we have the Hamiltonian:

$$\begin{aligned} H = py' - L &= p\left(\frac{p}{2}\right) - \left(\frac{p}{2}\right)^2 \\ &= \frac{p^2}{4} \end{aligned}$$

The Hamilton-Jacobi equation thus becomes:

$$\begin{aligned} H(y, p) + \frac{\partial \Phi}{\partial t} &\equiv H\left(q, \frac{\partial \Phi}{\partial y}\right) + \frac{\partial \Phi}{\partial x} = 0 \\ \Rightarrow \frac{1}{4} \left(\frac{\partial \Phi}{\partial y}\right)^2 + \frac{\partial \Phi}{\partial x} &= 0 \end{aligned}$$

*Ansatz:* The solution to the above equation is separable. We let

$$\Phi(x, y) = \varphi(x) + \psi(y)$$

Then,

$$\begin{aligned} -\frac{d\varphi}{dx} &= \frac{1}{4} \left(\frac{d\psi}{dy}\right)^2 = \text{const.} = k \text{ (say)} \\ \Rightarrow \varphi(x) &= -kx + l_1 \text{ and } \psi(y) = 2\sqrt{k}y + l_2 \end{aligned}$$

Thus,

$$\Phi(x, y) = 2\sqrt{k}y - kx + l$$

where  $k, l_1, l_2, l$  are arbitrary constants. This is the final solution to the Hamilton-Jacobi equation.

**Exercise 7.26:** The Lagrangian for the motion of a particle of unit mass in the plane under the action of uniform field is

$$L(t, \mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2}(\dot{q}_1^2 + \dot{q}_2^2) - gq_2$$

Derive the Hamilton-Jacobi equation and show that a solution to this equation is

$$\Phi = -\left(\frac{1}{2}\alpha^2 + gk\right)t + \alpha q_1 + \sqrt{2g}\frac{2}{3}(k - q_2)^{\frac{3}{2}}$$

where  $\alpha$  and  $k$  are arbitrary constants.

The generalised momenta are:

$$\begin{aligned} p_1 &= \frac{\partial L}{\partial \dot{q}_1} = \dot{q}_1 \\ p_2 &= \frac{\partial L}{\partial \dot{q}_2} = \dot{q}_2 \end{aligned}$$

The Hamiltonian is:

$$\begin{aligned} H &= p_1\dot{q}_1 + p_2\dot{q}_2 - L \\ &= \frac{1}{2}(p_1^2 + p_2^2) + gq_2 \end{aligned}$$

Therefore, the Hamilton-Jacobi equation becomes:

$$\begin{aligned} &H\left(q_1, q_2, \frac{\partial \Phi}{\partial q_1}, \frac{\partial \Phi}{\partial q_2}\right) + \frac{\partial \Phi}{\partial t} = 0 \\ \Rightarrow &\frac{1}{2}\left(\left(\frac{\partial \Phi}{\partial q_1}\right)^2 + \left(\frac{\partial \Phi}{\partial q_2}\right)^2\right) + gq_2 + \frac{\partial \Phi}{\partial t} = 0 \end{aligned}$$

Again,

*Ansatz:*  $\Phi(t, q_1, q_2) = \Phi_1(t) + \Phi_2(q_1) + \Phi_3(q_2)$ . We have,

$$\frac{1}{2}\left(\left(\frac{d\Phi_2}{dq_1}\right)^2 + \left(\frac{d\Phi_3}{dq_2}\right)^2\right) + gq_2 + \frac{d\Phi_1}{dt} = 0$$

Thus,

$$\begin{aligned} \frac{d\Phi_2}{dq_1} &= \text{const.} = \alpha \text{ (say)} \\ \Rightarrow \Phi_2(q_1) &= \alpha q_1 + c_1. \end{aligned}$$

And,

$$\begin{aligned}
\frac{1}{2} \left( \frac{d\Phi_3}{dq_2} \right)^2 + gq_2 &= \text{const.} = \beta \text{ (say)} \\
\Rightarrow \frac{d\Phi_3}{dq_2} &= \sqrt{2(\beta - gq_2)} \\
\Rightarrow \Phi_3(q_2) &= \sqrt{2g} \int \sqrt{\frac{\beta}{g} - q_2} dq_2 \\
&= \sqrt{2g} \frac{2}{3} (k - q_2)^{\frac{3}{2}} + c_2, \left[ \frac{\beta}{g} = k \text{ (say)} \right]
\end{aligned}$$

Therefore, we also get:

$$\begin{aligned}
\frac{d\Phi_1}{dt} &= -\frac{1}{2}\alpha^2 - \beta \\
&= -\frac{1}{2}\alpha^2 - gk \\
\Rightarrow \Phi_1(t) &= \left( -\frac{1}{2}\alpha^2 - gk \right) t + c_3
\end{aligned}$$

Hence, the final solution to the Hamilton-Jacobi equation is:

$$\begin{aligned}
\Phi(t, q_1, q_2) &= \Phi_1(t) + \Phi_2(q_1) + \Phi_3(q_2) \\
&= -\left( \frac{1}{2}\alpha^2 + gk \right) t + \alpha q_1 + \sqrt{2g} \frac{2}{3} (k - q_2)^{\frac{3}{2}} + c
\end{aligned}$$

Which satisfies the given solution in the question [ $c = 0$ ].

□

**Exercise 7.27:** Let

$$J(y) = \int_{x_0}^{x_1} xy'^2 dx.$$

Show that the transformation

$$X = x + \epsilon 2x \ln(x), Y = (1 + \epsilon)y$$

is a variational symmetry for  $J$ .

We have,

$$f(x, y, y') = xy'^2$$

We know that the transformation is variationally invariant if

$$J(Y_\epsilon) = J(y) + \mathcal{O}(\epsilon^2) + \dots$$

We compute  $J(Y_\epsilon)$ :

$$J(Y_\epsilon) = \int_{x_{0_\epsilon}}^{x_{1_\epsilon}} X \dot{Y}_\epsilon^2 dX$$

Now,

$$\begin{aligned} \dot{Y}_\epsilon &= \frac{dY_\epsilon}{dx} / \frac{dX}{dx} \\ &= (1 + \epsilon)y'(1 + \epsilon(2 \ln x + 2))^{-1} \\ &= (1 + \epsilon)y'(1 - \epsilon(2 \ln x + 2) + \mathcal{O}(\epsilon^2)) + \dots \\ &= y'(1 - \epsilon(1 + 2 \ln x)) + \mathcal{O}(\epsilon^2) + \dots \\ \Rightarrow \dot{Y}_\epsilon^2 &= y'^2(1 - 2\epsilon(1 + 2 \ln x)) + \mathcal{O}(\epsilon^2) + \dots \end{aligned}$$

Thus we get,

$$\begin{aligned} X \dot{Y}_\epsilon^2 &= (x + \epsilon 2x \ln x)y'^2(1 - 2\epsilon(1 + 2 \ln x)) + \mathcal{O}(\epsilon^2) + \dots \\ &= xy'^2 + \epsilon(2x \ln(x)y'^2 - 2x(1 + \ln(x))y'^2) + \mathcal{O}(\epsilon^2) + \dots \\ &= xy'^2 + \epsilon(-2xy'^2 - 2x \ln(x)y'^2) + \mathcal{O}(\epsilon^2) + \dots \end{aligned}$$

Also,

$$dX = (1 + \epsilon(2 \ln(x) + 2))dx$$

Therefore we have,

$$\begin{aligned} f(X, Y_\epsilon, \dot{Y}_\epsilon) dX &= (xy'^2 + \epsilon(-2xy'^2 - 2x \ln(x)y'^2))(1 + \epsilon(2 \ln(x) + 2))dx + \mathcal{O}(\epsilon^2) + \dots \\ &= (xy'^2)dx + \epsilon(\cancel{-2xy'^2} - \cancel{2x \ln(x)y'^2} + \cancel{2xy'^2} + \cancel{2x \ln(x)y'^2})dx + \mathcal{O}(\epsilon^2) + \dots \\ &= (xy'^2)dx + \mathcal{O}(\epsilon^2) + \dots \\ &= f(x, y, y')dx + \mathcal{O}(\epsilon^2) + \dots \end{aligned}$$

Hence,

$$J(Y_\epsilon) = J(y) + \mathcal{O}(\epsilon^2) + \dots$$

Thus, the given transformation is a variational symmetry for  $J$ . □

**Exercise 7.28:** Consider the functional from the previous exercise

$$J(y) = \int_{x_0}^{x_1} xy'^2 dx.$$

Show that the transformation

$$X = (1 + \epsilon)x, \quad Y = y$$

is a variational symmetry and establish the corresponding conservation law.

Here,

$$\begin{aligned}\dot{Y}_\epsilon &= \frac{dY_\epsilon}{dx} / \frac{dX}{dx} \\ &= y' \cdot \frac{1}{1+\epsilon}\end{aligned}$$

And also,

$$dX = (1 + \epsilon)dx$$

Therefore,

$$\begin{aligned}f(X, Y_\epsilon, \dot{Y}_\epsilon)dX &= X\dot{Y}_\epsilon^2 dX \\ &= \frac{x(1+\epsilon)y'^2(1+\epsilon)dx}{(1+\epsilon)^2} \\ &= xy'^2 dx \\ &= f(x, y, y')dx\end{aligned}$$

Thus,

$$J(Y_\epsilon) = J(y)$$

and the transformation is variationally symmetric. Now, the infinitesimal generators for the transformation are:

$$\xi(x, y) = x, \quad \eta(x, y) = 0$$

By Noether's theorem we have,

$$\begin{aligned}\eta \frac{\partial f}{\partial y'} + \xi \left( f - \frac{\partial f}{\partial y'} y' \right) &= \text{const.} \\ \Rightarrow x(xy'^2 - 2xy'^2) &= \text{const.} \\ \Rightarrow x^2 y'^2 &= \text{const.}\end{aligned}$$

Thus, the quantity  $x^2 y'^2$  is conserved for this Lagrangian.

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